

Noisy Expectations and Business Cycles

Identification with Local Projections

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Abstract

How noisy are expectations about future aggregate productivity, and does that noise matter for the business cycle? Applied to U.S. data, roughly one third of the new information in consumer-confidence signals is noise, yet noise shocks have only small and transitory effects on real activity, while news shocks generate persistent expansions. We obtain these answers from a simple and flexible local-projection framework that identifies either the Wold innovation of TFP growth or the structural news shock depending on whether surprise innovations are conditioned on, together with a finite-sample bias correction and model-free long-run statistics for the news content of both the signal level and the flow of new information, robust to gradual technology diffusion and multi-horizon news aggregation. An identification theorem underpins the framework: a structural impulse response is attainable from a finite lead window if and only if the window spans the lead support of the news shock in the innovations of the observables, which characterizes when full-history recoverability is essential. Thus, expectation signals can be substantially noisy without noise being an important source of business-cycle fluctuations.

Keywords: Total Factor Productivity, News Shocks, Noise, Moment Conditions.

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Introduction

Expectations about future productivity are a natural source of business-cycle fluctuations. But expectations need not reflect fundamentals alone. Agents may receive genuine information about future productivity, called news, as well as signals that ultimately prove unrelated to fundamentals, called noise. This distinction has motivated a large literature seeking to identify news and noise shocks and quantify their macroeconomic effects. Yet the literature has reached markedly different conclusions about the importance of noise. Some approaches attribute a substantial fraction of aggregate fluctuations to non-fundamental disturbances, while others find a much smaller role.

This paper shows that two distinct issues underlie these differences. The first is an identification issue. Existing methods exploit different information sets and therefore need not recover the same economic object. A shock identified from past and present observables is not necessarily the structural shock that could be recovered if future observables were also available. Comparing finite-lead and full-history procedures as alternative estimators of a common object can therefore be misleading: they may instead identify different points on an information-dependent identification frontier. The second issue is economic. The amount of noise contained in an expectation signal and the importance of noise shocks for aggregate fluctuations are conceptually distinct quantities. A signal can be substantially noisy even if the economy responds little to its non-fundamental component.

We address both questions for U.S. aggregate TFP with a deliberately transparent toolkit. A single local-projection framework identifies, on the same data and first-stage innovations, either the Wold innovation associated with the noise representation of Chahrour and Jurado (2018) or the structural news shock, depending on whether estimated surprise innovations are conditioned on; on the noise side, jointly partialling out future TFP realizations and surprise innovations isolates the structural noise shock. The choice between these specifications is therefore a choice of estimand rather than a horse race between estimators. We derive their probability limits and asymptotic distributions, characterize the leading $O(T^{-1})$ bias that arises when many estimated innovations serve as controls, and propose a residual-augmented correction whose magnitude is itself empirically diagnosable: it is negligible in our U.S. application but substantial in the Chahrour–Jurado benchmark in which the projection controls absorb almost all of the variation in the signal (Propositions 2–3).

An important practical advantage of the framework is its simplicity and flexibility. Once the relevant information set is specified, news and noise shocks are recovered through standard local projections augmented with TFP leads and estimated surprise innovations: ordinary regressions, run equation by equation with Newey–West standard errors. The approach therefore offers a substantially simpler implementation than the spectral approach of Chahrour and Jurado (2022) and the dynamic-rotation VAR procedure of Forni et al. (2017a), and adding outcome variables or estimated factors leaves the logic of the projections unchanged. This simplicity is not obtained by imposing stronger

identifying assumptions. Theorem 1 makes explicit the informational conditions under which the projections recover each structural object: the method is simple computationally, not loose conceptually. The framework is flexible on the shock structure itself. Chahrour and Jurado (2022) model technology as purely exogenous, with no separation between anticipated and unanticipated innovations, and Forni et al. (2017a) likewise operate without a structural surprise shock: their fundamental is driven by the delayed long-run shock alone. Both structures are nested here: imposing a zero surprise variance makes the surprise controls redundant and reproduces their estimands as special cases, while the general case identifies surprise, news, and noise jointly. This greater generality comes at no cost in implementation: all three shocks are identified within the same local-projection framework, by standard regressions. The practitioner can therefore maintain the two-shock structure of the existing literature or let the data speak on all three shocks. And the data do speak: TFP growth is dominated by its unanticipated component.

Our second set of tools asks a different question: how much genuine information does an expectation signal contain? We distinguish the news share in the *stock* of the signal from the news share in the *flow* of newly arriving information. We develop long-run covariance statistics for both objects that remain valid when productivity adjusts gradually and signals aggregate news arriving at multiple horizons (Section 4). This distinction matters because conventional finite-horizon measures can substantially overstate news content when signals are persistent or combine information about fundamentals at different dates. The same framework also yields a data-driven estimate of the horizon over which the signal’s information about future productivity materializes, which provides a candidate lead window for the identification procedure; whether that window suffices is then checked in the application against the observable implications of the identification theory.

Applied to U.S. productivity and consumer-confidence data, these tools lead to a simple but consequential conclusion. Consumer-confidence measures contain substantial information about future productivity, but that information is far from clean. Our preferred long-run statistic implies that roughly 37 percent of the quarterly flow of new information in both survey signals is noise, and a descriptive reading of the raw signal level suggests an even larger share. Yet this substantial signal noise does not translate into large noise-driven fluctuations. Across alternative identification procedures, news shocks generate persistent increases in productivity and real activity, whereas noise shocks have small and transitory effects and no permanent effect on productivity. By contrast, a VAR-implied forecast of future TFP contains essentially no detectable news once finite-sample projection noise is removed, highlighting the importance of using expectation measures that contain information beyond the econometrician’s macroeconomic information set.

These findings separate two questions that are often treated together. How noisy are expectations? And how important are noise shocks for the business cycle? Our evidence gives different answers: expectations can be substantially noisy without noise being an important source of aggregate fluctua-

tions.

These conclusions rest on an identification theory developed in the first part of the paper (Theorem 1). We characterize the minimum amount of future information required to identify structural impulse responses in environments with anticipated information: the structural news response is attainable from a finite lead window if and only if the window spans the *lead support* k^* of the news shock in the innovations of the observables. This organises the identification problem as a three-rung ladder, ordered by the information each rung requires: the Wold response, attainable from current information; the structural news and noise responses, attainable once the window spans k^* ; and the full-history recoverable responses of Chahrour and Jurado (2022). Identifying the impulse response does not require recovering the underlying shock: with one-shot materialization the response is identified after q periods even though the shock itself is never perfectly recovered. Under gradual diffusion with an insufficient information set the lead support is generically infinite and no finite-lead projection is exact. The result is complementary to Chahrour and Jurado (2018), who show that news and noise representations can be observationally equivalent from the agents' perspective, and to Chahrour and Jurado (2022), whose recoverability condition characterizes identification from the complete two-sided history: our result characterizes exactly when, and how far before that limit, structural responses can be identified from finite future information. It also explains why estimates from different news-shock procedures need not coincide even when each procedure is internally valid: the information sets they exploit determine the structural objects they can recover.

The paper contributes to three strands of the literature on news/noise identification. The dynamic-rotation SVAR approach of Forni et al. (2017a) exploits the asymptotic divergence of news and noise responses in a parametric SVAR; our method recovers the same noise-representation target object via a single local projection with explicit identifying assumptions. Chahrour and Jurado (2022) establish a general non-parametric identification result under recoverability and propose a two-sided spectral estimator; our local projection is the natural finite-lead counterpart, and the gap between the two estimators in the same data is itself a useful diagnostic for the importance of full-history information. Structural VARMA approaches impose theory-driven dynamic restrictions to recover shocks exactly in population; we instead exploit predictive relationships between observable signals and future TFP, sidestepping the need for parametric higher-order dynamics.

Although the empirical analysis focuses on aggregate TFP, the identification is fully general: it applies whenever agents observe a noisy signal about a forward-looking fundamental whose realisations the econometrician later records: dividend news ultimately observable in payments, or fiscal announcements that subsequently materialise in fiscal data. Projecting on the future realisation without surprise controls delivers the Chahrour and Jurado (2018) target, adding the surprise residuals isolates the structural news shock, and the model-free diagnostics carry over. Our focus on TFP is dictated by the empirical literature we engage with, not by any methodological constraint.

The remainder of the paper proceeds as follows. Section 1 introduces the environment, the information sets, and the structural objects. Section 2 presents the local-projection procedure and its finite-sample correction. Section 3 characterizes the finite-lead identification frontier. Section 4 develops the long-run measures of signal information content. Section 5 summarizes the simulation evidence, including the Chahrour–Jurado benchmark. Section 6 applies the methods to U.S. productivity and expectation data. Section 7 concludes. Detailed simulation designs, additional tables and figures, and the conventional empirical benchmarks are collected in the Online Appendix.

1 Framework and identification objects

Consider the following Structural Moving Average (SMA) representation for the $n \times 1$ vector of variables:

$$X_t = \Theta_1(L)u_t + \Theta_2(L)\varepsilon_t + \Theta_3(L)\nu_t + \Theta_4(L)\chi_t, \quad (1)$$

where the vector $X_t = (\Delta z_t, y_t, x_t')$ consists of Δz_t , total factor productivity (TFP); y_t , the variable of interest; and x_t , an $(n - 2) \times 1$ vector of other relevant variables. These variables are driven by the structural shocks u_t , the non-anticipated (surprise) technology shock; ε_t , the anticipated (news) technology shock; ν_t , the noise shock; and χ_t , other structural shocks. The polynomials $\Theta_i(L)$, for $i = 1, 2, 3, 4$, represent the moving average components applied to the respective shocks.

Define the vector of normalized structural shocks as:

$$e_t = (u_t, \varepsilon_t, \nu_t, \chi_t')'$$

with $E[e_t] = 0$ and $E[e_t e_t'] = I$.

Our setup consists of a stochastic process for total factor productivity (TFP), denoted z_t , and a noisy signal about its future changes. TFP follows the exogenous univariate process:

$$\Delta z_t = \Delta z_t^{(1)} + \Delta z_t^{(2)}, \quad (2)$$

where Δ denotes the first difference operator, with components

$$\Delta z_t^{(1)} = a(L)u_t, \quad (3)$$

$$\Delta z_t^{(2)} = b(L)\varepsilon_{t-q}. \quad (4)$$

The error terms u_t and ε_t are white-noise processes with variances σ_u^2 and σ_ε^2 , uncorrelated at all

leads and lags. The lag polynomials $a(L) = \sum_{i=0}^m a_i L^i$ and $b(L) = \sum_{i=0}^n b_i L^i$ have finite orders m and n , or infinite orders with square-summable coefficients, so that each component can follow, for instance, a stationary autoregression.

The integer $q \geq 0$ is the anticipation lag of TFP improvements: for $q \geq 1$, news announced at t leaves TFP unchanged until $t + q$, although agents anticipate its eventual materialisation. The specification does not force news to have zero impact effect: $q = 0$ is allowed, in which case we follow the definition of news shocks in Beaudry and Portier (2014) and Portier (2014): some b_i with $i > 0$ exceeds b_0 , or more generally $b(1) > b_0$, so that the long-run effect of the improvement exceeds its initial impact. This restriction captures gradual technology diffusion without imposing a functional form on it. The structural-news identification of Theorem 1 focuses on genuine anticipation, $q \geq 1$ (Assumption A.2).

The surprise polynomial is unrestricted except at the long run: we impose $a(1) = 0$, so that u_t has no permanent effect on the *level* of TFP while remaining free to be long-lasting. This is the restriction under which the long-run variance of TFP growth is attributable to news alone, the key input of the long-run statistics of Section 4, and it is consistent with the studies distinguishing unexpected from news components in aggregate TFP (Barsky and Sims, 2011; Forni and Gambetti, 2014; Fève and Guay, 2018; Bouakez and Kemoe, 2023), in which the non-news component is strongly mean-reverting. The intuition is mechanical: these studies use the Fernald (2014) series, which behaves like a random walk; if news diffuses slowly, $\Delta z_t^{(2)}$ is positively serially correlated, so the surprise component must be negatively serially correlated for TFP growth to remain unpredictable, exactly the configuration that a stationary $a(L)$ with $a(1) = 0$ accommodates.

Agents receive a noisy signal about the news shock:

$$s_t = \varepsilon_t + \nu_t, \tag{5}$$

where the noise ν_t is a white-noise process with variance σ_ν^2 , orthogonal to u_t and ε_t at all leads and lags. The noise captures extrinsic uncertainty, unrelated to TFP fundamentals (Blanchard et al., 2013; Forni et al., 2017b), and serves as a proxy for “animal spirits” or “sunspots” (Barsky and Sims, 2012). The whiteness of ν_t is a normalisation for the structural reading rather than a requirement of the procedure: the empirical implementation projects the signal on the date- $t-1$ information set and operates on the resulting innovation, which is serially uncorrelated by construction whatever the persistence of the noise; serial correlation in ν_t affects only the mapping from the estimated noise variance to $\text{Var}(\nu_t)$, in a known direction (Section 4). For now, the econometrician observes s_t without error; this assumption is relaxed in the practical implementation.¹

¹In the practical implementation of our approach, we discuss how to extract white noise for the signal and address the resulting implications.

Observables. For the identification analysis of structural IRFs (Sections 2–3), we maintain the simple signal (5); the TFP process is (2)–(4), with one notational adjustment: from here on the news polynomial is dated from the *announcement* rather than the *materialisation*: $b(L)$ denotes L^q times the polynomial of (4), so that $\Delta z_t = a(L) u_t + b(L) \varepsilon_t$ with $b_j = 0$ for $j < q$ by construction (Assumption A.2 below), and b_{q+i} corresponds to the b_i of (4). The variable of interest $y_{j,t}$ inherits from (1) the structural moving-average representation

$$y_{j,t} = \Theta_{j,1}(L) u_t + \Theta_{j,2}(L) \varepsilon_t + \Theta_{j,3}(L) \nu_t + \Theta_{j,4}(L)' \chi_t, \quad (6)$$

where $\Theta_{j,i}(L)$ is the row of $\Theta_i(L)$ in (1) corresponding to $y_{j,t}$, and χ_t is the vector of other structural shocks of (1), orthogonal to $(u_t, \varepsilon_t, \nu_t)$ at all leads and lags. Let $X_t = (\Delta z_t, s_t, w_t')'$ collect Δz_t , the signal s_t , and an auxiliary vector w_t of macroeconomic observables (possibly including principal components), and let $X_{t-1}^{(p)} = (X'_{t-1}, \dots, X'_{t-p})'$ collect p of its lags. Throughout, $\overline{\text{sp}}(\cdot)$ denotes the closed linear span, in L^2 , of the indicated set of variables.

The identification problem. Neither structural shock of interest is directly observable: the signal confounds news with noise, and TFP growth confounds surprises with materialising past news. Realised *future* TFP provides the missing link of Chahrour and Jurado (2018): the news shock materialises in Δz_{t+q} (Assumption A.2 below), so projecting on the future realisation can separate news from noise. But the realisation arrives mixed with the surprises $u_{t+1}, \dots, u_{t+q}$ that occur in between, and the residualised signal must be purged of them as well. The Wold innovation of Δz_t , by contrast, is an object of the observables' own information and is always attainable. This observation suggests a simple three-step procedure, presented in Section 2; Section 3 then characterises exactly how much future information each structural object requires.

1.1 Assumptions

A.1 (Stationarity.) X_t is jointly covariance-stationary with finite fourth moments and a strong-mixing condition sufficient for HAC inference.

A.2 (News structure.) The surprise polynomial satisfies $a_0 \neq 0$ (surprises hit on impact). The news polynomial satisfies $b_j = 0$ for $j < q$ and $b_q \neq 0$, where $q \geq 1$ is the news anticipation horizon: ε_t does not affect TFP for q periods after announcement, and materialises with non-zero loading b_q in Δz_{t+q} .

A.3 (Informational sufficiency for surprises.) The lagged observables span the predictable part of TFP growth, so that its innovation is the current surprise: as $p \rightarrow \infty$, $\Delta z_t - P[\Delta z_t \mid X_{t-1}^{(p)}] \xrightarrow{L^2} 0$

$a_0 u_t$ (in particular, $u_t \in \overline{\text{sp}}(X_s : s \leq t)$). The first-stage residual $\hat{u}_t$ denotes the sample counterpart.

A.3' (*Informational sufficiency for news.*) The news shocks are recoverable from the current and past history of the observables: $\varepsilon_t \in \overline{\text{sp}}(X_s : s \leq t)$ for every t .

A.4 (*Non-degenerate news content.*) Partialling the surprises and the past out of the future realisation leaves exactly the current news: as $p \rightarrow \infty$, $\Delta z_{t+q} - P[\Delta z_{t+q} \mid u_{t:t+q}, X_{t-1}^{(p)}] \xrightarrow{L^2} b_q \varepsilon_t$. The condition fails when the controls absorb the news content, as when the signal confounds news with surprise innovations that materialise contemporaneously.

Assumptions **A.2–A.4** have direct empirical content: each generates an observable diagnostic, implemented in Section 6. Their primitive underpinnings, the relation of the three-shock structure to the environments of Forni et al. (2017a) and Chahrour and Jurado (2022), and the margins by which each assumption holds in the data are discussed in Section 3.1.

2 A Simple and Flexible Local-Projection Procedure

The identification problem of Section 1 calls for a procedure that requires nothing beyond ordinary least squares. It consists of three steps.

- Step 1** (*Surprise.*) Recover the unanticipated TFP innovation $\hat{u}_t$ as the residual of current TFP growth on past information.
- Step 2** (*News.*) Use the future TFP realisation Δz_{t+q} as the focus regressor and partial out the estimated surprise innovations and past information: by Frisch–Waugh–Lovell, the residualised focus isolates the anticipated component and identifies the news response.
- Step 3** (*Noise.*) Use the expectation signal as the focus regressor and partial out future TFP realisations, the surprise innovations, and past information: the residualised signal isolates the noise component.

Each step is implemented as an otherwise standard local projection,

$$y_{j,t+h} = \beta_h r_t + \Gamma_h' C_t + e_{j,t+h}, \quad (7)$$

run for any outcome variable with Newey–West standard errors: the focus regressor r_t and the conditioning set C_t , always including the lag block $X_{t-1}^{(p)}$, determine the shock-specific estimand. The

method thus nests the shock structures underlying the leading alternatives while allowing for the empirically relevant coexistence of surprise, news, and noise shocks. This greater generality does not come at the cost of a more elaborate estimator: all three shocks are identified within a common local-projection framework using standard regressions. We refer to the procedure as *BFG* throughout the simulations and the empirical application. This section displays the regressions of each step, presents the menu of estimands obtained by varying the conditioning information, states the formal identification results, and derives the finite-sample correction. Section 3 then locates these estimands on a three-rung identification ladder and characterises exactly how much future information each rung requires.

2.1 Step 1: Recovering the Surprise Shock

Suppose the TFP process is governed by equations (2)–(4).

The vector X_t of relevant variables in equation (1) may include business cycle factors, such as those identified by McCracken and Ng (2016), to account for potential missing information about the business cycle. This approach helps mitigate the risk of non-fundamental representation issues (see Forni and Gambetti, 2014; Beaudry et al., 2019).

In the first step, the surprise shock is identified using the first equation of the DGP (1). Specifically, we regress Δz_t on the past values of X_t up to time $t - 1$:

$$\Delta z_t = \Psi_z(L)X_{t-1} + u_t,$$

where $\Psi_z(L)$ captures the lag structure of X_{t-1} , and u_t represents the non-anticipated (surprise) TFP shock.

Under the assumptions of Forni et al. (2017a) and Chahrour and Jurado (2022), this first step becomes unnecessary since TFP depends on a single structural shock, which, in our framework, corresponds to the news shock.

2.2 Step 2: Identifying the News Shock

The variable of interest admits the structural moving-average representation (6); the objective is to trace its impulse responses to the surprise shock u_t , the news shock ε_t , and the noise shock ν_t .

For the news shock, we project on the future realisation Δz_{t+q} , cleaned of the intervening surprises by the first-stage residuals of Step 1:

$$y_{j,t+h} = \beta_h^\varepsilon \Delta z_{t+q} + \sum_{k=0}^q \delta_{h,k} \hat{u}_{t+k} + \gamma_h^{\varepsilon'} X_{t-1}^{(p)} + e_{j,t+h}^\varepsilon. \quad (8)$$

The Frisch–Waugh–Lovell logic (residualising Δz_{t+q} on the surprise block and the lag block leaves $b_q \varepsilon_t$, so that $b_q \hat{\beta}_h^\varepsilon$ estimates the structural news IRF) is formalised in Proposition 1 below. In the case where no non-anticipated shocks affect TFP, as in Forni et al. (2017a) and Chahrour and Jurado (2022), the same specification applies with the $\hat{u}$ block omitted.

An important question is how to choose the lead-time q of Δz_{t+q} to avoid weak identification: the value of q should be set to a horizon where the impact of the news shock on TFP is significant. Section 4 provides precisely this procedure: the plateau selector $h^* = \arg \max_h \theta_h^{\text{stat,adj}}$ of (27) estimates the full materialisation horizon of the news from the data, is by construction the minimum lead window that purges the news content of the signal, and requires no auxiliary model. In our empirical sample it selects $h^* = 20$ for all three signals, the lead window used throughout Section 6.

2.3 Step 3: Identifying the Noise Shock

On the noise side the focus regressor is the signal itself, with the future-TFP block *and* the surprise residuals among the controls:

$$y_{j,t+h} = \beta_h^{\varepsilon,\nu} s_t + \sum_{k=0}^q \delta_{h,k}^\varepsilon \Delta z_{t+k} + \sum_{k=0}^q \eta_{h,k} \hat{u}_{t+k} + \gamma_h^{\varepsilon,\nu'} X_{t-1}^{(p)} + e_{j,t+h}^{\varepsilon,\nu}. \quad (9)$$

Corollary 1 below establishes that this combined specification is the consistent estimator of the noise IRF, whose residualised signal converges to ν_t , and the empirical application implements it.

In practice the specification may be augmented with a contemporaneous block $\tilde{X}_t$ ensuring that s_t is not influenced by structural shocks other than news and noise: Forni et al. (2017a) propose an ordering that includes the federal funds rate to filter out signals related to monetary policy.

For a signal of the form adopted by Chahrour and Jurado (2022), $s_t = \sum_{i=0}^\infty \varepsilon_{t+i} + \nu_t$, the lead block extends accordingly, $\sum_{i=0}^h \delta_i \Delta z_{t+i}$, the multi-horizon case covered by the general structure (16), with the lead window selected by h^* of Section 4.

2.4 One Framework, Alternative Estimands

Dropping the surprise residuals changes the estimand, not the machinery. Without surprise controls, the news-side projection focuses on current TFP growth,

$$y_{j,t+h} = \beta_h^W \Delta z_t + \gamma_h^{W'} X_{t-1}^{(p)} + e_{j,t+h}^W, \quad (10)$$

and targets the Wold innovation of Δz_t , the technology-side Chahrour and Jurado (2018) target $\mathcal{O}_1$ (and, under the zero-impact reading of Barsky and Sims (2011), the surprise IRFs). The leads-only

noise projection,

$$y_{j,t+h} = \beta_h^{W,\nu} s_t + \sum_{k=0}^q \delta_{h,k}^W \Delta z_{t+k} + \gamma_h^{W,\nu'} X_{t-1}^{(p)} + e_{j,t+h}^{W,\nu}, \quad (11)$$

omits the surprise residuals; its estimand is a news/noise mixture whenever $\sigma_u^2 > 0$, and consistent exactly in the $\sigma_u^2 = 0$ case (Corollary 1(ii)). Table 1 summarises the menu. The first stage, the information set, and the projection machinery are common to all four rows; the focus regressor and the presence of the surprise block select the estimand, and adding an outcome variable changes nothing to the identification. The chain of reasoning that runs through the menu deserves to be stated once in full: the leading alternatives impose $\sigma_u^2 = 0$; the framework allows $\sigma_u^2 > 0$; in the data the unanticipated component dominates TFP growth; ignoring it contaminates the noise estimand (Corollary 1(ii), and its information-set instability documented in Section 6); and yet the generalisation costs only one additional block of controls in a standard local projection.

Specification	Focus	Controls beyond $X_{t-1}^{(p)}$	Estimand
LP_W (10)	Δz_t	—	Wold innovation of Δz_t : $\mathcal{O}_1$
LP_ε (8)	Δz_{t+q}	$\hat{u}_{t:t+q}$	structural news IRF: $\mathcal{O}_2$
LP_ε ^ν (9)	s_t	$\Delta z_{t:t+q}, \hat{u}_{t:t+q}$	structural noise IRF: $\mathcal{O}_2$
LP_W ^ν (11)	s_t	$\Delta z_{t:t+q}$	news/noise mixture (Cor. 1(ii))

Table 1: The local-projection menu. All four specifications share the first stage, the information set, and the lag block; the focus regressor and the surprise block select the estimand. The structural readings of the LP_ε and LP_ε^ν rows hold under the conditions of Theorem 1; LP_W^ν is consistent for the noise IRF exactly when $\sigma_u^2 = 0$, the case of Forni et al. (2017a) and Chahrour and Jurado (2022).

2.5 Identification: formal statements

Relation to the Chahrour–Jurado representations. The specifications implement the “missing link” of Chahrour and Jurado (2018): their news and noise representations generate the same joint law of fundamentals and beliefs, so what realised future TFP recovers is the news/noise *decomposition*: the part of the signal that materialises in future TFP is news, the orthogonal residual is noise. The distinction between the specifications is between estimands, not estimators.

The mapping to the Chahrour and Jurado (2018) noise representation is worth stating once, because the framework implements two *distinct* objects of that representation and a single unqualified name would conflate them. In the noise representation, the joint law of $(\Delta z_t, s_t)$ is generated by a technology-side disturbance, the innovation to realised TFP, and a belief-side disturbance orthogonal to the entire realised TFP path. The TFP-focus projection (10) recovers the response to the

former: the Wold innovation of Δz_t , rung $\mathcal{O}_1$ of the identification ladder of Section 3. We call this the *technology-side CJ-2018 target*. The signal-focus noise projections (11)–(9) recover the response to the residualised signal, the finite-lead counterpart of the belief-side disturbance, orthogonalised against realised TFP up to the finite lead window rather than the entire path. We call this the *noise-side CJ-2018 target*. These two names are used consistently in the Monte Carlo and empirical sections below.

Relative to Chahrour and Jurado (2022), who identify the news/noise pair by spectral factorisation of the joint (z_t, b_t) system with $b_t = \mathbb{E}_t[z_{t+h}]$ a VAR forecast, the Frisch–Waugh–Lovell argument below provides a finite-lead time-domain counterpart to the separation achieved by their two-sided factorisation, *under the informational conditions of Theorem 1*; when those conditions fail, the two-sided procedure is strictly more informative (Remark R2). We exploit this correspondence in Application 3 of Section 6, replacing the survey signal with a Chahrour and Jurado (2022)-style VAR forecast.

We now state the formal identification results for the four specifications and establish the asymptotic properties of the corresponding estimators.

Proposition 1 (Joint identification of the Wold target and the structural news shock). *Under Assumptions A.1–A.3:*

(i) *The OLS estimator $\widehat{\beta}_h^W$ in (10) satisfies*

$$\widehat{\beta}_h^W \xrightarrow{p} \beta_h^W \equiv \frac{\text{Cov}(y_{j,t+h}, \tilde{\Delta} z_t)}{\text{Var}(\tilde{\Delta} z_t)}, \quad \tilde{\Delta} z_t = \Delta z_t - P\left[\Delta z_t \mid X_{t-1}^{(p)}\right],$$

where β_h^W is the impulse response of $y_{j,t+h}$ to the Wold innovation of Δz_t in the BFG VAR, the technology-side Chahrour and Jurado (2018) target.

If, in addition, Assumption A.4 holds, together with either (a) the single-horizon news loading $b_j = 0$ for $j > q$, or (b) Assumption A.3':

(ii) *The OLS estimator $\widehat{\beta}_h^\varepsilon$ in (8) satisfies*

$$\widehat{\beta}_h^\varepsilon \xrightarrow{p} \frac{\Theta_{j,2}(h)}{b_q},$$

where $\Theta_{j,2}(h)$ is the structural impulse response of $y_{j,t+h}$ to a unit news innovation ε_t and b_q is the news-impact loading defined in A.2. Equivalently, the rescaled estimator $b_q \widehat{\beta}_h^\varepsilon$ is consistent for $\Theta_{j,2}(h)$.

(iii) (Asymptotic normality.) *Under standard local-projection regularity conditions (Jordà, 2005; Plagborg-Møller and Wolf, 2021), $\sqrt{T}(\widehat{\beta}_h^W - \beta_h^W)$ and $\sqrt{T}(\widehat{\beta}_h^\varepsilon - \Theta_{j,2}(h)/b_q)$ are asymptotically normal with HAC-consistent variances.*

Proof sketch. By the Frisch–Waugh–Lovell theorem, the OLS slope $\widehat{\beta}_h^W$ is the slope of $y_{j,t+h}$ on $\tilde{\Delta}z_t = \Delta z_t - P[\Delta z_t \mid X_{t-1}^{(p)}]$. Under **A.1**, $\tilde{\Delta}z_t$ converges in L^2 to the Wold innovation of Δz_t , establishing (i). For (ii), partialling out $(\widehat{u}_t, \dots, \widehat{u}_{t+q})$ from Δz_{t+q} and using the TFP process (2)–(4) removes the surprise contributions $\sum_{i=0}^q a_i u_{t+q-i}$. Under **A.3**, the linear span of $\widehat{u}_{t:t+q}$ asymptotically coincides with that of $u_{t:t+q}$, and past surprises are absorbed by the lag block. Past news enters Δz_{t+q} only through the diffusion tail $\{b_j\}_{j>q}$: under condition (a) that tail is absent, and under condition (b) the lag block spans past news by **A.3'**; in either case the residual is asymptotically equal to $b_q \varepsilon_t$, with **A.4** guaranteeing that it survives the partialling. The probability limit of the slope is then $\text{Cov}(y_{j,t+h}, b_q \varepsilon_t) / \text{Var}(b_q \varepsilon_t) = \Theta_{j,2}(h)/b_q$; multiplying by the regressor variance recovers the stated probability limit. Item (iii) follows from Jordà (2005). Full details are in Appendix A. $\square$

Comments on Proposition 1. Statement (i) does *not* require Assumptions **A.2** or **A.4**: the Wold target is identifiable whenever X_t is informationally sufficient (**A.1**, **A.3**). This is the population content of the Chahrour and Jurado (2018) noise-representation framework. Statement (ii) additionally requires **A.2** (a genuine anticipation horizon $q \geq 1$, so that the news polynomial vanishes for $j < q$ and $b_q \neq 0$), **A.4** (non-degeneracy of the residualised news content), and one of the two conditions (a)/(b) that control the *past-news leakage* into Δz_{t+q} . The role of (a)/(b) is not a technicality: when the news diffuses beyond its arrival period ($b_j \neq 0$ for some $j > q$) and the information set is too poor for **A.3'** (e.g., bivariate $(\Delta z, s)$), the lag block removes past news only partially, the residualised regressor retains a past-news component, and the probability limit of $b_q \widehat{\beta}_h^\varepsilon$ blends the news IRF at adjacent horizons: in a population calculation with a three-period diffusion tail, the distortion reaches ± 12 – 17% of the target. This is the local-projection counterpart of the diffusion bias λ of the level statistic (25): finite-lead identification of structural news objects requires either that news materialises in one shot, or that the observables span past news. The BLL DGP of Section 5.2 *violates* **A.2** ($b_0 \neq 0$), no choice of q restores an anticipation structure, and only Statement (i) is then recoverable, precisely the weaker claim the BLL experiment verifies under the canonical adversarial benchmark of the noisy-news literature.

Corollary 1 (Noise-side identification). *Let Assumptions **A.1**–**A.4** hold together with either (a) the single-horizon news loading or (b) Assumption **A.3'**, as in Proposition 1.*

- (i) (The combined specification is consistent.) *The estimator $\widehat{\beta}_h^{\varepsilon, \nu}$ of the surprise-augmented specification (9) satisfies*

$$\widehat{\beta}_h^{\varepsilon, \nu} \xrightarrow{p} \Theta_{j,3}(h),$$

the impulse response of $y_{j,t+h}$ to a unit noise innovation ν_t in the SMA representation (6).

(ii) (The leads-only specification is not.) *The estimator $\widehat{\beta}_h^{W,\nu}$ of (11) satisfies*

$$\widehat{\beta}_h^{W,\nu} \xrightarrow{p} \frac{\sigma_\nu^2 \Theta_{j,3}(h) + \text{Cov}(y_{j,t+h}, \check{\varepsilon}_t)}{\sigma_\nu^2 + \text{Var}(\check{\varepsilon}_t)}, \quad \check{\varepsilon}_t \equiv \varepsilon_t - P\left[\varepsilon_t \mid \{\Delta z_{t+0:q}\}, X_{t-1}^{(p)}\right],$$

a mixture of the noise IRF and a smeared news response. The limit equals $\Theta_{j,3}(h)$ for all h and j if and only if $\text{Var}(\check{\varepsilon}_t) = 0$, i.e., iff the controls span the current news shock, which fails whenever $\sigma_u^2 > 0$, because the TFP leads are contaminated by the future surprises $u_{t+1:t+q}$, which lie outside the span of the controls.

Proof sketch. By Frisch–Waugh–Lovell, each estimator is the OLS slope of $y_{j,t+h}$ on the residualised signal $\tilde{s}_t = s_t - P[s_t \mid \text{controls}]$. Since ν_t is orthogonal to every control (the TFP leads carry only u and ε ; the lag block predates t), $\tilde{s}_t = \check{\varepsilon}_t + \nu_t$ exactly in population, with $\check{\varepsilon}_t$ the residual of ε_t on the relevant control set and $\check{\varepsilon}_t \perp \nu_t$. The slope is therefore $[\text{Cov}(y_{j,t+h}, \check{\varepsilon}_t) + \Theta_{j,3}(h)\sigma_\nu^2]/[\text{Var}(\check{\varepsilon}_t) + \sigma_\nu^2]$ for both specifications: they differ only through $\check{\varepsilon}_t$. For (9), the controls include $\widehat{u}_{t:t+q} \rightarrow u_{t:t+q}$ (A.3); partialling them out of Δz_{t+q} leaves $b_q \varepsilon_t$ plus past news, and past news is eliminated under (a) or absorbed by the lag block under (b), so ε_t lies in the span of the controls, $\check{\varepsilon}_t = 0$, $\tilde{s}_t \xrightarrow{L^2} \nu_t$, and statement (i) follows. For (11), the surprise content $u_{t+1:t+q}$ of the TFP leads cannot be removed by any control in the set (future innovations are orthogonal to the lag block), so the projection of ε_t on the leads is attenuated, $\text{Var}(\check{\varepsilon}_t) > 0$, and the limit is the stated mixture. Full details are in Appendix A. $\square$

Comments on Corollary 1. The corollary makes the surprise-residual block *essential* rather than cosmetic: the combined specification is the consistent estimator, while the leads-only one contaminates the noise IRF with a smeared news response whose weight grows with σ_u^2 , quantitatively first-order (in a population calculation under the single-horizon DGP, 0.55 at the peak against a true noise response of 0.20). The mixture also explains why Forni et al. (2017a) need no surprise controls: their TFP innovation decomposes into news and noise only, so the mixture weight is zero. It is the presence of the transitory surprise in TFP that makes the surprise-augmented specification essential.

2.6 Finite-sample bias and a residual-augmented correction

Proposition 1 establishes consistency of both LP slopes. The simulations of Section 5.1 and Appendix D document a non-trivial finite-sample bias: on the order of 15–20% downward at the peak for $T = 284$ in the BLL DGP, growing with the projection horizon h . We characterise the leading-order bias and propose an analytic correction.

Three sources contribute to the leading-order bias: first-stage estimation of $\widehat{u}_t$ (B.1), the direct $O(p/T)$ local-projection bias from the lagged controls (B.2) (Kilian and Kim, 2011; Herbst and Jo-

hannsen, 2024), and a many-controls term of order $O((q+1)/T)$ from the $q+1$ surprise controls (B.3); closed-form expressions are in Appendix A.

Proposition 2 (Finite-sample bias of the LP estimators). *Under Assumptions A.1–A.4 and standard regularity conditions (uniform mixing, finite eighth moments, $T \rightarrow \infty$ with p, q fixed), the LP estimators satisfy*

$$\mathbb{E}[\widehat{\beta}_h^W] = \beta_h^W + \frac{1}{T} \delta_h^W + O(T^{-3/2}), \quad (12)$$

$$\mathbb{E}[\widehat{\beta}_h^\varepsilon] = b_q \Theta_{j,2}(h) + \frac{1}{T} \delta_h^\varepsilon + O(T^{-3/2}), \quad (13)$$

where the leading-order bias coefficients admit the additive decomposition

$$\delta_h^* = \delta_h^{*,1st} + \delta_h^{*,LP} + \delta_h^{*,Q}, \quad * \in \{W, \varepsilon\}, \quad (14)$$

corresponding to sources (B.1), (B.2), (B.3) respectively. Closed-form expressions are derived in Appendix A. Three qualitative properties hold:

- (i) (Sign.) If Δz_t is positively autocorrelated, $\delta_h^* < 0$ and the LP estimator is biased toward zero.
- (ii) (Horizon.) $|\delta_h^*|$ is non-decreasing in h over the range where the cumulative effect of the structural shock is non-decreasing.
- (iii) (Specification.) For (8), $|\delta_h^\varepsilon| > |\delta_h^W|$ when q is large relative to T : surprise controls reduce asymptotic bias (toward $\Theta_{j,2}$ rather than β^W) but inflate finite-sample bias through (B.1) and (B.3).

Noise-side bias. The same decomposition extends to the noise-side estimators, with the focus s_t replacing Δz_{t+q} and a noise-side partial- R^2 diagnostic $\widehat{\xi}^{Q,\nu}$ in place of $\widehat{\xi}^Q$ (Appendix A). In our sample $\widehat{\xi}^{Q,\nu} \approx 0.18$ for both surveys, implying a correction factor of about 1.02 that leaves the noise IRFs visually unchanged.

A residual-augmented correction. The bootstrap bias-correction $\widehat{\theta}_h^{BC} = 2\widehat{\beta}_h^* - \frac{1}{B} \sum_b \widehat{\beta}_h^{*,b}$ over-corrects when the bootstrap re-estimator is closer to truth than the original, as documented in Appendix D.5. Proposition 2 suggests an alternative *residual-augmented* correction:

1. Estimate $\delta_h^{*,1st}, \delta_h^{*,LP}, \delta_h^{*,Q}$ from the closed-form expressions of Appendix A, evaluated at consistent estimates of the second-moment matrix and the AR coefficients.
2. Form the bias-corrected estimator $\widehat{\beta}_h^{*,cor} = \widehat{\beta}_h^* - \frac{1}{T} \widehat{\delta}_h^*$.

Proposition 3 (Properties of the residual-augmented correction). *Under the conditions of Proposition 2:*

- (i) $\mathbb{E}[\widehat{\beta}_h^{*,cor}] = \beta_h^* + O(T^{-3/2})$ (unbiased to order $O(T^{-1})$).
- (ii) $\widehat{\beta}_h^{*,cor}$ inherits the asymptotic distribution of $\widehat{\beta}_h^*$ since the correction is $O_p(T^{-1})$, dominated by the $O_p(T^{-1/2})$ sampling fluctuation.
- (iii) (Stability.) *Unlike the bootstrap correction, the residual-augmented correction does not over-correct in the presence of informationally weak controls (small σ_ξ).*

The signal in practice. In empirical applications the signal need not be directly observed: it may be proxied by direct expectation measures, as in Forni et al. (2017a), or constructed as a forecast from observables, as in Chahrour and Jurado (2022). Section 6 implements both.

3 The finite-lead identification frontier

The ladder of identification objects. Let $\mathcal{I}_t = \overline{\text{sp}}(X_s : s \leq t)$ denote the contemporaneous information set, $\mathcal{I}_t^{(k)} = \overline{\text{sp}}(X_s : s \leq t + k)$ its extension by a finite lead window of k periods, and $\mathcal{I}_t^{2s} = \overline{\text{sp}}(X_s : s \in \mathbb{Z})$ the two-sided span, all linear, consistently with the second-order scope of the analysis. We call an identification scheme *finite-lead* when its estimand is measurable with respect to $\mathcal{I}_t^{(k)}$ for some finite $k \geq 0$ (conditioning on realisations a fixed number of periods past the shock date, as our projections on Δz_{t+q} do and as any implementation of the Chahrour and Jurado (2018) missing link must), and *full-history* when it requires the entire future. The substantive question is then the *size of the lead window* each rung requires: none for $\mathcal{O}_1$; for $\mathcal{O}_2$, the lead support k^* characterised in Theorem 1, equal to the anticipation horizon q under the conditions of the theorem; and, in general, the full two-sided span for $\mathcal{O}_3$. Chahrour and Jurado (2018) establish the starting identification problem: innovations in agents' expectations, the signal shock, are observationally equivalent under their news and noise representations, so no information available to the agents themselves can separate news from noise. The question is therefore which structural object becomes attainable as the econometrician enlarges the information set with realised TFP. This motivates a three-rung identification ladder:

- $\mathcal{O}_1$: the impulse response of y_j to the Wold innovation of Δz_t , the *technology-side Chahrour and Jurado (2018) target*, attainable from $\mathcal{I}_t$;
- $\mathcal{O}_2$: the impulse responses of y_j to the structural news innovation ε_t and, on the noise side, to the noise innovation ν_t , attainable from $\mathcal{I}_t^{(k)}$ if and only if $k \geq k^*$ (Theorem 1);

$\mathcal{O}_3$: the structural news and noise impulse responses attainable from the complete two-sided history $\mathcal{I}_t^{2s}$ under recoverability; Chahrour and Jurado (2022) provide a spectral procedure that identifies them.

Each rung is an information-indexed object, and the rungs are ordered by the information they require: $\mathcal{I}_t \subset \mathcal{I}_t^{(k)} \subset \mathcal{I}_t^{2s}$. We write $\mathcal{O}_i(h)$ for the value at horizon h . The ladder fixes the objects; the theorem of the next section characterises which rungs a finite-lead projection attains, and under exactly which informational conditions.

The Wold rung $\mathcal{O}_1$ requires no future information at all (Proposition 1(i)); the substantive question concerns $\mathcal{O}_2$: how much future information does the structural rung require? We characterise the answer for the class of estimands this paper studies: probability limits of linear-projection slopes whose focus regressor and controls are measurable with respect to a finite lead window $\mathcal{I}_t^{(k)}$, with the lag block $X_{t-1}^{(p)}$ among the controls. The key object is the representation of the news shock in the *innovations basis* of the observables. Let ι_t denote the population innovation of X_t with respect to its own history, with $F = \text{Var}(\iota_t)$ nonsingular (guaranteed by $\sigma_u^2, \sigma_v^2 > 0$), and define

$$c_m = \mathbb{E}[\varepsilon_t \iota'_{t+m}], \quad m \geq 0, \quad k^* = \sup\{m \geq 0 : c_m \neq 0\}. \quad (15)$$

The integer k^* is the *lead support* of the news shock: the last date at which the arrival of new observable information still revises the linear estimate of ε_t ; equivalently, the lead support of the Wiener–Kolmogorov fixed-point smoother of ε_t .

Theorem 1 (The finite-lead identification frontier). *Consider the benchmark signal $s_t = \varepsilon_t + \nu_t$ of (5) under Assumption A.I, and say that $\mathcal{O}_2$ is attainable on $\mathcal{I}_t^{(k)}$ if some projection specification in the class above has slopes converging to $\gamma \Theta_{j,2}(h)$ for a common $\gamma \neq 0$, every system variable j , and every horizon $h \geq 0$.*

- (i) (Characterisation.) $\mathcal{O}_2$ is attainable on $\mathcal{I}_t^{(k)}$ if and only if $k \geq k^*$. The attaining residualised regressor is unique up to scale and equals the finite-lead smoother $P[\varepsilon_t | \mathcal{I}_t^{(k)}]$.
- (ii) (Sufficiency revisited.) Under the single-horizon loading (a), $k^* = q$, even in the bivariate system $X_t = (\Delta z_t, s_t)'$ where A.3 fails and the shock path is never recovered ($\text{Var}(\varepsilon_t - P[\varepsilon_t | \mathcal{I}_t^{(k)}]) \geq \sigma_\varepsilon^2 \sigma_\nu^2 / (\sigma_\varepsilon^2 + \sigma_\nu^2 + b_q^{-2} \sigma_u^2) > 0$ for every k): attainment of the IRF parameter does not require recovery of the shock path. Under A.3' (b), and likewise when $\sigma_\nu = 0$, $k^* = 0$.
- (iii) (Impossibility.) In the bivariate system with $\sigma_u^2, \sigma_\nu^2 > 0$ and a nondegenerate diffusion tail ($b_j \neq 0$ for some $j > q$), $k^* = \infty$ for all parameter values outside a closed set with empty interior: no finite lead window attains the structural news IRF. The window- k smoother specification matches $\Theta_{j,2}(h)$ exactly at horizons $h \leq k$ and generates spurious news responses at horizons

$h > k$, equal to the truncated smoother tail and decaying geometrically at the contraction rate of the associated filter.

Proof sketch. Partialling the lag block orthogonalises every admissible regressor into the span of the current and future innovations $\iota_t, \dots, \iota_{t+k}$. Matching the slopes to the structural news IRFs at every horizon is then a forward-triangular system in the Wold coefficients of the observables ($\Psi_0 = I$, $F \succ 0$): solving it forces the regressor weights to equal the smoother weights $F^{-1}c'_m$ up to a common scale at every $m \leq k$, and forces $c_m = 0$ at every $m > k$. The mechanism behind (iii) is *joint learning*. With one-shot news, every observation involves a single news shock, so the filter's estimation errors are uncorrelated across shocks: information about ε_t stops arriving at $t + q$, and $c_m = 0$ beyond. With gradual diffusion, each TFP realisation mixes adjacent news shocks; the filter learns them jointly, its estimation error for ε_t contaminates its forecasts at every later date, and every future innovation retains a nonzero loading on ε_t . The full proof is in Appendix A. $\square$

Corollary 2 (Under Gaussianity, the frontier is an identification impossibility). *Let the shocks be jointly Gaussian. Then for any square-integrable function g of the window- k observables, the covariance of any system variable with g equals its covariance with the L^2 -projection of g onto $\overline{\text{sp}}(\iota_t, \dots, \iota_{t+k})$, so the slopes on g reduce to the slopes on a linear regressor of Theorem 1. Consequently, when $k < k^*$ no function of the finite-lead information, linear or not, attains the structural news IRF: statement (iii) is an identification impossibility result for finite-lead conditioning, not merely a limitation of linear projection methods. Under non-Gaussian shocks, the proposition reads as the second-order frontier, and higher-order moments may shift it (see the Scope remark below).*

Remark 1 (Quantifying the truncation). In a population benchmark with $q = 2$, diffusion tail $(b_q, b_{q+1}, b_{q+2}) = (1, 0.6, 0.3)$ and unit shock variances, the window $k = q$ delivers the news profile $(1, 0.41, 0.20)$ against the true $(1, 0.6, 0.3)$; the window $k = q + 2$ is exact on the diffusion support but generates spurious responses of 3.1% and 0.6% of the impact response at the two subsequent horizons; and longer windows shrink the spurious tail geometrically without ever eliminating it. This is the frontier in numerical form: finite-lead projections approximate $\mathcal{O}_2$ at a geometric rate in the window length, and attain it exactly only in the cases of statement (ii). The complementarity with Chahrour and Jurado (2022) is then precise: recoverability characterises what the complete two-sided history permits; Theorem 1 characterises exactly what finite-lead projections deliver before that limit, and k^* is the minimal window at which the two coincide.

Remark 2 (Signal aggregation as the signal-side twin of diffusion). The characterisation extends to the general multi-horizon signal (16) of Section 4 through the same innovations basis: nothing in the triangular argument uses the simple structure of (5). What changes is the content of the lead support, which becomes governed *jointly* by the supports of the diffusion polynomial $b(L)$ and of the loading polynomial $\ell(L)$: each signal observation then mixes adjacent news shocks, the filter learns them

jointly, and the mechanism of statement (iii) operates from the signal side even when the TFP loading is one-shot. In a population computation with one-shot TFP loading and a two-horizon signal ($H = 1$, $\ell = (1, 0.5)$), the lead loadings c_m decay geometrically (at rate ≈ 0.16 ; ≈ 0.46 for a three-horizon signal) instead of terminating at $m = q$: signal aggregation is the signal-side twin of technology diffusion. The window that identification requires under (16) is accordingly the full materialisation horizon $q + H + n$, precisely the window that the data-driven selector h^* of Section 4 delivers and that the empirical application uses. The *scale* of the structural loadings inherits the set identification of the roots of $\ell(L)$ discussed there; the news/noise split itself does not.

Scope: second moments and Gaussianity. All statements in this paper are second-order: the objects on the ladder are defined by linear projections, and Assumptions **A.1–A.4** impose only moment and orthogonality conditions. Under Gaussianity the frontier binds for all finite-lead conditioning, linear or not (Corollary 2).² The consistency of our estimators and statistics is unaffected either way: they estimate their stated second-order targets under **A.1** regardless of the distribution of the shocks.

Remarks: the rest of the ladder. Three further placements are useful but are either definitional or belong to the cited papers, and we state them as remarks rather than theorem items. (R1) The Beaudry and Portier (2006) long-run identification is consistent for the permanent-component IRF of TFP when its long-run zero restriction is correctly specified, which in our DGP requires $a(1) = 0$ so that only the news shock has a permanent TFP effect. (R2) The Chahrour and Jurado (2022) spectral estimator is consistent for $\mathcal{O}_3$ under their recoverability conditions; when the structural shocks are moreover invertible from $\mathcal{I}_t$, $\mathcal{O}_3 = \mathcal{O}_2$ and the full-history procedure buys nothing beyond the finite-lead one. When the shocks are recoverable but *not* invertible, the full-history information is strictly valuable: conditions (a)/(b) of the theorem fail, finite-lead projections attain $\mathcal{O}_1$ but not $\mathcal{O}_2$, and the distance between the two is the empirical content of the *recoverability ceiling*. (R3) $\mathcal{O}_1$ and $\mathcal{O}_2$ are non-nested objects, and the theorem makes the distinction sharp: in the limiting case $\sigma_\nu = 0$ of statement (ii), the TFP-focus Wold projection $\hat{\beta}_h^W$ recovers the *surprise* response $\Theta_{j,1}(h)$ (the Wold innovation of Δz_t is then $a_0 u_t$, all news being anticipated), while the signal-focus projection recovers the *news* response $\Theta_{j,2}(h)$: two different structural objects from two different finite-lead projections on the same data. The choice between $\hat{\beta}_h^W$ and $\hat{\beta}_h^\varepsilon$ is a choice of estimand, not a ranking. Finally, we offer no formal result for the max-FEV identification of Barsky and Sims (2011): the max-FEV

²Under non-Gaussian independent shocks, higher-order moments can in principle detect non-invertible representations from finite-lead data (the identification route developed for SVARs through higher unconditional moments in Guay (2021) and the related literature on non-causal and independent-component-based systems), so the frontier should then be read as the frontier of *linear* finite-lead methods. The same qualification applies at the top of the ladder: recoverability in Chahrour and Jurado (2022) is defined by two-sided *linear* spans, so under non-Gaussian shocks the entire ladder, not only its finite-lead frontier, would be redrawn by higher-order information. This extension is the subject of ongoing work.

shock generically loads news and noise innovations jointly, and deriving the implied mixture weight would require their full system and criterion, beyond our scope.

What is new in Theorem 1. The essential novelty is twofold. First, the frontier is characterised *exactly*: within the class of finite-lead linear projections, attainability of the structural news IRF is governed by the lead support k^* : finite under conditions (a)/(b), generically infinite under gradual diffusion in the bivariate system (Theorem 1). Nothing sharper is claimed for that class, nothing weaker proved. Second, the surprise controls perform a separation that the $\sigma_u^2 = 0$ environments of Forni et al. (2017a) and Chahrour and Jurado (2022) never face: they make the surprise-augmented noise specification consistent where the leads-only one converges to a mixture, with the contaminated limits characterised when the conditions fail (Corollary 1(ii)). The closure of the frontier as noise vanishes ($k^* = 0$ when $\sigma_\nu = 0$) and the observable diagnostics complete the picture. The theorem builds on Chahrour and Jurado (2018) for the equivalence of representations and on Chahrour and Jurado (2022) for recoverability (Remark R2); none of these results appears in, or follows from, those papers.

Implications. The identification extends the projection-based literature in two ways: a single local-projection framework delivers $\mathcal{O}_1$ when surprise controls are omitted and $\mathcal{O}_2$ when they are included; and the residual-augmented correction of Proposition 3 delivers a finite-sample-corrected estimator of $\mathcal{O}_2$ without the parametric DSGE structure of Chahrour and Jurado (2018) or the full-history information set of Chahrour and Jurado (2022). When conditions (a)/(b) fail, full-history information has strictly positive value (Remark R2), and the comparison of $\hat{b}_q \hat{\beta}_h^\varepsilon$ with $\hat{\theta}_h^{\text{CJ-S}}$ in the same data is a direct diagnostic for the empirical importance of the recoverability gap.

3.1 Discussion of the assumptions

With the theorem in hand, the assumptions of Section 1 can be assessed for what each contributes and how each is checked.

A.1 bundles three requirements. *Covariance stationarity* is comfortable for Δz_t and formally defensible for the bounded survey indices, but the signals' effective persistence is high ($V_{LR}(s_t)/\text{Var}(s_t) \approx 7\text{--}15$; Section 6), which is why the measurement statistics are conducted on the signal innovation $\eta_t^s \equiv s_t - \text{Proj}[s_t \mid X_{t-1}^{(p)}]$ rather than the raw level; the persistent level series of the auxiliary block (hours, the federal funds rate) enter only as controls. *Finite fourth moments* are all that consistency and HAC inference require (Gaussianity enters only the informational reading of the frontier, Corollary 2) and are comfortably supported in the data.³ *Mixing sufficient for HAC inference* interacts

³The signal innovations display modest excess kurtosis (about 1 for BUS12, whose asymmetry drives a Jarque–Bera

with the long horizons ($H = 20$): the remedy recommended by Montiel Olea and Plagborg-Møller (2021), lag augmentation, is built in, since the lag block that the identification requires doubles as the augmentation that robustifies inference.

A.2 is the structural restriction, best understood against the environments of the leading alternatives. In Chahrour and Jurado (2022) technology is purely exogenous with a single innovation, and in Forni et al. (2017a) the fundamental is driven solely by the delayed long-run shock, their “surprise” being explicitly a reduced-form object: in our notation, $\sigma_u^2 = 0$ in both cases. Assumption **A.2** ($b_0 = 0$ with $q \geq 1$) is what gives the three-shock structure empirical content, and the nesting is exact: imposing $\sigma_u^2 = 0$ makes the surprise controls redundant and the leads-only noise specification consistent (the mixture weight of Corollary 1(ii) vanishes, the reason FGLS need no surprise controls), while under $\sigma_u^2 > 0$, the empirically relevant case since TFP growth is nearly unpredictable, the surprise controls are what keeps the noise estimand pure. Within **A.2**, $a_0 \neq 0$ is definitional; the load-bearing zero is $b_0 = 0$, whose violation the BLL benchmark of Section 5.2 exhibits (only the Wold target survives there); the interior zeros protect the surprise-augmented specification, the general b -support being covered by k^* ; and $b_q \neq 0$ at a known q is the practical margin, handled by the data-driven selection of q and probed by the robustness exercises of Section 6. **A.2** is directly testable: under $b_0 = 0$ the signal is orthogonal to the current TFP surprise, $\text{Cov}(s_t, \hat{u}_t) = 0$, the check reported in Section 6.

A.3 rests on three primitive conditions: *exclusivity at impact* (u_t is the sole time- t shock entering Δz_t , delivered by $q \geq 1$, the reason the surprise side is easier than the news side); *readability of materialised news* (the anticipated component of Δz_t must be spanned by $X_{s \leq t-1}$, a lagged and weakened version of **A.3'**, which requires readability at *announcement*, so **A.3'** implies it but not conversely; both fail in the bivariate system with $\sigma_\nu > 0$ and are restored by forward-looking auxiliary variables, exactly as in FGLS); and *invertibility in levels* ($\tilde{a}(L) \equiv a(L)/(1-L)$ minimum-phase, since $a(1) = 0$ places a root of $a(L)$ on the unit circle). **A.3** is thus a *partial* recoverability condition (one shock, not fundamentalness of the whole system) and testable: $\hat{u}_t$ must be unpredictable from omitted information, the orthogonality test of Section 6. Where **A.3'** matters, the single-horizon loading ($b_j = 0$ for $j > q$) is always offered as the alternative sufficient condition.

A.4 is stated at the level of the residualised objects, deliberately so: under **A.2–A.3** together with (a) or (b) it is *derived* (the residual converges to $b_q \varepsilon_t$ exactly), and its independent bite lies in two failure modes, *absorption* (BLL: the failure of **A.2** puts news content into the surprise residuals themselves) and *contamination* (a diffusion tail without **A.3'**: positivity holds, exactness fails). Its margin is monitored by the diagnostic $\hat{\xi}_Q$ of Section 2.6: $\hat{\xi}_Q \approx 0.99$ in the BLL benchmark (near-degeneracy, the regime where the finite-sample correction is decisive) against $\hat{\xi}_Q \approx 0.18$ in our

rejection of normality without threatening moment existence) and no conditional heteroskedasticity (ARCH-LM(4) p -values of 0.23 and 0.15 for the two surveys).

application, a comfortable margin.

4 Measuring the information content of expectation signals

The paper’s first economic question, *how noisy are expectations?*, is a measurement question rather than an identification one, and this section builds the instrument. Sections 2–3 establish what the identified shocks do; this section measures how noisy the signal itself is. We define two targets, the news share in the *stock* of the signal and in the *flow* of newly arriving information, and construct an exact long-run statistic for each, valid under any diffusion polynomial $b(L)$ and any signal loadings; a third statistic serves exclusively as a data-driven selector of the lead window. The flow statistic is the headline instrument of the empirical analysis; conventional alternatives and the full comparison are collected in Appendix B.

The simple signal structure (5) carries a single fresh news shock. In practice, survey-based and VAR-implied signals (the Michigan ICS, or the rational forecast $b_t = \mathbb{E}_t[z_{t+h}]$ used in Section 6) aggregate news received at several dates. We therefore consider the more general structure

$$s_t = \sum_{i=0}^H \ell_i \varepsilon_{t+i} + \nu_t, \quad (16)$$

where $\{\ell_i\}_{i=0}^H$ are the signal loadings on the current and future news shocks, and ν_t is white noise orthogonal to all news shocks and to the surprise process. Each component $\ell_i \varepsilon_{t+i}$ materialises in TFP from $t + q + i$ onward, so all the news carried by the signal lies ahead of its materialisation and all signal–TFP cross-covariances lie at non-negative leads.⁴ The simple structure (5) is the case $\ell_0 = 1$, $H = 0$; throughout, $\ell(x) \equiv \sum_{i=0}^H \ell_i x^i$ denotes the loading polynomial, distinct from the TFP lag polynomials $a(L)$ and $b(L)$.

Under (16) two structural targets are natural. The first is the news variance share of the signal *level*, the stock of news embedded in the signal:

$$\theta \equiv \frac{\text{Var}\left(\sum_{i=0}^H \ell_i \varepsilon_{t+i}\right)}{\text{Var}(s_t)} = \frac{\sum_i \ell_i^2 \sigma_\varepsilon^2}{\sum_i \ell_i^2 \sigma_\varepsilon^2 + \sigma_\nu^2}, \quad (17)$$

which generalises $\alpha^2 = \sigma_\varepsilon^2 / \sigma_s^2$ from the simple-news case and answers: “*of the signal’s total variance, what fraction is genuine news about future TFP?*”. The second is the news share of the *flow* of new information. Let $x_t \equiv \sum_{i=0}^H \ell_i \varepsilon_{t+i}$ denote the news component of the signal and

⁴If an empirical signal also reflected already-materialised news (non-zero correlation with past TFP growth), the one-sided sums below would be replaced by two-sided sums $\sum_{j=-h}^h$; nothing else changes.

$\Omega_x \equiv \text{Var}(x_t - P[x_t \mid x_{t-1}, x_{t-2}, \dots])$ its one-step (Wold) innovation variance, and define

$$\tilde{\theta} \equiv \frac{\Omega_x}{\Omega_x + \sigma_\nu^2}, \quad (18)$$

which answers: “*of what the agent newly learns at time t , what fraction is news?*”. The two targets coincide in the simple-news case ($x_t = \varepsilon_t$ is white, $\Omega_x = \sigma_\varepsilon^2$) and differ when the signal aggregates news across horizons. When the loading polynomial is minimum-phase, $\Omega_x = \ell_0^2 \sigma_\varepsilon^2$ and $\tilde{\theta}$ has the structural reading $\ell_0^2 \sigma_\varepsilon^2 / (\ell_0^2 \sigma_\varepsilon^2 + \sigma_\nu^2)$.

Two remarks frame what follows. First, $\tilde{\theta}$ is a *structural* object: the innovation in (18) is that of the noise-free news component x_t , not that of the observable signal, $\eta_t^s \equiv s_t - \text{Proj}[s_t \mid X_{t-1}^{(p)}]$. The latter is a signal-extraction residual (past signals reveal the anticipated shocks only mixed with past noise) and retains news content beyond Ω_x (by nine percent in a population calculation with the loadings of our Monte Carlo designs). Second, estimation is complicated on the TFP side: under gradual diffusion the signal–TFP cross-covariances are convolutions of b with ℓ , from which neither factor is separately identified. The identity below cuts through both difficulties.

Identification at frequency zero. All the exactness results below rest on one identity. Let $V_{LR}(s_t) \equiv \sum_{k=-\infty}^{\infty} \text{Cov}(s_t, s_{t+k})$ denote the long-run variance of the signal, defined exactly as $V_{LR}(\Delta z_t)$ and estimated by the same Newey–West HAC device. At frequency zero the sum of a convolution factorises, $\sum_{j=0}^h \text{Cov}(s_t, \Delta z_{t+j}) = \sigma_\varepsilon^2 b(1) \ell(1)$ for $h \geq q + H + n$, while $V_{LR}(\Delta z_t) = \sigma_\varepsilon^2 b(1)^2$ because the surprise has no long-run effect ($a(1) = 0$, Assumption **A.3**). Squaring the first and dividing by the second eliminates $b(L)$ entirely; and because ν_t is white noise, it contributes its full variance to $V_{LR}(s_t) = \sigma_\varepsilon^2 \ell(1)^2 + \sigma_\nu^2$. The noise variance is therefore identified from long-run moments alone:

$$\sigma_\nu^2 = V_{LR}(s_t) - \left[\sum_{j=0}^h \text{Cov}(s_t, \Delta z_{t+j}) \right]^2 / V_{LR}(\Delta z_t), \quad h \geq q + H + n, \quad (19)$$

for any diffusion polynomial $b(L)$ and any loadings. Identity (19) has a transparent spectral reading: since the surprise vanishes at frequency zero, whatever fraction of the signal’s long-run variance is incoherent with TFP growth at frequency zero is noise, and whiteness converts that long-run noise variance into $\text{Var}(\nu_t)$.

The stock statistic. Substituting (19) into $\theta = 1 - \sigma_\nu^2 / \text{Var}(s_t)$ gives the *long-run stock statistic*

$$\theta_h^{LR} \equiv 1 - \frac{V_{LR}(s_t) - \left[\sum_{j=0}^h \text{Cov}(s_t, \Delta z_{t+j}) \right]^2 / V_{LR}(\Delta z_t)}{\text{Var}(s_t)}. \quad (20)$$

Proposition 5 (stated in Appendix B, proved in Appendix A) establishes that $\theta_h^{LR} = \theta$ *exactly* for $h \geq q + H + n$, for any diffusion polynomial and any loadings. Empirically, however, the level share turns out to be a descriptive object (the survey levels fail the specification test of the third remark below), and the flow statistic is the headline instrument.

The flow statistic. The identity (19) also unlocks the flow target. For $k \neq 0$ the autocovariances of the signal are free of noise, $\text{Cov}(s_t, s_{t+k}) = \sigma_\varepsilon^2 \sum_i \ell_i \ell_{i+|k|}$, and (19) supplies the missing lag-zero term: the full spectral density of the news component, $S_s(\omega) - \sigma_\nu^2$ with $S_s(\omega) \equiv \sum_k \text{Cov}(s_t, s_{t+k}) e^{-i\omega k}$, is identified at every frequency. By Szegő–Kolmogorov’s formula, the one-step innovation variance of the news component is a functional of that spectrum alone,

$$\Omega_x = \exp\left\{\frac{1}{2\pi} \int_{-\pi}^{\pi} \log[S_s(\omega) - \sigma_\nu^2] d\omega\right\} \equiv \Omega_s, \quad (21)$$

and the *Kolmogorov flow statistic* is

$$\tilde{\theta}^{LR} \equiv \frac{\Omega_s}{\Omega_s + \sigma_\nu^2}, \quad (22)$$

with σ_ν^2 from (19). No assumption on the roots of $\ell(L)$ is involved: the target $\tilde{\theta}$ is defined through the Wold innovation of the news component, and (21) is that innovation variance.

Corollary 3 (Exactness of the flow statistic). *Assume the TFP process (2)–(4) with $a(1) = 0$ and $b(1) \neq 0$, the general signal (16) with ν_t white noise, and mutual orthogonality of u , ε , and ν at all leads and lags. Then $\tilde{\theta}^{LR} = \tilde{\theta}$ for any diffusion polynomial $b(L)$ and any signal loadings $\{\ell_i\}_{i=0}^H$. If, in addition, $\ell(L)$ is minimum-phase, $\Omega_s = \ell_0^2 \sigma_\varepsilon^2$ and $\tilde{\theta}$ has the structural-loading reading.*

Proof. See Appendix A. □

The respective roles of the two conditions deserve emphasis. The flow share $\tilde{\theta}$, the Wold-innovation news share, is point-identified and estimated exactly by (22) with no assumption on the roots of $\ell(L)$. What the minimum-phase condition buys is only the further *structural* reading $\Omega_x = \ell_0^2 \sigma_\varepsilon^2$: a loading polynomial with a root inside the unit circle generates the same signal spectrum, so the individual loadings are set-identified, the signal-side counterpart of the non-invertibility issue that Chahrour and Jurado (2022) emphasise on the TFP side. The stock target is immune even to that: $\sum_i \ell_i^2$ is invariant to root flips.

Scope and practice. Three remarks. First, the whiteness of ν_t is *not* needed for the identity itself: (19) uses only the orthogonality of the noise to the TFP process, and therefore delivers the zero-frequency spectrum $S_\nu(0)$ of the noise component whatever its persistence. Whiteness matters only for reading $S_\nu(0)$ as the noise *variance*. For AR(1) noise with parameter ρ_ν , $S_\nu(0) = \text{Var}(\nu_t) (1 +$

$\rho_\nu)/(1 - \rho_\nu)$, so under the empirically plausible case of positively autocorrelated belief noise the white-noise reading *overstates* the noise variance, and the implied noise share is an upper bound. The sensitivity is computable: pinning the noise spectrum $S_\nu(\omega; \rho_\nu)$ at frequency zero by the identity and re-solving the flow decomposition, the noise share of the surveys falls monotonically as ρ_ν rises (Section 6 reports the numbers). Persistent noise thus tightens, rather than threatens, the transmission-side conclusion; the pre-whitening of Section 6 remains good practice because it keeps the residual noise autocorrelation small. Second, the only estimation inputs are HAC long-run variances and (cross-)covariances (no auxiliary regression, no surprise residuals, no knowledge of $b(L)$), with the truncation lag set near the process memory $q + H + n$, the bandwidth the selector h^* below delivers. Third, when the empirical signal is a persistent *level* series (survey indices are), the finite-memory structure (16) should be imposed by applying the statistics to the signal *innovation* η_t^s rather than to the raw level. On a well-whitened signal $V_{LR}(\eta^s) \approx \text{Var}(\eta^s)$, the level and flow constructions coincide, and both estimate the flow share $\tilde{\theta}$; the raw-level computation then serves as a built-in *specification test*: a ratio $V_{LR}(s_t)/\text{Var}(s_t)$ far above one and growing with the bandwidth, or $\mathbf{nsr}_h^{LR} \gg 1$, diagnoses persistence that the model (16) does not accommodate, and the long-run identification of the *level* share is then infeasible without pre-whitening. Monte Carlo experiments under the four configurations $\{\text{single-horizon vs. diffused TFP loading}\} \times \{\text{simple vs. multi-horizon signal}\}$ confirm the exactness of θ_h^{LR} and $\tilde{\theta}^{LR}$ in all four cases.

The sum-of-squared-covariances statistic and the lead-window selector. The third statistic is a diagnostic rather than a headline estimator. Define

$$\theta_h^{\text{stat}} \equiv \frac{\sum_{j=0}^h [\text{Cov}(s_t, \Delta z_{t+j})]^2}{V_{LR}(\Delta z_t) \cdot \text{Var}(s_t)}. \quad (23)$$

Under the *single-horizon* TFP-news loading $b(L) = b_q$ (each news shock materialises in TFP growth within one period) and Assumption **A.3**,

$$\theta_h^{\text{stat}} \xrightarrow{h \geq q+H} \frac{\sum_i \ell_i^2 \sigma_\varepsilon^2}{\sum_i \ell_i^2 \sigma_\varepsilon^2 + \sigma_\nu^2} = \theta, \quad (24)$$

exactly: the mutual orthogonality of news shocks across arrival dates makes the squared covariances aggregate the news variances additively, term by term. Under the general diffusion polynomial, however, the cross-covariances become convolutions, $\text{Cov}(s_t, \Delta z_{t+j}) = \sigma_\varepsilon^2 \sum_k b_k \ell_{j-q-k}$, and the

population limit of θ_h^{stat} is $\lambda \theta$ with

$$\lambda \equiv \frac{\sum_j (\sum_k b_k \ell_{j-k})^2}{b(1)^2 \sum_i \ell_i^2}, \quad (25)$$

where $\ell_i \equiv 0$ outside $[0, H]$. For same-signed diffusion coefficients, Young's inequality gives $\lambda \leq 1$, strictly so whenever the diffusion spreads over two or more periods: under diffusion, θ_h^{stat} *understates* θ , and the gap $\theta_h^{LR} - \theta_h^{\text{stat}}$ is a direct estimate of the diffusion discount $1 - \lambda$.

The practical value of θ_h^{stat} lies in its monotone-then-plateau shape in h : the statistic rises as the cumulative leads capture more of the signal's news content, and plateaus once h reaches the full materialisation horizon $q + H + n$. In finite samples each squared covariance contributes a positive bias of order $\gamma_{zz}(0)/[T_{\text{eff}} V_{LR}(\Delta z)]$ even when its population value is zero, so the statistic drifts upward beyond the true plateau. The *adjusted* statistic subtracts this contribution term by term, in the spirit of the adjusted R^2 :

$$\theta_h^{\text{stat,adj}} \equiv \theta_h^{\text{stat}} - (h + 1) \cdot \frac{\hat{\gamma}_{zz}(0)}{T_{\text{eff}} \cdot \hat{V}_{LR}(\Delta z)}. \quad (26)$$

Beyond the true plateau every extra lead adds the penalty but no signal, so $\theta_h^{\text{stat,adj}}$ peaks, and

$$h^* \equiv \arg \max_{h \in \mathcal{H}} \theta_h^{\text{stat,adj}} \quad (27)$$

estimates the materialisation horizon $q + H + n$ without an ad-hoc tolerance (under diffusion the plateau *level* reads $\lambda \theta$, but the argmax logic is unaffected). The selector feeds directly into the identification: the noise regressions must include at least $h^* + 1$ TFP leads for the residualised signal to be purged of news (fewer leads bias the noise IRFs; more are asymptotically harmless but costly through the many-controls term of Proposition 2(B.3)), and the same h^* is the recommended truncation for (20) and (22). In the one-shot environment of Theorem 1(ii), where the materialisation horizon coincides with the lead support k^* , h^* is thus the minimum lead window for an unbiased noise IRF. Empirically, $\theta_h^{\text{stat,adj}}$ peaks at $h^* = 20$ for all three signals (exactly the lead window used throughout Section 6, where the q -robustness exercises corroborate it), and the adjustment shrinks the level news shares to 0.42 (BUS5), 0.46 (BUS12), and 0.03 (b_t , whose unadjusted 0.13 is essentially accumulated finite-sample projection noise).

Relation to conventional measures. Several familiar statistics estimate the same targets under stronger conditions, and we use them as benchmarks rather than headline measures. The simple news-to-signal ratio built on (B.1) is exact in the single-arrival case $H = 0$ (Appendix B, Proposition 6) but overstates the news content of multi-horizon signals; the conditional R^2 estimator of the flow share is likewise exact in the simple-news case but overstates the flow share when the signal

aggregates news across horizons, because the lag-block projection removes the anticipated content only partially; and partialling-based R^2 measures are further inflated in finite samples by the lag-predictability of persistent signals. Appendix B defines each statistic, derives its population limit, and tabulates the comparison; the empirical implementation is in Section 6 (Table 2).

5 Two Monte Carlo Illustrations

This section summarizes two sets of Monte Carlo experiments that validate the local-projection identification of Section 2 and demonstrate the simplicity-and-flexibility property of the method; complete designs, parameter values, tables, and figures are in Online Appendices C–D.

5.1 RBC-style DGP: both structural objects are recovered

The first experiment uses a canonical RBC-style data-generating process with a clean separation between transient surprise and persistent news shocks, so that Assumptions **A.1**–**A.4** hold by construction and statements (i)–(ii) of Proposition 1 are both testable. The results confirm the estimand dichotomy of Section 2.5 in numerical form: toggling the surprise innovations $\hat{u}_{t+j}$ in or out of the projection switches the estimand between the Wold-innovation IRF (the technology-side Chahrour and Jurado (2018) target) and the structural news IRF, and BFG recovers each object essentially exactly in population ($T = 10,000$), with finite-sample 68% bands that remain centred on the truth at $T = 250$. A principal-components experiment anticipates the informational-sufficiency discipline of the empirical section: when the simulated information set omits the forward-looking variables that span the news shock, the estimated news IRFs are biased, and adding two principal components restores sufficiency in the sense of Forni and Gambetti (2014) and Beaudry et al. (2019), the population analogue of the sufficiency test implemented on the U.S. data in Section 6.

5.2 The BLL benchmark: the recoverability ceiling in numbers

The second experiment is deliberately adversarial: data are generated from the estimated medium-scale DSGE of Chahrour and Jurado (2022), which embeds the Blanchard et al. (2013) (BLL) information structure. This DGP *violates* Assumption **A.2** (the permanent technology shock hits TFP on impact, so no anticipation horizon exists and the structural news IRF is not identifiable from any finite lead window), while **A.1** and **A.3** hold. Only statement (i) of Proposition 1 is therefore testable, and the experiment verifies it under the canonical benchmark of the noisy-news literature. Because the signal is informative about TFP at every future horizon through a persistent transitory component, the lead window must be large: $q = 40$ reduces the residual anticipated-TFP contamination to about one

percent, at the finite-sample cost captured by the many-controls diagnostic $\widehat{\xi}_Q \approx 0.99$, the regime in which the residual-augmented correction of Proposition 3 is decisive, in sharp contrast with the U.S. data where $\widehat{\xi}_Q \approx 0.18$ and the correction is negligible.

In population ($T = 100,000$), BFG with $q = 40$ reproduces the Wold-innovation IRF, the technology-side Chahrour and Jurado (2018) target, almost exactly for both the noise and the technology shocks, and the finite-sample distributions at $T = 284$ remain centred on it. The structural permanent-shock IRF ε^μ is a distinct object, with a markedly different hump-then-dip profile, that the finite-lead projection does not recover even in population. This gap is the recoverability ceiling of Theorem 1 in numerical form: finite-lead data attain $\mathcal{O}_1$ but not $\mathcal{O}_2$, and reading the two experiments together traces the ladder of Chahrour and Jurado (2022): when SW-style separability holds, both target objects are identified; when it fails, the technology-side target survives and the structural news IRF does not.

6 Empirical Application

6.1 Data and identification diagnostics

We use quarterly U.S. data over the sample period 1960:Q1–2019:Q4. Total factor productivity (TFP) is the utilization-adjusted series constructed by Fernald (2014). As forward-looking signals, we consider two components of the University of Michigan Consumer Survey: (i) the Index of Consumer Expectations over the next 5 years (ICS 5-Year, denoted BUS5), and (ii) the Expected Business Conditions over the next 12 months (ICS 12-Month, denoted BUS12). These variables have been used as proxies for noisy signals about future TFP in Forni et al. (2017a) and related studies.

To control for potential non-fundamentalness, we augment the information set with 7 factors extracted from a large panel of 127 macroeconomic series using the FRED-MD dataset of McCracken and Ng (2016) (March-2026 vintage). The factors are estimated via principal component analysis with the EM algorithm for missing data, following the methodology in McCracken and Ng (2016). Monthly factors are aggregated to a quarterly frequency by averaging. The same factor set is used throughout the empirical exercises that follow (news-to-signal ratio, local projections, FGLS, and the q-robustness diagnostics).

The headline results of this section come from the local-projection identification of Sections 2–3, applied to the Michigan confidence signals; the conventional benchmarks (news-to-signal ratio, FGLS) are summarised below and reported in full in Online Appendix E. Throughout, the structural interpretation of the news and noise IRFs is conditional on the informational-sufficiency conditions of Theorem 1; the FRED-MD factor block in the lag set and the $\widehat{\xi}_Q$ diagnostic are our empirical response to that requirement. A direct check supports it: following the informational-sufficiency test of Forni

and Gambetti (2014), the innovations of the baseline system ($\Delta z_t, s_t$, seven FRED-MD factors, $p = 4$ lags) are unpredictable from lags of an eighth factor excluded from the system (F -test p -values: 0.46 for the BUS5 signal innovation, 0.69 for BUS12, 0.87 for the TFP surprise residual). The test is not vacuous in this design: when the factor block is truncated to four factors, the BUS12 innovation becomes predictable from the excluded factors ($p = 0.021$), so the data do reject sufficiency when the information set is too poor. The zero-impact restriction of Assumption A.2 is checkable on the same objects: under $b_0 = 0$ the signal is orthogonal to the current TFP surprise, and in the data the correlation between the signal and the surprise residual $\hat{u}_t$ is -0.002 (HAC $t = -0.03$) for BUS5 and -0.003 ($t = -0.04$) for BUS12, with the same conclusion for the signal innovations (-0.005 and -0.009). The purpose is twofold: (i) to trace the impulse responses of the main macroeconomic aggregates to news and noise shocks using a transparent single-equation approach, and (ii) to compare the news-to-signal ratio across the two natural Michigan confidence horizons (the 5-year index BUS5 and the 12-month index BUS12).

Sample and variables. The sample runs from 1960:Q1 to 2019:Q4 ($T = 240$ quarters). The information set contains $n = 6$ core variables: (1) TFP growth Δz_t (Fernald utilization-adjusted series); (2) the consumer confidence signal s_t (BUS5 or BUS12); (3) real GDP per capita growth Δy_t ; (4) real consumption per capita growth Δc_t ; (5) hours worked per capita in levels N_t ; and (6) the federal funds rate R_t . The VAR/LP information set is augmented with $k = 7$ McCracken–Ng principal components extracted from the FRED-QD dataset.

Local-projection specification. Following Jordà (2005), we estimate level IRFs directly from LPs. For each variable j and horizon $h = 0, 1, \dots, H$, the *news* LP projects the variable on a block of $q + 1$ TFP leads, with focus coefficient on the furthest lead Δz_{t+q} :

$$y_{j,t+h} - y_{j,t-1} = \beta_{h,j}^{\text{news}} \Delta z_{t+q} + \sum_{k=0}^{q-1} \alpha_{h,j,k} \Delta z_{t+k} + \Psi(L) X_{t-1} + v_{t+h,j}.$$

The intermediate leads $\Delta z_t, \dots, \Delta z_{t+q-1}$ are partialled out so that, by Frisch–Waugh–Lovell, $\beta_{h,j}^{\text{news}}$ identifies the response of y_j to the part of Δz_{t+q} orthogonal to lower-order leads and to lagged controls X_{t-1} , i.e. to the q -step-ahead news component of TFP growth.⁵ The lead horizon q is selected by maximising the t -statistic of the signal coefficient in the auxiliary regression of Δz_{t+q} on s_t and lags, capped at $q_{\max} = 20$, mirroring the selection criteria introduced in Section 2.3. The *noise* LP adds the current signal as the focus regressor and includes the entire block of TFP leads $\Delta z_t, \Delta z_{t+1}, \dots, \Delta z_{t+q}$

⁵For $j = 1$ (TFP itself), the block specification cannot be applied without partialling out the dependent variable. We therefore replace the news LP with the simpler projection $\Delta z_{t+h} = \beta_{h,1}^{\text{news}} s_t + \Psi(L) X_{t-1} + v_{t+h,1}$, focus coefficient on s_t , which directly identifies the TFP response to the part of the signal orthogonal to the lagged controls.

as controls:

$$y_{j,t+h} - y_{j,t-1} = \beta_{h,j}^{\text{noise}} s_t + \sum_{k=0}^q \delta_{h,j,k} \Delta z_{t+k} + \Psi(L) X_{t-1} + v_{t+h,j}.$$

The coefficient $\beta_{h,j}^{\text{noise}}$ then captures the response of variable j to the part of s_t orthogonal to current and future realised TFP up to horizon q . All projections use $p = 4$ lags, $H = 20$ quarters, Newey–West HAC standard errors, and 90% confidence bands.

Choice of the lead window q in the empirical application. The empirical BFG implementation sets $q = 20$, so that the $q + 1 = 21$ leads $\Delta z_t, \Delta z_{t+1}, \dots, \Delta z_{t+20}$ enter the noise regression as controls. The justification differs from the BLL Monte Carlo of Section 5.2, even though both share approximately white-noise TFP growth. In BLL, the signal’s predictive content for future Δz_{t+k} decays geometrically as ρ^k through the persistent latent state z , with no finite cutoff; the lead window must be large enough that the residual contamination is negligible (which forces $q = 40$). In the data, the predictive horizon of the signal is instead *survey-motivated*: BUS5 asks respondents about expected conditions five years ahead, i.e. 20 quarters, so the signal is designed to carry predictive content within a 20-quarter window. Setting $q = 20$ matches this survey horizon, provides a uniform window across the BUS5 and BUS12 specifications, and coincides with the data-driven search ceiling $q_{\max} = 20$ used in the t -statistic selection step. With $T = 240$ in our sample, going substantially beyond $q = 20$ would amplify the many-controls term of order $O((q + 1)/T)$ from Proposition 2, item (B.3), so $q = 20$ is the natural empirical choice.

Theorem 1 disciplines this choice: the window that identification requires is not the survey’s materialisation horizon but the lead support k^* of the news shock, and the two coincide only in the one-shot case: under gradual diffusion with an insufficient information set, k^* is generically infinite, and no survey-design bound would rescue a finite window. Three empirical observations support treating k^* as finite and spanned by $q = 20$ here. First, the near-white-noise property of Fernald’s utilization-adjusted Δz_t (largest autocorrelation 0.12 over twenty lags, Ljung–Box p -values 0.43–0.73; Section F.1) bears directly on the diffusion tail that drives $k^* = \infty$: a tail $b_j \neq 0$ beyond the arrival horizon generates moving-average autocorrelation in Δz_t at lags up to the tail length, and none is detected: the environment is statistically indistinguishable from the one-shot case of Theorem 1(ii), in which the lead window equals the materialisation horizon. (This reading assumes the tail is not masked by a proportionally large surprise variance; the next two checks do not rely on it.) Second, the informational-sufficiency test reported above supports the rich-information case of the proposition, in which k^* collapses regardless of diffusion. Third, the proposition makes the failure of a too-short window *observable*: a window below k^* generates spurious news responses at horizons beyond the window and leaves news content in the residualised signal, both decaying geometrically in the window length. The q -robustness exercise below exhibits precisely the absence of these truncation signatures

at $q = 20$: the news IRFs are stable from $q \geq 10$ onward rather than drifting as the window grows, and the noise-on-TFP response, the sharpest contamination diagnostic, is numerically zero only at $q = 20$. The survey horizon thus motivates the candidate window; the white-noise property of TFP growth, the sufficiency test, and the absence of the truncation signatures of Theorem 1 are what support the identification.

Residual-augmented correction in the empirical application. We applied the residual-augmented correction of Proposition 3 to the empirical noise IRFs. The plug-in diagnostic $\hat{\xi}_Q$, the partial- R^2 of s_t on the $q + 1$ TFP-lead controls given the lag block, equals 0.176 for BUS5 and 0.198 for BUS12 in our sample. The implied correction factor $1/[1 - (q + 1)/T \cdot \hat{\xi}_Q]$ is 1.018 and 1.020 respectively, well within the 90% HAC bands reported below. The correction therefore leaves the empirical noise IRFs visually unchanged and does not affect any of the qualitative or quantitative conclusions of Section 6.

The contrast with the BLL Monte Carlo of Section 5.2 (where $\hat{\xi}_Q \approx 0.99$ produces a correction factor of ≈ 1.20) is informative: $\hat{\xi}_Q$ is itself a directly interpretable diagnostic of how much the many-controls bias of Proposition 2 (B.3) actually matters in any given application. In a regime where the signal carries information that is largely orthogonal to current and future realised TFP (as in the data, where Fernald’s Δz_t is approximately white noise and the consumer-confidence series does not span a persistent transitory technology component), the $q + 1$ TFP leads absorb only a small fraction of s_t ’s variance, the implied bias is small, and the correction is quantitatively negligible. In a regime where the signal is heavily correlated with future TFP through a persistent latent state, as in BLL, the leads absorb almost all of s_t ’s variance and the correction is decisive. Reporting $\hat{\xi}_Q$ alongside any finite-lead LP estimate is therefore a simple, model-free way to inform the reader whether the small-sample correction will materially move the IRFs.

6.2 Do noise shocks drive business cycles?

News and noise IRFs across estimators. Figures 1–4 report the news- and noise-shock IRFs for both signals (BUS5 and BUS12) for the six macroeconomic variables (TFP, signal, GDP, consumption, hours, federal funds rate). Each panel overlays three estimators on a common information set: LP_W (the leads-only specification, with the $q + 1$ TFP leads as controls in both the news and noise regressions: technology-side Chahrour and Jurado (2018) target in the news regression, noise-side target in the noise regression); LP_ε (the structural news IRF $\Theta_{j,2}/b_q$ under Assumption A.4: in the *news* regression the $q + 1$ surprise residuals $\hat{u}_{t+0:q}$ are used as controls and the focus is the lead Δz_{t+q} ; in the *noise* regression the focus is s_t and the controls combine the $q + 1$ TFP leads $\Delta z_{t+0:q}$ and the $q + 1$ surprise residuals $\hat{u}_{t+0:q}$, so that, by Frisch–Waugh–Lovell, the news and surprise components of future TFP are jointly partialled out of s_t and the slope identifies the noise component alone);

and FGLS (the dynamic-rotation SVAR identification of Barsky and Sims, 2011; Forni et al., 2017a). Shaded 90% HAC bands are reported only for LP_W to keep the panels readable; the LP_ε and FGLS bands are typically of comparable width.

For the news shock the three estimators broadly agree on TFP (all three converge near 1% at long horizons) and on the qualitative profile of GDP, consumption, hours, and the federal funds rate. The signal itself responds positively under all three methods. The LP_ε profile is somewhat smaller on TFP at long horizons and broadly tracks LP_W on real activity, with FGLS sitting closer to the LP_W trajectory. For the noise shock, the TFP rows of LP_W and LP_ε coincide by construction (both reduce to the same regression for $j = 1$, since including any future-TFP or surprise-residual control would partial out the dependent variable from itself), while FGLS delivers a near-zero TFP response throughout. On real activity (GDP, consumption, hours) the noise IRFs are hump-shaped and transitory, peaking around $h \approx 4\text{--}6$ and decaying back toward zero, consistent with agents acting on misinformation and then learning the truth.

Robustness and stability. Three families of checks, reported in full in Online Appendix F, discipline the structural reading. First, the IRFs are stable in the lead window: news IRFs cluster tightly for $q \geq 10$, and the noise-on-TFP response, the sharpest contamination diagnostic, is numerically zero only at $q = 20$, the absence of the truncation signatures that Theorem 1 associates with a too-short window. Second, the estimated structural news IRFs stabilise as the information set grows: re-estimating LP_ε with 0, 2, 4, and 7 factors, the deviations from the seven-factor baseline roughly halve with each enlargement (to 9–18 percent of the peak for real activity at four factors), while the leads-only LP_W news IRFs are essentially invariant. The leads-only *noise* IRFs shrink toward the baseline from above by a factor of three to six, the observable signature of the mixture result of Corollary 1(ii). Read together, these exercises reproduce the ordering of the identification ladder in the data.

Application 1: Michigan ICS 5-Year (BUS5)

The news- and noise-shock IRFs for BUS5 are reported in Figures 1 and 2 above.

A news shock in the BUS5 specification generates a hump-shaped and permanent increase in TFP, with the peak response around $h = 12\text{--}16$ quarters and 90% bands strictly above zero. Real GDP, consumption, and hours worked all respond positively, with hump-shaped dynamics peaking at medium horizons; the federal funds rate rises moderately, consistent with a systematic monetary response to the anticipated productivity boom. The confidence signal itself responds strongly and persistently to news, providing a clear anticipation channel. By contrast, the noise shock has *no* permanent effect on TFP (the response is flat and indistinguishable from zero), but it moves the signal significantly on impact and generates transitory, mostly modest movements in GDP, consumption,

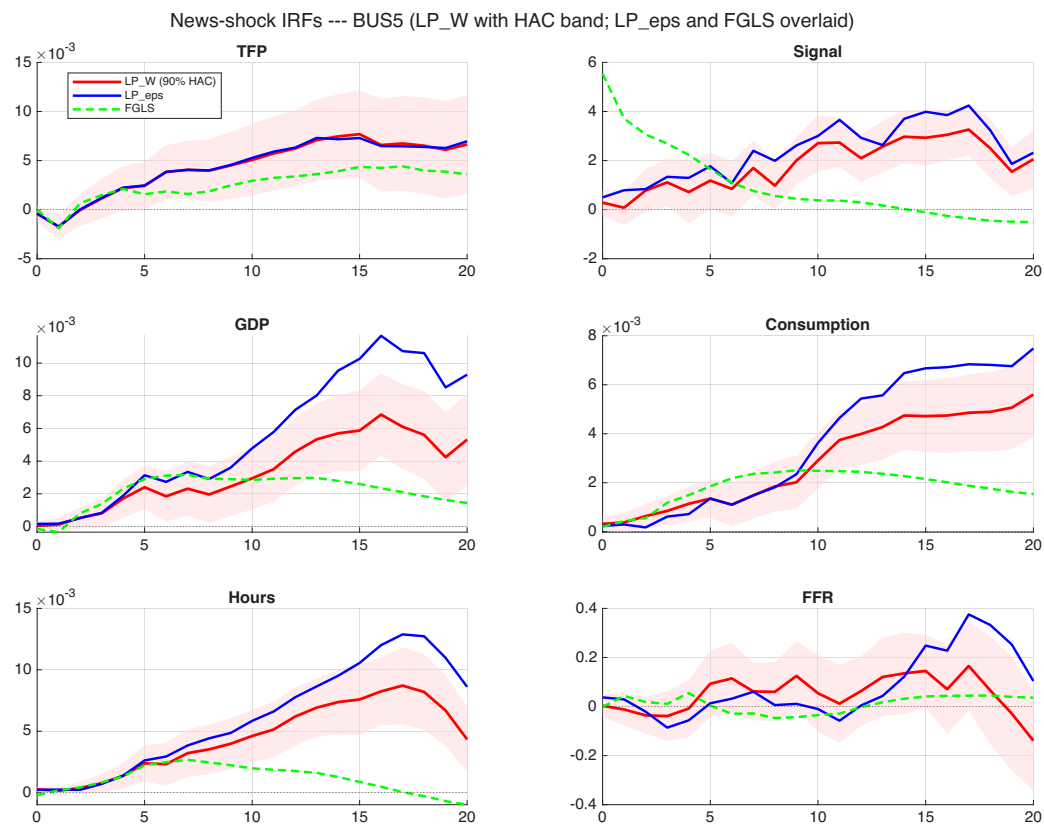


Figure 1: News-shock IRFs to BUS5: three estimators on the same 13-variable information set (TFP, signal, GDP, consumption, hours, federal funds rate, plus seven FRED-MD principal components), $p = 4$ lags, $H = 20$ quarters. LP_W (red, with 90% Newey–West HAC band); LP_ε (blue solid); FGLS (green dashed). Sample: 1960:Q1–2019:Q4.

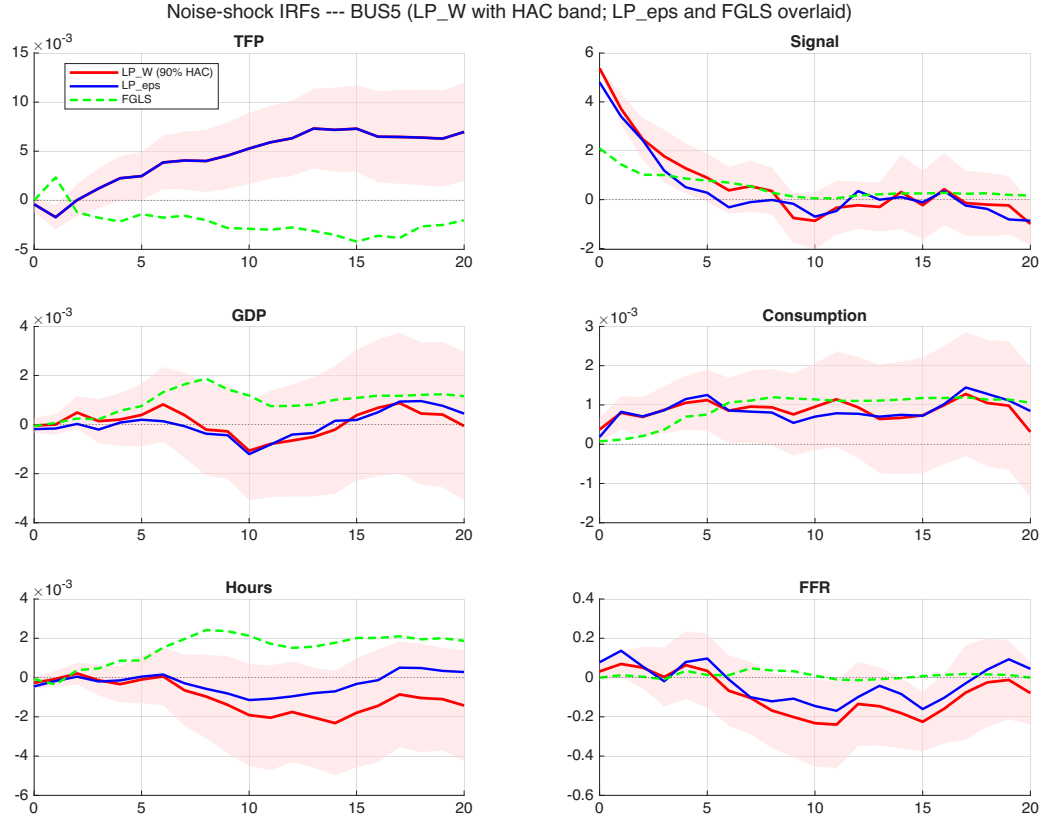


Figure 2: Noise-shock IRFs to BUS5: same conventions as Figure 1. The TFP row coincides between LP_W and LP_ε because both reduce to the regression $\Delta z_{t+h} = \beta s_t + \gamma' X_{t-1}^{(p)} + e$ for $j = 1$ (the dependent variable would otherwise enter the lead block as a control).

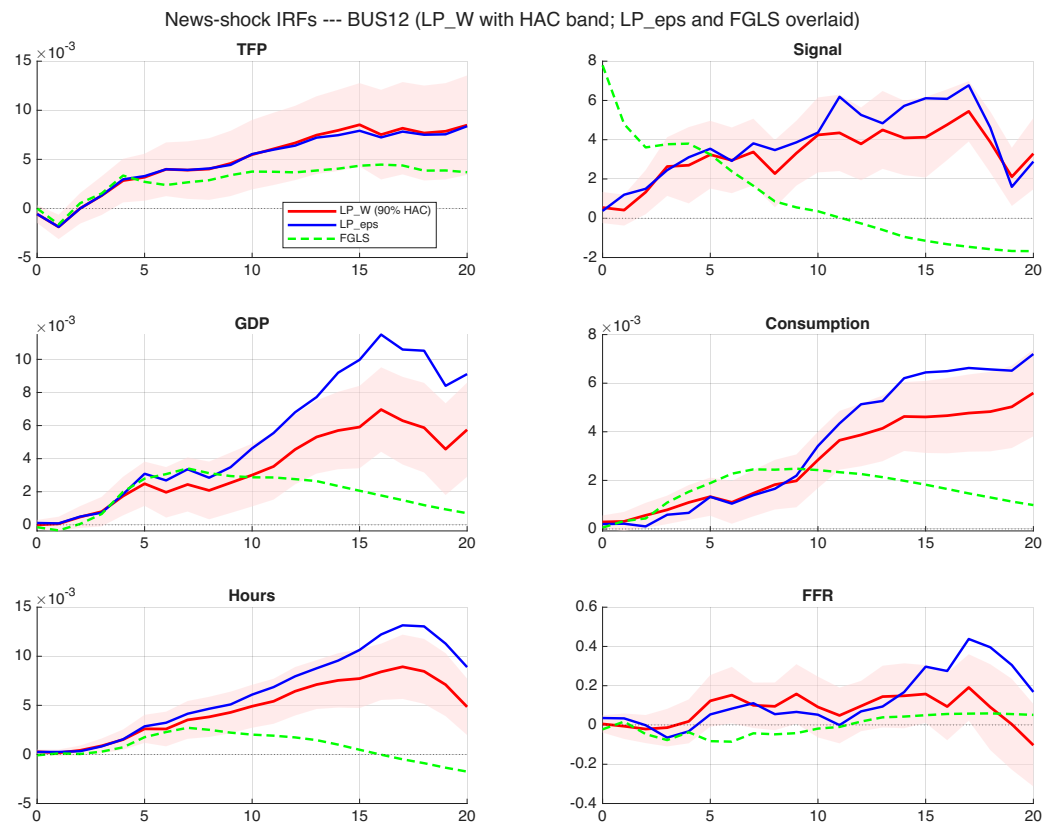


Figure 3: News-shock IRFs to BUS12: same conventions as Figure 1.

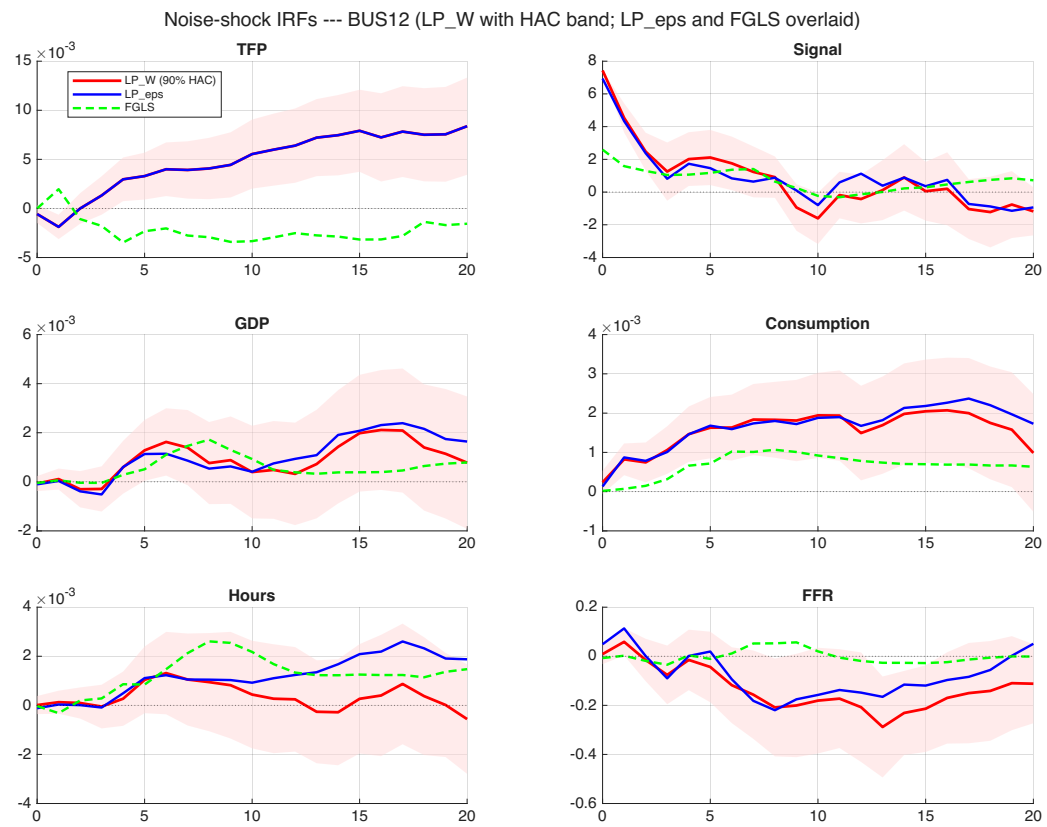


Figure 4: Noise-shock IRFs to BUS12: same conventions as Figure 2.

and hours. The federal funds rate response to noise is small and statistically weak. These patterns confirm the theoretical prediction that noise in the signal can contaminate short-run macro dynamics without altering the long-run productivity trend.

Application 2: Michigan ICS 12-Month (BUS12)

The news- and noise-shock IRFs for BUS12 are reported in Figures 3 and 4 above.

The qualitative picture is very similar to the BUS5 case. The news shock again produces a permanent, hump-shaped increase in TFP, a persistent signal response, and positive hump-shaped responses in output, consumption, and hours. The noise shock has a flat TFP response by construction, a large on-impact movement in the signal itself, and small but mostly positive transitory responses in GDP and consumption. Relative to BUS5, the noise responses of real variables under BUS12 are slightly larger in magnitude and more often above zero at the 90% level, suggesting that the shorter-horizon confidence index carries more transient noise-driven information about near-term activity. The monetary policy response (federal funds rate) is comparable across the two specifications.

A VAR-implied signal as additional validation. As an additional validation, Online Appendix F replaces the survey signal by a VAR-implied rational forecast of TFP, $b_t = \mathbb{E}_t[z_{t+20}] - z_t$, in the spirit of Chahrour and Jurado (2022). The qualitative IRF picture is unchanged, and once finite-sample projection noise is removed the constructed forecast contains essentially no detectable news, consistent with survey beliefs incorporating information outside the econometrician’s macroeconomic information set (the b_t rows of Table 2 below). The appendix also discusses the choice of signal under the random-walk behaviour of measured TFP.

6.3 How noisy are expectations?

Model-free news-to-signal ratios. Table 2 reports two model-free invariant statistics from Section 4: θ_h^{stat} (23), exact for the level news share θ absent TFP diffusion and a lower bound $\lambda\theta$ under gradual diffusion, and $\tilde{R}_h^{2,LR}$ (B.2), exact for the innovation news share $\tilde{\theta}$ under the system-invertibility condition on the lag block, together with the partialling-based estimator $\mathbf{nsr}_h^{\varepsilon, \text{adj}}$ of θ as a comparison. The diffusion-robust long-run statistics θ_h^{LR} (20) and $\tilde{\theta}^{LR}$ (22) require only the additional HAC estimate $\hat{V}_{LR}(s_t)$; the gap between θ_h^{LR} and θ_h^{stat} then measures the diffusion discount directly. Five observations stand out.

(1) *Level news share, primary estimator.* Under θ_h^{stat} , the implied noise variance share of the signal level equals approximately 48% for BUS5, 44% for BUS12, and 82% for b_t at $h = 20$ (and plateaus at 47%, 42%, 82% respectively at $h = 40$). Absent TFP diffusion this formulation is unbiased under the general signal model, invariant to the transitory shock σ_u^2 , and computable without partialling

out the surprise residuals; under gradual diffusion it understates the news share by the factor λ of (25), so the noise-share readings above are upper bounds; and given the specification test of observation (5), the readings on the raw levels should be viewed as descriptive projection statistics rather than model-based shares. The bias-corrected variant $\theta_h^{\text{stat,adj}}$ (26) (which removes the per-term finite-sample inflation of θ_h^{stat} in the spirit of the adjusted R^2) attains a clear maximum at $h^* = 20$ for all three signals, with values $\theta_{20}^{\text{stat,adj}} = 0.42$ (BUS5), 0.46 (BUS12), and 0.03 (b_t). The argmax matches exactly the lead window $q = 20$ used in the local-projection identifications, providing an internally consistent data-driven choice. The reading: the surveys carry approximately equal parts of genuine news about future TFP and idiosyncratic noise, while the VAR-implied b_t has a much smaller stock of news per unit signal variance (much of b_t is contemporaneous estimation noise from the rational projection, as the bias correction makes vivid: the unadjusted plateau of ≈ 0.13 shrinks to ≈ 0.03 once the cumulative finite-sample bias of 21 covariance terms is removed).

(2) *Innovation news share.* Under $\tilde{R}_h^{2,LR}$, the corresponding noise share for the contemporaneous innovation equals approximately 49% for BUS5, 48% for BUS12, and 88% for b_t at $h = 20$. The closeness to the level estimator θ_h^{stat} ($\sim 51\%$, $\sim 53\%$, $\sim 12\%$ respectively for the news share) suggests that the empirical signal structure is moderately multi-horizon: most of the news content in s_t loads the contemporaneous news shock, with smaller contributions from anticipated shocks at longer horizons (which the lag block removes only partially). The level and innovation readings converge.

(3) *Comparison with the partialling-based estimator.* The auxiliary $\mathbf{nsr}_h^{\varepsilon,\text{adj}}$ (the augmented R^2 with surprise residuals as controls) yields a much higher level news share of $\approx 90\%$ for the surveys and $\approx 96\%$ for b_t . This is the squared-version of the original news-to-signal ratio of Forni et al. (2017a) interpreted as an R^2 ; we report it for transparency but warn the reader against treating it as a clean news variance share. As shown analytically in Remark 3 and verified by Monte Carlo, the partialling-based R^2 is inflated by two distinct mechanisms: the square-of-sum overcount of multi-horizon news with same-signed loadings (double-counting news arrivals at different dates), and a finite-sample lag-prediction contribution from the auxiliary $X_{t-1}^{(p)}$ block (the signal s_t is persistent, so a large share of its variance is mechanically explained by its own lags whether or not those lags carry news). θ_h^{stat} avoids both, but under gradual TFP diffusion it errs in the opposite direction ($\lambda \leq 1$), so the two estimators bracket the level news share; the diffusion-robust θ_h^{LR} of Section 4 adjudicates.

(4) *Why b_t has a near-zero adjusted news share: TFP is a near-random walk.* The very small bias-corrected value $\theta_h^{\text{stat,adj}} \approx 0.03$ for the VAR-implied forecast b_t reflects a well-documented property of Fernald (2014) TFP: over our sample, Δz_t is statistically indistinguishable from a white noise. All sample autocorrelations satisfy $|\rho_k| \leq 0.12$ for $k \leq 20$, the Ljung–Box statistic does not reject the white-noise null at any standard horizon (p -values of 0.45, 0.65, 0.73, 0.43 at $L = 4, 8, 12, 20$ respectively), and the Newey–West long-run-to-short-run variance ratio $V_{LR}(\Delta z_t)/\hat{\gamma}_{zz}(0) = 0.94$ is essentially unity. When TFP in levels is well approximated by a random walk, the population

variance of the VAR-implied cumulative forecast $b_t = \sum_{k=1}^{20} \mathbb{E}_t[\Delta z_{t+k} \mid \mathcal{I}_t^{\text{VAR}}]$ tends toward zero, so its empirical variance is dominated by VAR estimation noise: the unadjusted plateau $\theta_h^{\text{stat}} \approx 0.13$ for b_t is mostly the accumulated estimation noise of the rational projection over $h+1 = 21$ leads, which the bias correction strips out, bringing the adjusted news share to ≈ 0.03 . The surveys BUS5 and BUS12 retain a substantial adjusted news share ($\approx 0.42\text{--}0.46$) because they draw on information sources outside the econometrician’s VAR: consumer-side beliefs and micro signals that are not summarised by past macro time series. This signal-side contrast is consistent with the recoverability framework of Chahrour and Jurado (2022), which articulates the conditions under which a candidate signal carries identifying content about the news shock; we return to this point in Section F.1 below.

(5) *The diffusion-robust flow statistic: one third of the information flow is noise.* The long-run statistics of Section 4 are implemented on the signal innovation η_t^s (the residual on the same lag block as the other statistics), as prescribed in Section 4: applied to the raw survey *levels*, the long-run identification fails its own specification test: $V_{LR}(s_t)/\text{Var}(s_t)$ ranges from 7 to 15 and grows with the HAC bandwidth, and $\text{nsr}_h^{LR} \approx 10\text{--}11 \gg 1$, confirming that the survey levels carry persistence that the finite-memory signal model does not accommodate. On the innovation, everything is well behaved: η_t^s is nearly white (first autocorrelation below 0.01 in absolute value, $V_{LR}(\eta^s) \approx \text{Var}(\eta^s)$), and the Kolmogorov flow statistic delivers $\tilde{\theta}^{LR} = 0.63$ for BUS5 and 0.63 for BUS12, stable across HAC bandwidths 8–20 (range 0.62–0.65), with the two independent long-run constructions (θ^{LR} applied to η^s and the spectral formula (22)) agreeing to two decimal places, as they must when the innovation is well whitened. The reading is also robust to the auxiliary specification: adding the eighth FRED-MD factor to the lag block leaves $\tilde{\theta}^{LR}$ unchanged at 0.63 for both surveys, and across lag orders $p \in \{2, 4, 8\}$ with the full factor block the statistic ranges from 0.43 to 0.63, so the implied noise share of the information flow stays between roughly one third and one half in every configuration. Truncating the factor block to four factors instead pushes the statistic to the $\tilde{\theta}^{LR} = 1$ boundary ($\hat{\sigma}_\nu^2 \leq 0$) in most configurations, consistent with the informational-sufficiency test of Section 6.1, which rejects the under-sized information set. Persistence of the structural noise, finally, is a one-directional sensitivity rather than a threat: the identity (19) delivers $S_\nu(0)$ regardless of the noise’s serial correlation (Section 4), and positively autocorrelated noise makes the white-noise reading an overstatement: re-solving the decomposition under AR(1) noise, the flow noise share falls monotonically from 37–38 percent at $\rho_\nu = 0$ to 25, 16, and 9 percent at $\rho_\nu = 0.2, 0.4$, and 0.6, for both surveys. The headline share is therefore an upper bound under the empirically plausible sign of belief-noise persistence. The flow noise share of the surveys is therefore approximately 37%, somewhat below the $\approx 48\%$ implied by $\tilde{R}_h^{2,LR}$ (the difference is the $V_{LR}(\eta^s) < \text{Var}(\eta^s)$ correction). For the VAR-implied b_t , $\tilde{\theta}^{LR} \approx 0.00$: the news innovation variance recovered by the Kolmogorov formula is indistinguishable from zero, in full agreement with the bias-corrected $\theta_{20}^{\text{stat,adj}} = 0.03$ of observation (4). The headline reading: *roughly one third of what consumers newly learn each quarter is noise, while the*

VAR-implied forecast carries essentially no detectable news about subsequently realised TFP. A caution on vocabulary: for b_t , what the statistics classify as “noise” is in large part VAR estimation error rather than the belief shock ν_t of the model: the precise statement is the absence of detectable news content, not a measurement of agents’ noise.

Reconciling noisy signals with small noise IRFs. The empirical noise variance shares of the surveys are substantial: approximately half of the variance of BUS5 and BUS12 is structurally orthogonal to realised future TFP. Yet, as the overlay IRF figures (Figures 1–36) show, the response of real activity to the noise component of these signals is small and transitory: noise IRFs on GDP, consumption and hours peak below 0.002 in absolute value and decay back toward zero by $h = 20$. The two readings are jointly consistent and informative. The signals carry non-trivial noise, but that noise does not propagate strongly to macroeconomic outcomes. This pattern is consistent with filtering of the signal by agents or with attenuated transmission of belief shocks; our identification establishes the fact, not the mechanism, and we do not attempt to discriminate between these explanations. The empirical conclusion stands regardless: the macroeconomic effects of the noise component of confidence signals are limited, even when the noise itself is large.

Relation to the sentiment-shock literature. Our model-free statistic places the noise share of consumer-confidence signals at $\theta_{20}^{\text{stat,adj}} \approx 0.42\text{--}0.46$ (noise share $\approx 54\text{--}58\%$). This order of magnitude is broadly compatible with the SVAR-based analysis of Fève and Guay (2018), who identify sentiment shocks, the non-fundamental component of belief signals, in a structural VAR with consumer-confidence surveys. The two exercises are methodologically distinct: Fève and Guay (2018) target the contribution of sentiment shocks to macroeconomic aggregates through SVAR identifying restrictions, whereas $\theta_h^{\text{stat,adj}}$ measures the noise share in the signal itself in a model-free way. The objects are not directly comparable, but the two readings of the data (one model-free on the signal, one structural on transmission) do not contradict one another, and both suggest that the non-fundamental component of consumer-confidence signals is non-trivial.

6.4 Why do previous estimates differ?

Our findings assign noise a small macroeconomic role. Forni et al. (2017a) reach the opposite headline: in their baseline six-variable specification, noise shocks account for approximately 16 percent of the forecast-error variance of GDP and 22 percent of investment at the one-year horizon, and 20–25 percent of GDP, consumption, and investment at the two-year horizon when TFP replaces potential output (their Tables 2–3; sample 1960:I–2010:IV, VAR in levels with four lags, stock prices as the signal). The two readings come from the same class of data, so the discrepancy demands an explanation

Signal	Statistic	value at horizon h						at $h = 20$	
		0	4	8	12	16	20	value	noise share
BUS5	θ_h^{stat} (23)	0.026	0.155	0.281	0.411	0.476	0.516	0.516	$\approx 48\%$
	$\theta_h^{\text{stat,adj}}$ (26)	0.022	0.133	0.241	0.351	0.397	0.418*	0.418	$\approx 58\%$
	$\tilde{R}_h^{2,LR}$ (B.2)	0.000	0.073	0.136	0.333	0.386	0.513	0.513	$\approx 49\%$
	$\tilde{\theta}^{LR}$ (22)	—	—	—	—	—	0.63	0.63	$\approx 37\%$
	$\mathbf{nsr}_h^{\varepsilon,\text{adj}}$ (B.6)	0.862	0.895	0.898	0.898	0.896	0.896	0.896	$\approx 10\%$
BUS12	θ_h^{stat} (23)	0.019	0.139	0.267	0.411	0.502	0.558	0.558	$\approx 44\%$
	$\theta_h^{\text{stat,adj}}$ (26)	0.015	0.116	0.226	0.351	0.423	0.460*	0.460	$\approx 54\%$
	$\tilde{R}_h^{2,LR}$ (B.2)	0.000	0.088	0.124	0.306	0.370	0.525	0.525	$\approx 48\%$
	$\tilde{\theta}^{LR}$ (22)	—	—	—	—	—	0.63	0.63	$\approx 37\%$
	$\mathbf{nsr}_h^{\varepsilon,\text{adj}}$ (B.6)	0.900	0.921	0.919	0.920	0.919	0.918	0.918	$\approx 8\%$
b_t	θ_h^{stat} (23)	0.000	0.036	0.062	0.085	0.110	0.132	0.132	$\approx 87\%$
	$\theta_h^{\text{stat,adj}}$ (26)	−0.005	0.013	0.020	0.024	0.029	0.031*	0.031	$\approx 97\%$
	$\tilde{R}_h^{2,LR}$ (B.2)	0.015	0.007	0.008	0.053	0.112	0.119	0.119	$\approx 88\%$
	$\tilde{\theta}^{LR}$ (22)	—	—	—	—	—	0.00	0.00	$\approx 100\%$
	$\mathbf{nsr}_h^{\varepsilon,\text{adj}}$ (B.6)	0.955	0.956	0.960	0.960	0.960	0.961	0.961	$\approx 4\%$

Table 2: Five model-free R^2 -style news-to-signal ratios from Section 4. The statistics θ_h^{stat} (23) and $\tilde{R}_h^{2,LR}$ (B.2) target the level and innovation news shares θ and $\tilde{\theta}$ respectively; both are invariant to the transitory shock σ_u^2 and computable without partialling out the surprise residuals, with $V_{LR}(\Delta z_t)$ estimated by Newey–West HAC. θ_h^{stat} is exact absent TFP diffusion and reads as a lower bound $\lambda\theta$ on the news share under gradual diffusion; $\tilde{R}_h^{2,LR}$ is exact in the simple-news case and overstates the flow share under multi-horizon aggregation (Appendix B). The bias-corrected variant $\theta_h^{\text{stat,adj}}$ (26) subtracts the per-term finite-sample bias of θ_h^{stat} (analogous to the adjusted R^2); the asterisk * marks the argmax horizon h^* , which lands at $h = 20$ for all three signals, exactly the lead window $q = 20$ used in the local-projection identifications. The fourth statistic $\mathbf{nsr}_h^{\varepsilon,\text{adj}}$ (B.6) is the augmented R^2 with surprise-residual partialling (the partialling-based level estimator); it is reported for comparison but is empirically inflated by the lag-predictability of s_t and by the square-of-sum multi-horizon factor (see Remark 3). The row $\tilde{\theta}^{LR}$ (22) is the diffusion-robust Kolmogorov flow statistic, computed on the signal innovation η_t^s (same lag block) with the full $h = 20$ covariance window; it is a single number per signal (shown under $h = 20$ and $h = 40$), stable across HAC bandwidths 8–20 (range 0.62–0.65 for the surveys), unchanged when an eighth factor is added to the lag block, and ranging 0.43–0.63 across lag orders $p \in \{2, 4, 8\}$ with the full factor block. Lag block: $X_{t-1}^{(p)} = (\Delta z_{t-1:t-p}, s_{t-1:t-p}, F_{1,t-1:t-p}, \dots, F_{7,t-1:t-p})$ with $p = 4$ lags. Sample: 1960:Q1–2019:Q4 for BUS5 and BUS12; 1962:Q1–2019:Q4 for b_t . Noise share reported at $h = 20$. The primary unadjusted statistics agree on $\sim 50\%$ noise for the surveys and $\sim 85\%$ noise for b_t ; the bias-corrected variant shifts these to $\sim 55\%$ for the surveys and, for b_t , to an essentially undetectable news content ($\approx 3\%$).

rather than a shrug, and the identification theory of Section 3 supplies a precise one: *informational sufficiency*.

We first reproduce their finding on our own sample. Table 3 reports the share of forecast-error variance attributed to the noise shock when the FGLS dynamic rotation is applied to our data in their configuration: a six-variable system in levels with four lags, with the consumer-confidence signal in place of stock prices. The noise shock then accounts for 13–21 percent of the forecast-error variance of GDP at the one-to-two-year horizons for BUS12 (14 percent for BUS5), with comparable shares for consumption and hours: the published magnitudes emerge on our data, from our signal. Augmenting the *same* system with the seven FRED-MD factors collapses the noise share of GDP, consumption, and hours to at most 4 percent at every horizon, for both signals and for lag orders 4 and 8. Two candidate explanations are thereby ruled out: the signal (we obtain their magnitudes with consumer confidence, and their own robustness exercise with the E5Y confidence component leaves their conclusions unchanged) and the estimand (on the factor-augmented information set the FGLS rotation and our local projections agree that the noise footprint is small, as the overlay figures show). The sample plays at most a minor role. What separates the two readings is the information set.

The informational-sufficiency test of Forni and Gambetti (2014) adjudicates between the two information sets on our data. The innovations of the six-variable system are strongly predictable from lags of the FRED-MD factors (F -statistics between 2.3 and 12.1, with p -values below 0.001 for every equation and both signals), so the configuration that generates the large noise shares fails the orthogonality requirement on our sample; the factor-augmented system passes it (Section 6.1). The mechanism is the one our theory isolates: on an insufficient information set the rotated “noise” shock absorbs news and surprise content that the econometrician cannot separate (the mixture logic of Corollary 1(ii)), and restoring sufficiency reclassifies that content, shrinking the measured macroeconomic footprint of noise by an order of magnitude while leaving the signal itself as noisy as the model-free statistics of Table 2 say it is. Forni et al. (2017a) implemented the same orthogonality test and did not reject sufficiency of their six-variable systems on their sample and vintage (their Table 1); on our longer sample the analogous configuration is rejected decisively. The reconciliation is therefore not a verdict on the dynamic-rotation procedure, which is a valid identification under its conditions; it is a demonstration that the answer to “does noise drive business cycles?” runs through an informational-sufficiency requirement that is testable, and that, once imposed, overturns the answer.

Conventional benchmarks. For comparison with the prior literature, Online Appendix E reports the two conventional benchmarks on the same information set. The semi-parametric news-to-signal ratio $\mathbf{nsr}_h^{\text{news}}$ of Appendix B rises monotonically with the horizon and stabilises near $\alpha \approx 0.90$ at $h = 20$ for both signals, robustly across factor counts and lag orders (Tables 11–12); the FGLS dynamic-rotation estimates yield $\alpha \approx 0.87$ – 0.88 in the level specification (Table 13). Their higher implied

Signal	Variable	Six variables (FGLS configuration)			Six variables + 7 factors		
		1-year	2-years	4-years	1-year	2-years	4-years
BUS5	GDP	0.14	0.14	0.10	0.00	0.02	0.02
	Consumption	0.10	0.12	0.10	0.01	0.02	0.03
	Hours	0.09	0.12	0.10	0.00	0.02	0.04
BUS12	GDP	0.19	0.21	0.13	0.00	0.01	0.01
	Consumption	0.13	0.16	0.13	0.01	0.02	0.02
	Hours	0.13	0.20	0.18	0.00	0.02	0.03
Forni et al. (2017a), GDP		0.16	0.14	0.10	—	—	—
GDP (TFP spec.)		0.24	0.24	0.22	—	—	—

Table 3: Share of forecast-error variance attributed to the noise shock by the FGLS dynamic rotation, on our sample (1960:Q1–2019:Q4). Left block: six-variable system in log levels (TFP, signal, GDP, consumption, hours, federal funds rate), four lags, the Forni et al. (2017a) configuration with the consumer-confidence signal in place of stock prices. Right block: the same system augmented with the seven FRED-MD principal components (cumulated to levels). Shares are computed from the rotated orthogonalised IRFs at horizons 4, 8, and 16 quarters. Memo rows: the published noise shares of GDP in Forni et al. (2017a), Table 2 (baseline, potential output) and Table 3 (TFP specification), at their one-, two-, and four-year horizons. The six-variable system fails the informational-sufficiency test on our sample (all $p < 0.001$); the factor-augmented system passes it.

news shares relative to Table 2 reflect the estimands and bias properties documented in Appendix B: the correlation-type ratio is exact only in the single-arrival benchmark and overstates news content under multi-horizon news aggregation. Both benchmarks are consistent with the transmission-side findings above.

Summary. The empirical findings deliver a consistent two-part message. *On the signal side, the expectation signals are substantially noisy.* The headline number is the diffusion-robust Kolmogorov flow statistic: $\tilde{\theta}^{LR} = 0.63$ for both BUS5 and BUS12, stable across HAC bandwidths, lag orders, and factor counts, and an upper bound under serially correlated structural noise: *approximately 37% of the quarterly information flow of consumer confidence is noise.* On the stock dimension, the sum-of-squared-covariances statistic reads a noise share of up to one half for the raw signal *level* (a descriptive figure, since the finite-memory signal model is rejected for the levels by the specification test of observation (5)), while the 10% reading of the partialling-based estimator is attributable to its documented lag-predictability inflation (Table 2). *On the transmission side, the noise that reaches TFP and real activity is small.* The semi-parametric **nsr** statistic converges to approximately 0.90 at long horizons, the FGLS dynamic rotation in levels yields $\alpha \approx 0.90$, the Chahrour-Jurado spectral factorization produces TFP noise responses at machine zero, and the local projection confirms that the noise shock has zero contemporaneous effect on TFP. These findings hold for both the 5-year and 12-month consumer confidence indexes.

7 Conclusion

The paper delivers three results. *Identification:* a finite-lead projection can recover the structural news and noise impulse responses if and only if its lead window spans the lead support k^* of the news shock in the innovations of the observables: finite under one-shot materialisation or informational sufficiency, generically infinite under gradual diffusion with an insufficient information set. The result orders the identification ladder (Wold, structural, full-history) by the information each rung requires, with k^* the threshold between the two structurally relevant rungs. Recoverability of the shock path is not required: the response can be identified even where the shock is not. *Method:* the answer is implemented by three ordinary least-squares projections (surprise, news, noise) in which the estimated surprise innovations switch the estimand from the Wold object to the structural news and noise responses without changing the projection machinery; the shock structures of Forni et al. (2017a) and Chahrour and Jurado (2022) are nested as the zero-surprise-variance case. *Economics:* applied to U.S. data over 1960–2019, roughly one third of the quarterly flow of new information in consumer-confidence signals is noise (an upper bound under persistent belief noise), yet noise shocks have only small and transitory effects on real activity, while news shocks generate persistent

expansions. The discrepancy with published estimates of a large noise share is fully accounted for by informational sufficiency: the sparse systems that generate large noise shares fail the sufficiency test on our sample, and enriching the information set collapses them.

One direction extends naturally from the framework. The gap between the finite-lead estimates and their full-history counterparts (computable in the same data by comparing $\widehat{b}_q \widehat{\beta}_h^\varepsilon$ with the spectral estimator of Chahrour and Jurado (2022)) is itself an informative diagnostic of recoverability, and could be reported routinely in applied work whenever non-invertibility is a concern.

U.S. expectation signals are substantially noisy, yet noise is not an important source of business-cycle fluctuations, and both facts are recoverable from a simple, flexible, and theoretically disciplined local-projection framework.

Appendix

A Discussion and Proofs

Attainability of the local-projection estimands: proof details

This subsection collects proof details for the attainability statements: Proposition 1, Corollary 1, and the closure case $\sigma_\nu = 0$ (statement (ii) of Theorem 1). Statement numbers below refer to the original grouped formulation of these results.

Statements (i) and (ii) are Proposition 1(i)–(ii), proved below in this appendix under the stated conditions; statement (iii) is Corollary 1(i), proved in the noise-side block of this appendix. For statement (iv): with $\sigma_\nu = 0$, $s_t = \varepsilon_t$ is white noise, orthogonal to every element of $X_{t-1}^{(p)}$, so the signal innovation is $\eta_t^s \equiv s_t - P[s_t | X_{t-1}^{(p)}] = \varepsilon_t$; the population slope of $y_{j,t+h}$ on η_t^s is $\text{Cov}(y_{j,t+h}, \varepsilon_t) / \sigma_\varepsilon^2 = \Theta_{j,2}(h) = \mathcal{O}_2(h)$. $\square$

Our discussion focuses on three crucial aspects of the identification of noisy news TFP shocks: permanent/transitory shocks, measurement errors in the signal, and measurement errors in TFP.

Permanent/Transitory Effects of Unexpected TFP Shocks We use a pedagogical and simple example to highlight the importance of our assumptions, without undermining their relevance.⁶ To illustrate this issue, consider the following representation of TFP growth:

$$\Delta z_t = u_t + a_1 u_{t-1} + \varepsilon_t + b_1 \varepsilon_{t-1},$$

where, for simplicity, we normalize the variances of u_t and ε_t to unity.

Under the random walk assumption for z_t , we impose:

$$a_1 + b_1 = 0.$$

In this example, the surprise shock u_t has a transitory effect if $a_1 = -1$, while the news shock ε_t has a permanent effect if $b_1 = 1$.

Measurement Error in the Signal Suppose the signal s_t may be subject to measurement errors or that other shocks in the economy may affect the signal:

$$\tilde{s}_t = s_t + e_t,$$

where s_t is defined as in (4) and the measurement error e_t is a white noise with variance σ_e^2 . As before, this shock is orthogonal to all other shocks for all leads and lags. Let ω denote the share of the variance of the observed signal $\tilde{s}_t$ explained by the news and noise shocks, i.e., $V(s_t)/V(\tilde{s}_t)$. We have the following property:

⁶As noted previously, and in references therein, our statistical setup for TFP is supported by actual data.

Proposition 4. For $h > q + n$, $\mathbf{nsr}_h^{\text{news}} = \alpha\sqrt{1 - \omega} \leq \alpha$.

Proof. See Appendix A. □

Measurement Error in TFP Consider the following decomposition:

$$\Delta z_t = \Delta z_t^{(1)} + \Delta z_t^{(2)} + \Delta z_t^{(3)}, \quad (\text{A.1})$$

where the components $\Delta z_t^{(1)}$ and $\Delta z_t^{(2)}$ are as defined previously (see equations 3). We add a measurement equation for the third component:

$$\Delta z_t^{(3)} = c(L)\Delta \zeta_t, \quad (\text{A.2})$$

where $c(L)$ is either a finite-order lag polynomial or square-summable, and ζ_t is a zero-mean white noise with variance σ_ζ^2 . This shock can have a persistent effect on TFP.

Initially, we consider ζ_t as a measurement error orthogonal to all other shocks for all leads and lags. This specification aligns with a classical errors-in-variables view of measured TFP, under which measurement errors, although relevant in the short run, have no long-run effect on TFP.

Note: To ensure a well-defined long-run covariance matrix, it is necessary to impose absolute summability on $c(L)$.

Under this condition, we state the following proposition:

Proof of Proposition 6

Using equations (2)–(5), we obtain the moments

$$\begin{aligned} E(s_t \Delta z_t) &= 0 & h < q \\ E(s_t \Delta z_{t+q}) &= \sigma_\varepsilon^2 b_0 & h = q \\ E(s_t \Delta z_{t+q+1}) &= \sigma_\varepsilon^2 b_1 & h = q + 1 \\ &\vdots \\ E(s_t \Delta z_{t+q+n}) &= \sigma_\varepsilon^2 b_n & h = q + n \\ E(s_t \Delta z_{t+q+n}) &= 0 & h > q + n \end{aligned}$$

After replacement, we obtain

$$\sum_{j=0}^h E(s_t \Delta z_{t+j}) = \sigma_\varepsilon^2 \sum_{j=0}^{\min(h-q, n)} b_j.$$

If $h \geq q + n$, then the sum is equal to $\sigma_\varepsilon^2 b(1)$. The long-run variance of Δz_t depends on the two components $z_t^{(1)}$ and $z_t^{(2)}$:

$$V_{LR}(\Delta z_t) = \sigma_u^2 \left(\sum_{j=0}^m a_j \right)^2 + \sigma_\varepsilon^2 \left(\sum_{j=0}^n b_j \right)^2 \equiv \sigma_u^2 a(1)^2 + \sigma_\varepsilon^2 b(1)^2$$

The variance of the signal is immediately deduced $V(s_t) = \sigma_\varepsilon^2 + \sigma_\nu^2$. Now, inserting these three moments into (B.1), we get

$$\begin{aligned} \mathbf{nsr}_h^{\text{news}} &= \frac{\sigma_\varepsilon^2 b(1)}{\sqrt{\sigma_u^2 a(1)^2 + \sigma_\varepsilon^2 b(1)^2} \sqrt{\sigma_\varepsilon^2 + \sigma_\nu^2}} \\ &= \frac{1}{\sigma_u^2 \frac{a(1)^2}{\sigma_\varepsilon^2 b(1)^2} + 1} \frac{\sigma_\varepsilon}{\sigma_s} \end{aligned}$$

If $\frac{\sigma_u^2 a(1)^2}{\sigma_\varepsilon^2 b(1)^2} = 0$, i.e. the surprise shock has non long-run effect on TFP, we obtain $\mathbf{nsr}_h^{\text{news}} = \alpha$. This completes the proof.

Proof of Proposition 4

The moments conditions $E(s_t \Delta z_t)$, the partial sum $\sum_{j=0}^h E(s_t \Delta z_{t+j})$ and the long variance of TFP $V_{LR}(\Delta z_t)$ are identical to section A. The only change is the variance of the noise, because it is subject to a measurement error. The statistics is now given by:

$$\begin{aligned} \text{nsr}_h^{\text{news}} &= \frac{\sigma_\varepsilon^2 \sum_{j=0}^h b_j}{\sqrt{\sigma_\varepsilon^2 \left(\sum_{j=0}^h b_j \right)^2} \sqrt{\sigma_\varepsilon^2 + \sigma_\nu^2 + \sigma_e^2}} \\ &= \frac{\sigma_\varepsilon \sigma_s}{\sigma_s \sigma_s} \\ &= \alpha \sqrt{1 - \omega} \end{aligned}$$

This completes the proof.

Proof of Proposition 5

Throughout, ℓ_i (for $i = 0, \dots, H$, with $\ell_i \equiv 0$ outside this range) denotes the signal loadings of (16) and $\ell(1) \equiv \sum_{i=0}^H \ell_i$ their sum; $a(L)$ and $b(L) = \sum_{k=0}^n b_k L^k$ are the TFP lag polynomials of (3)–(4).

Step 1: cross-covariances. Since u is orthogonal to (ε, ν) at all leads and lags, only the news component of Δz covaries with the signal:

$$\text{Cov}(s_t, \Delta z_{t+j}) = \text{Cov}\left(\sum_{i=0}^H \ell_i \varepsilon_{t+i}, \sum_{k=0}^n b_k \varepsilon_{t+j-q-k}\right) = \sigma_\varepsilon^2 \sum_{k=0}^n b_k \ell_{j-q-k},$$

which is non-zero only for $q \leq j \leq q + H + n$: all cross-covariances lie at non-negative leads, so the one-sided sum below captures them all. Hence, for $h \geq q + H + n$, exchanging the order of summation,

$$\sum_{j=0}^h \text{Cov}(s_t, \Delta z_{t+j}) = \sigma_\varepsilon^2 \sum_{k=0}^n b_k \sum_{i=0}^H \ell_i = \sigma_\varepsilon^2 b(1) \ell(1).$$

Step 2: long-run variances. As in the proof of Proposition 6, $V_{LR}(\Delta z_t) = \sigma_u^2 a(1)^2 + \sigma_\varepsilon^2 b(1)^2 = \sigma_\varepsilon^2 b(1)^2$ under $a(1) = 0$, and $b(1) \neq 0$ ensures this is positive. Therefore

$$\frac{\left[\sum_{j=0}^h \text{Cov}(s_t, \Delta z_{t+j}) \right]^2}{V_{LR}(\Delta z_t)} = \frac{\sigma_\varepsilon^4 b(1)^2 \ell(1)^2}{\sigma_\varepsilon^2 b(1)^2} = \sigma_\varepsilon^2 \ell(1)^2,$$

free of $b(L)$. For the signal, the autocovariances are $\text{Cov}(s_t, s_{t+k}) = \sigma_\varepsilon^2 \sum_i \ell_i \ell_{i+|k|} + \sigma_\nu^2 \mathbf{1}\{k = 0\}$, using the whiteness of ν_t and its orthogonality to ε . Summing over all $k \in \mathbb{Z}$ (the sum of all autocovariances of a finite moving average equals the squared sum of its coefficients times the innovation variance),

$$V_{LR}(s_t) = \sigma_\varepsilon^2 \ell(1)^2 + \sigma_\nu^2.$$

Step 3: assembly. Subtracting the two previous displays,

$$V_{LR}(s_t) - \frac{\left[\sum_{j=0}^h \text{Cov}(s_t, \Delta z_{t+j}) \right]^2}{V_{LR}(\Delta z_t)} = \sigma_\nu^2,$$

which is identity (19), so that, with $\text{Var}(s_t) = \sigma_\varepsilon^2 \sum_i \ell_i^2 + \sigma_\nu^2$,

$$\theta_h^{LR} = 1 - \frac{\sigma_\nu^2}{\text{Var}(s_t)} = \frac{\sigma_\varepsilon^2 \sum_i \ell_i^2}{\sigma_\varepsilon^2 \sum_i \ell_i^2 + \sigma_\nu^2} = \theta.$$

This completes the proof. Note where each assumption enters: $a(1) = 0$ and $u \perp (\varepsilon, \nu)$ only through $V_{LR}(\Delta z_t)$ and the cross-covariances; the whiteness of ν_t only through $V_{LR}(s_t)$; $h \geq q + H + n$ only to capture all non-zero cross-covariances in the one-sided sum (if an empirical signal also carries already-materialised news, use the two-sided sum $\sum_{j=-h}^h$); and no restriction is placed on $b(L)$ beyond $b(1) \neq 0$ and square-summability.

Proof of Corollary 3

Notation as in the proof of Proposition 5; write $\ell(x) = \sum_{i=0}^H \ell_i x^i$ for the loading polynomial.

Step 1: identification of the news spectrum. For $k \neq 0$, the whiteness of ν_t and its orthogonality to ε give $\text{Cov}(s_t, s_{t+k}) = \sigma_\varepsilon^2 \sum_i \ell_i \ell_{i+|k|}$, free of σ_ν^2 . The lag-zero autocovariance is $\text{Var}(s_t) = \sigma_\varepsilon^2 \sum_i \ell_i^2 + \sigma_\nu^2$, and σ_ν^2 is identified by (19) (Step 3 of the previous proof). Hence the spectral density of the news component of the signal, in the sum-of-autocovariances normalisation,

$$S_s(\omega) - \sigma_\nu^2 = \sigma_\varepsilon^2 |\ell(e^{-i\omega})|^2,$$

is identified at every frequency from observable second moments of $(s_t, \Delta z_t)$.

Step 2: Szegő–Kolmogorov. The news component $x_t = \sum_i \ell_i \varepsilon_{t+i}$ is covariance-stationary with spectral density $S_s(\omega) - \sigma_\nu^2$ in the sum-of-autocovariances normalisation. By the Szegő–Kolmogorov theorem, its one-step Wold innovation variance is

$$\Omega_x = \exp \left\{ \frac{1}{2\pi} \int_{-\pi}^{\pi} \log [S_s(\omega) - \sigma_\nu^2] d\omega \right\},$$

which is precisely Ω_s in (21). Substituting $\Omega_s = \Omega_x$ and the identified σ_ν^2 into (22) yields $\tilde{\theta}^{LR} = \Omega_x / (\Omega_x + \sigma_\nu^2) = \tilde{\theta}$, for any loadings. For the structural-loading reading, factor $\ell(x) = \ell_0 \prod_{j=1}^H (1 - x/r_j)$; by Jensen's formula $\frac{1}{2\pi} \int_{-\pi}^{\pi} \log |1 - e^{-i\omega}/r_j|^2 d\omega = 0$ whenever $|r_j| > 1$, so under minimum phase $\Omega_x = \ell_0^2 \sigma_\varepsilon^2$. This completes the proof.

Note that $b(L)$ never enters beyond its role in Proposition 5, so the result holds for any diffusion polynomial. The flow share $\tilde{\theta}$ is point-identified without any root condition, because its definition runs through the Wold innovation. Only the structural reading $\Omega_x = \ell_0^2 \sigma_\varepsilon^2$ requires minimum phase: a root inside the unit circle generates the same spectrum, so the individual loadings are set-identified, each admissible root configuration giving one value. The stock target θ is immune even to that, since $\sum_i \ell_i^2 = \frac{1}{2\pi} \int |\ell(e^{-i\omega})|^2 d\omega$ is invariant to root flips.

Proof of Proposition 1

We give the proof in three blocks corresponding to statements (i)–(iii).

Block 1: Wold-target identification (statement (i))

Apply the Frisch–Waugh–Lovell (FWL) theorem to the local projection (10). The OLS estimator of the slope on Δz_t is

$$\hat{\beta}_h^W = \frac{\sum_t \tilde{\Delta z}_t \tilde{y}_{j,t+h}}{\sum_t \tilde{\Delta z}_t^2}, \quad (\text{A.3})$$

where $\tilde{\Delta z}_t = \Delta z_t - \hat{P}[\Delta z_t | X_{t-1}^{(p)}]$ and $\tilde{y}_{j,t+h} = y_{j,t+h} - \hat{P}[y_{j,t+h} | X_{t-1}^{(p)}]$ are the OLS residuals of Δz_t and $y_{j,t+h}$ regressed on the lag block $X_{t-1}^{(p)}$.

Under **A.1**, X_t is jointly covariance-stationary with finite fourth moments and a strong-mixing condition. By the ergodic theorem and the LLN for mixing sequences, the sample-moment matrix $T^{-1} \sum_t X_{t-1}^{(p)} X_{t-1}^{(p)'}$ converges in probability to its expectation $\Sigma_{XX}^{(p)}$, and similarly for the cross-moments with Δz_t and $y_{j,t+h}$. Hence the projection coefficients are consistent and

$$\tilde{\Delta z}_t \xrightarrow{L^2} \Delta z_t - P[\Delta z_t | X_{t-1}^{(p)}] =: \tilde{\Delta z}_t^*, \quad \tilde{y}_{j,t+h} \xrightarrow{L^2} y_{j,t+h} - P[y_{j,t+h} | X_{t-1}^{(p)}].$$

The continuous-mapping theorem applied to (A.3) gives

$$\widehat{\beta}_h^W \xrightarrow{p} \frac{\text{Cov}(\tilde{\Delta z}_t^*, y_{j,t+h})}{\text{Var}(\tilde{\Delta z}_t^*)} \equiv \beta_h^W,$$

which is the population IRF of $y_{j,t+h}$ to the residual $\tilde{\Delta z}_t^*$. As $p \rightarrow \infty$, $P[\Delta z_t | X_{t-1}^{(p)}]$ converges in L^2 to the projection on the closed linear span $\overline{\text{sp}}(X_s : s \leq t-1)$, so $\tilde{\Delta z}_t^*$ converges in L^2 to the Wold innovation of Δz_t relative to that information set — the Chahrour and Jurado (2018) technology-side Chahrour and Jurado (2018) target. This establishes statement (i). $\square$

Block 2: Structural-news identification (statement (ii))

Apply FWL to the local projection (8). The OLS estimator of the slope on Δz_{t+q} equals the slope of $\tilde{y}_{j,t+h}$ on $\tilde{\Delta z}_{t+q}$, where the residuals are now obtained by partialling out the augmented control vector $W_t \equiv (\hat{u}_t, \hat{u}_{t+1}, \dots, \hat{u}_{t+q}, X_{t-1}^{(p)})'$.

Step 1: structural decomposition of Δz_{t+q} . Substituting the TFP process (2)–(4),

$$\begin{aligned} \Delta z_{t+q} &= a(L) u_{t+q} + b(L) \varepsilon_{t+q} \\ &= \underbrace{\sum_{i=0}^q a_i u_{t+q-i}}_{=: U_t^{(q)}} + \underbrace{\sum_{i>q} a_i u_{t+q-i}}_{=: U_t^{\text{past}}} + \underbrace{b_q \varepsilon_t}_{=: V_t^{(q)}} + \underbrace{\sum_{j>q} b_j \varepsilon_{t+q-j}}_{=: V_t^{\text{past}}}, \end{aligned} \quad (\text{A.4})$$

where the third equality uses Assumption **A.2** ($b_j = 0$ for $j < q$, so no future news $\varepsilon_{t+1}, \dots, \varepsilon_{t+q-1}$ contributes to Δz_{t+q}).

Step 2: partialling out W_t removes $U_t^{(q)}$, U_t^{past} , V_t^{past} asymptotically.

1. Under **A.3**, $T^{-1} \sum_t (\hat{u}_t - u_t)^2 \rightarrow_p 0$, so the linear span of $(\hat{u}_t, \dots, \hat{u}_{t+q})$ converges in L^2 to the linear span of $(u_t, \dots, u_{t+q})$. Partialling out the former asymptotically equals partialling out the latter, which removes $U_t^{(q)}$ entirely.
2. For p large, the lag block $X_{t-1}^{(p)}$ contains $\Delta z_{t-1}, \dots, \Delta z_{t-p}$, $s_{t-1}, \dots, s_{t-p}$, and the auxiliary block $w_{t-1}, \dots, w_{t-p}$. Under **A.3**, $u_{t-1}, u_{t-2}, \dots$ lie in the closed linear span of $X_{t-1}^{(p)}$ as $p \rightarrow \infty$ (the first-stage projection defining $\hat{u}$ is precisely this recovery), so partialling out $X_{t-1}^{(p)}$ removes $U_t^{\text{past}} = \sum_{i>q} a_i u_{t+q-i}$ asymptotically. For the past-news term $V_t^{\text{past}} = \sum_{j>q} b_j \varepsilon_{t+q-j}$, two cases. Under condition (a), $b_j = 0$ for $j > q$ and $V_t^{\text{past}} = 0$ identically. Under condition (b), **A.3'** gives $\varepsilon_{t-1}, \varepsilon_{t-2}, \dots \in \overline{\text{sp}}(X_s : s \leq t-1)$, so partialling out $X_{t-1}^{(p)}$ removes V_t^{past} asymptotically. Absent both conditions the removal is only partial — past news is observed through the noisy signal, a signal-extraction problem — the residual retains a past-news component, and the probability limit of the slope is contaminated; see the comments following the proposition.
3. Assumption **A.4** states that the residualised regressor converges in L^2 to $b_q \varepsilon_t$: the news content survives the partialling, and the projection below is non-degenerate since $b_q \neq 0$ (**A.2**) and $\sigma_\varepsilon^2 > 0$.

Step 3: the residualised regressor converges to $b_q \varepsilon_t$. Combining Steps 1–2,

$$\tilde{\Delta z}_{t+q} = \Delta z_{t+q} - P[\Delta z_{t+q} | W_t] \xrightarrow{L^2} b_q \varepsilon_t,$$

and the dependent-variable residual converges to

$$\tilde{y}_{j,t+h} = y_{j,t+h} - P[y_{j,t+h} | W_t] \xrightarrow{L^2} y_{j,t+h} - P[y_{j,t+h} | u_{t:t+q}, \overline{\text{sp}}(X_s)_{s \leq t-1}].$$

Under the orthogonality of $\{\varepsilon_t\}$ to $\{u_s\}$ and to $\{\varepsilon_s\}_{s < t}$, the inner product $\mathbb{E}[\tilde{y}_{j,t+h} \varepsilon_t]$ collapses to $\Theta_{j,2}(h) \sigma_\varepsilon^2$, where $\Theta_{j,2}(h)$ is the structural IRF of y_j to a unit news shock at horizon h from (6).

Step 4: probability limit of the OLS slope. By the continuous-mapping theorem,

$$\begin{aligned}\widehat{\beta}_h^\varepsilon &\xrightarrow{p} \frac{\text{Cov}(\tilde{y}_{j,t+h}, \tilde{\Delta} z_{t+q})}{\text{Var}(\tilde{\Delta} z_{t+q})} = \frac{\text{Cov}(y_{j,t+h}, b_q \varepsilon_t)}{\text{Var}(b_q \varepsilon_t)} \\ &= \frac{b_q \Theta_{j,2}(h) \sigma_\varepsilon^2}{b_q^2 \sigma_\varepsilon^2} = \frac{\Theta_{j,2}(h)}{b_q},\end{aligned}$$

which is statement (ii). The rescaling $b_q \widehat{\beta}_h^\varepsilon \rightarrow_p \Theta_{j,2}(h)$ recovers the structural IRF, with b_q identifiable separately by projecting Δz_{t+q} on $\widehat{u}_{t:t+q}$ and the lag block and reading off the residual variance. $\square$

Block 3: asymptotic normality (statement (iii))

The proof reduces to showing that, under **A.1** and the additional moment conditions cited, the centered-and-rescaled estimator $\sqrt{T}(\widehat{\beta}_h^* - \beta_h^*)$ for $* \in \{W, \varepsilon\}$ admits a Bahadur expansion of the standard local-projection form

$$\sqrt{T}(\widehat{\beta}_h^* - \beta_h^*) = \frac{1}{\sqrt{T}} \sigma_*^{-2} \sum_t \tilde{\Delta} z_t^* \tilde{e}_{j,t+h}^* + o_p(1),$$

where $\sigma_*^2 = \text{Var}(\tilde{\Delta} z_t^*)$ is the long-run variance of the residualised regressor and $\tilde{e}_{j,t+h}^*$ are the residualised LP innovations. The leading term is a scaled sample average of a strong-mixing sequence with finite fourth moments. By the central limit theorem for mixing processes (Davidson, 2002, Theorem 14.6) and the long-run-variance HAC framework of Jordà (2005); Plagborg-Møller and Wolf (2021), the leading term is asymptotically normal with HAC-consistent variance

$$\Omega_* = \sigma_*^{-4} \Lambda_*, \quad \Lambda_* = \sum_{\ell=-\infty}^{\infty} \mathbb{E}[\tilde{\Delta} z_t^* \tilde{e}_{j,t+h}^* \tilde{\Delta} z_{t-\ell}^* \tilde{e}_{j,t-\ell+h}^*].$$

Newey–West-type estimators of Λ_* with bandwidth $b_T = o(T^{1/2})$ are consistent under **A.1**; combining with the $\sqrt{T}$ -consistency of $\widehat{\sigma}_*^2$ yields a feasible HAC variance estimator. This establishes statement (iii). $\square$

Proof of the bias decomposition (Proposition 2)

The proof proceeds in four steps. Steps 1–3 derive the three leading-order bias terms in (14); Step 4 verifies the qualitative properties (i)–(iii) of Proposition 2.

Step 1: First-stage bias $\delta_h^{*,1\text{st}}$ (source B.1)

The first-stage projection of Δz_t on $X_{t-1}^{(p)}$ has population coefficient $\Pi = \Sigma_{XX}^{-1} \Sigma_{X,\Delta z}$ where $\Sigma_{XX} = \mathbb{E}[X_{t-1}^{(p)} X_{t-1}^{(p)'}]$ and $\Sigma_{X,\Delta z} = \mathbb{E}[X_{t-1}^{(p)} \Delta z_t]$. The OLS estimator $\widehat{\Pi}$ admits the standard Marriott–Pope-type expansion (Kilian, 1998, Lemma 3.1):

$$\widehat{\Pi} - \Pi = \frac{1}{T} \Sigma_{XX}^{-1} \kappa(\Pi) + O_p(T^{-3/2}), \quad (\text{A.5})$$

where $\kappa(\Pi)$ is the explicit Kilian-bias correction term for stationary AR processes. Under stationarity (**A.1**), $\kappa(\Pi)$ is bounded; for an $\text{AR}(p)$ companion matrix A , the leading element is $-(I + A + A^2 + \dots + A^{p-1}) \Sigma_{XX}^{-1} \mathbb{E}[X_{t-1}^{(p)} u_t]$ in the standard derivation.

The first-stage residual estimator is $\widehat{u}_t = \Delta z_t - \widehat{\Pi}' X_{t-1}^{(p)}$, which satisfies

$$\widehat{u}_t - u_t = (\Pi - \widehat{\Pi})' X_{t-1}^{(p)} = -\frac{1}{T} \kappa(\Pi)' \Sigma_{XX}^{-1} X_{t-1}^{(p)} + O_p(T^{-3/2}).$$

Substituting into the news LP (8), the first-stage bias propagates into $\widehat{\beta}_h^\varepsilon$ via the Frisch–Waugh–Lovell projection of $\widehat{u}_t$ on

$X_{t-1}^{(p)}$. Taking expectations and collecting $O(1/T)$ terms,

$$\delta_h^{\varepsilon, 1st} = - \sum_{k=0}^q \mathbb{E} \left[\frac{\partial \hat{\beta}_h^{\varepsilon}}{\partial \hat{u}_{t+k}} \right] \cdot \kappa(\Pi)' \Sigma_{XX}^{-1} \mathbb{E}[X_{t+k-1}^{(p)} u_{t+k}], \quad (\text{A.6})$$

where the partial-derivative weights are obtained by FWL on the augmented control matrix. For the no-controls specification (10), $\hat{u}_{t+k}$ does not enter, and (A.6) simplifies to a single term involving the projection of Δz_t on lags. In both cases, $\delta_h^{\varepsilon, 1st} = O(1/T)$ and is bounded under **A.1**.

Step 2: Direct LP bias $\delta_h^{*,LP}$ (source B.2)

For the no-controls LP (10), the OLS slope can be expanded around its probability limit using the Bahadur representation. Define the residualised quantities $\tilde{\Delta} z_t$ and $\tilde{y}_{j,t+h}$ as in Block 1 of the proof of Proposition 1. Standard finite-sample expansions of OLS for serially correlated data (Kilian, 1998; Kilian and Kim, 2011; Herbst and Johansson, 2024) give

$$T \mathbb{E} [\hat{\beta}_h^W - \beta_h^W] = -\sigma_W^{-2} \mathbb{E} [\tilde{\Delta} z_t \tilde{\Delta} z_t' \Sigma_{XX}^{-1}] \kappa(\Pi) + \tau_h^W + O(T^{-1/2}), \quad (\text{A.7})$$

where $\sigma_W^2 = \text{Var}(\tilde{\Delta} z_t)$ is the long-run variance of the Wold residual and

$$\tau_h^W = \sigma_W^{-2} \sum_{\ell=1}^{\infty} \mathbb{E} [\tilde{\Delta} z_t \tilde{y}_{j,t+h-\ell}]$$

is a horizon-dependent term reflecting the cumulative autocorrelation of the partialled-out residuals. Under **A.1** (positive autocorrelation of Δz_t), $\tau_h^W \leq 0$ and grows in magnitude with h . The combined leading-order coefficient is

$$\delta_h^{W,LP} = -(p \rho_h^W + \tau_h^W),$$

where $\rho_h^W = \mathbb{E}[\tilde{\Delta} z_t \tilde{\Delta} z_{t-1}] / \sigma_W^2$ is the first-order autocorrelation of the partialled-out regressor.

For the news LP (8), the same derivation applies with $\tilde{\Delta} z_{t+q}$ in place of $\tilde{\Delta} z_t$ and the augmented control vector. Define analogously $\sigma_\varepsilon^2, \rho_h^\varepsilon, \tau_h^\varepsilon$ and obtain

$$\delta_h^{\varepsilon,LP} = -(p \rho_h^\varepsilon + \tau_h^\varepsilon).$$

Step 3: Many-controls bias $\delta_h^{\varepsilon,Q}$ (source B.3)

The news LP (8) includes $q+1$ surprise controls beyond the lag block. By a finite-sample F -test calculation (cf. Herbst and Johansson, 2024, § 3.4), the additional bias from each linearly independent control is of order $1/T$ and proportional to the partial covariance with the dependent variable. Aggregating across the $q+1$ surprise controls,

$$\delta_h^{\varepsilon,Q} = -(q+1) \frac{\text{Cov}(y_{j,t+h}, \varepsilon_t)}{\text{Var}(\varepsilon_t)} \xi^Q(b(L), \rho), \quad (\text{A.8})$$

where the scalar $\xi^Q(b(L), \rho) > 0$ depends on the news polynomial $b(L)$ and the autocorrelation structure ρ of X_t via the partial-leverage formula

$$\xi^Q(b(L), \rho) = \frac{1}{q+1} \sum_{k=0}^q \frac{\text{Cov}(\hat{u}_{t+k}, \tilde{\Delta} z_{t+q})^2}{\text{Var}(\hat{u}_{t+k}) \text{Var}(\tilde{\Delta} z_{t+q})}.$$

For the no-controls specification (10), $q = -1$ effectively and $\delta_h^{W,Q} = 0$.

Step 4: Qualitative properties

Combining (A.6), (A.7), and (A.8) via the additive decomposition (14),

$$\delta_h^* = \delta_h^{*,1st} + \delta_h^{*,LP} + \delta_h^{*,Q}, \quad * \in \{W, \varepsilon\}.$$

Property (i): sign. Under **A.1** (positive autocorrelation of Δz_t), $\rho_h^* \geq 0$, $\tau_h^* \leq 0$, and $\xi^Q \geq 0$. The first-stage term $\delta_h^{*,1st}$ has the sign of $-\kappa(\Pi)$, which is non-positive under positive autocorrelation (Kilian, 1998, Theorem 1). Hence $\delta_h^* < 0$, i.e., the LP estimator is biased toward zero in finite samples.

Property (ii): horizon. The term $|\tau_h^*|$ is monotone in h over the range where the cumulative impulse response of $\tilde{y}$ is monotone, because each additional autocovariance lag contributes a same-sign increment under **A.1**. The other two terms ($\delta_h^{*,1st}$ and $\delta_h^{*,Q}$) are first-order constant in h to leading order. Hence $|\delta_h^*|$ is non-decreasing in h over the same range.

Property (iii): specification. The many-controls term $\delta_h^{\varepsilon,Q}$ scales with $q + 1$ in (A.8), while $\delta_h^{W,Q} = 0$. The first-stage term $\delta_h^{\varepsilon,1st}$ also accumulates over the $q + 1$ surprise controls, while $\delta_h^{W,1st}$ involves a single projection. Hence, for any q large enough relative to T , $|\delta_h^{\varepsilon}| > |\delta_h^W|$. The asymptotic bias-variance trade-off is therefore: surprise controls reduce the asymptotic bias by targeting $\Theta_{j,2}$ instead of β^W , but inflate the finite-sample bias through B.1 and B.3. $\square$

Implementation of the residual-augmented correction

The closed-form expressions (A.6), (A.7), and (A.8) are functions of population quantities $(\Pi, \Sigma_{XX}, \rho_h^*, \tau_h^*, \xi^Q)$, all of which admit consistent sample analogues. The feasible bias-corrected estimator of Proposition 3 is constructed by plugging in:

- $\widehat{\Pi}, \widehat{\Sigma}_{XX}$ from the first-stage VAR;
- $\widehat{\rho}_h^*$ from the autocorrelation of the residualised LP regressor;
- $\widehat{\tau}_h^*$ from a Newey–West estimator of the relevant cross-covariances;
- $\widehat{\xi}^Q$ from the partial-leverage formula evaluated at the sample residuals;
- $\widehat{\kappa}(\widehat{\Pi})$ from the closed-form Kilian-bias correction.

These plug-in estimates yield $\widehat{\delta}_h^*$, and the bias-corrected estimator is

$$\widehat{\beta}_h^{*,cor} = \widehat{\beta}_h^* - \frac{1}{T} \widehat{\delta}_h^*.$$

By the continuous-mapping theorem and the consistency of each plug-in, $\widehat{\delta}_h^* - \delta_h^* = o_p(1)$, so $\widehat{\beta}_h^{*,cor}$ is unbiased to order $O(1/T)$ as stated in Proposition 3(i). Statement (ii) of Proposition 3 (asymptotic distribution) follows because the correction is $O_p(T^{-1})$ and is thus dominated by the $O_p(T^{-1/2})$ sampling fluctuation. Statement (iii) of Proposition 3 (stability under weak controls) follows because each plug-in estimator depends only on the second-moment structure of X_t and does not require resampling-based extrapolation, whereas the bootstrap correction relies on the bootstrap re-estimator reproducing the original finite-sample bias — a property that fails when the bootstrap simulates from an estimated VAR with weak first-stage controls (cf. the diagnostic in Appendix D.5).

Proof of the noise-side bias decomposition (Corollary 1)

The noise-side estimators (11)–(9) admit the same three-component bias decomposition as their news-side counterparts (10)–(8), with the focus regressor s_t in place of Δz_t or Δz_{t+q} and the future TFP block $\{\Delta z_{t+0:q}\}$ in the role of either “no controls” (LP_W news) or “surprise residuals” (LP_ε news). We record the noise-side counterparts of Steps 1–3 of the news-side proof; the qualitative properties (sign, horizon dependence, specification ordering) follow from the same arguments as Step 4.

Step 1^ν: First-stage bias $\delta_h^{*,\nu,1st}$ (B.1)

For (11), no first-stage estimation is required: the TFP-lead controls $\{\Delta z_{t+0:q}\}$ are observable and do not depend on the VAR residuals. Consequently $\delta_h^{W,\nu,1st} = 0$. For (9), the surprise residual block introduces a first-stage bias structurally identical to (A.6) but evaluated at the noise-side projection coefficients:

$$\delta_h^{\varepsilon,\nu,1st} = - \sum_{k=0}^q \mathbb{E} \left[\frac{\partial \widehat{\beta}_h^{\varepsilon,\nu}}{\partial \widehat{u}_{t+k}} \right] \cdot \kappa(\Pi)' \Sigma_{XX}^{-1} \mathbb{E} \left[X_{t+k-1}^{(p)} u_{t+k} \right].$$

The qualitative properties (sign and horizon dependence) match the news-side first-stage term: $\delta_h^{\varepsilon, \nu, 1st} \leq 0$ under positive autocorrelation of Δz_t , and $|\delta_h^{\varepsilon, \nu, 1st}|$ accumulates linearly in $q + 1$ over the surprise control block.

Step 2 $^\nu$: Direct LP bias $\delta_h^{*, \nu, LP}$ (B.2)

Define the noise-side residualised focus regressor $\tilde{s}_t = s_t - P[s_t \mid \text{controls}]$, where the control set is that of the specification considered: $\{\Delta z_{t+0:q}\}$, $X_{t-1}^{(p)}$ for (11), augmented by $\hat{u}_{t:t+q}$ for (9). By Corollary 1, $\tilde{s}_t \xrightarrow{L^2} \nu_t$ for the *combined* specification under **A.1–A.4** plus condition (a) or (b); for the leads-only specification $\tilde{s}_t = \check{\varepsilon}_t + \nu_t$ with $\text{Var}(\check{\varepsilon}_t) > 0$, and the decomposition below applies to the mixture limit. Repeating the Bahadur expansion of Step 2 of the news-side proof with $\tilde{s}_t$ replacing Δz_t ,

$$\delta_h^{*, \nu, LP} = -(p \rho_h^{*, \nu} + \tau_h^{*, \nu}), \quad * \in \{W, \varepsilon\},$$

where $\rho_h^{*, \nu} = \mathbb{E}[\tilde{s}_t \tilde{s}_{t-1}] / \text{Var}(\tilde{s}_t)$ is the first-order autocorrelation of the partialled-out signal and $\tau_h^{*, \nu} = \sigma_\nu^{-2} \sum_{\ell=1}^{\infty} \mathbb{E}[\tilde{s}_t \tilde{y}_{j, t+h-\ell}]$ is the noise-side counterpart of τ_h^* . Under **A.1**, both quantities are bounded. When ν_t is i.i.d. (the canonical noise-shock assumption underlying (5)), $\rho_h^{*, \nu} = 0$ and $\delta_h^{*, \nu, LP}$ collapses to $-\tau_h^{*, \nu}$.

Step 3 $^\nu$: Many-controls bias $\delta_h^{*, \nu, Q}$ (B.3)

The TFP-lead block contributes $q + 1$ controls beyond the lag block in both (11) and (9). Aggregating the partial-leverage contribution across the lead block (the noise-side analog of Step 3),

$$\delta_h^{*, \nu, Q} = -(q + 1) \frac{\text{Cov}(y_{j, t+h}, \nu_t)}{\text{Var}(\nu_t)} \xi^{Q, \nu},$$

where the noise-side many-controls scalar is

$$\xi^{Q, \nu} = \frac{1}{q + 1} \sum_{k=0}^q \frac{\text{Cov}(\Delta z_{t+k}, \tilde{s}_t^{[lags]})^2}{\text{Var}(\Delta z_{t+k}) \text{Var}(\tilde{s}_t^{[lags]})} > 0,$$

with $\tilde{s}_t^{[lags]} = s_t - P[s_t \mid X_{t-1}^{(p)}]$ the lag-residualised signal. Equivalently, $\xi^{Q, \nu}$ is the partial- R^2 of s_t on $\{\Delta z_{t+0:q}\}$ given $X_{t-1}^{(p)}$, which is the diagnostic statistic invoked in the main text. For the combined specification (9), the surprise-residual block is *not* redundant in population — it is what removes the future-surprise content of the TFP leads and delivers consistency (Corollary 1(i)) — and it contributes up to $q + 1$ additional controls: the many-controls count in (9) is $2(q + 1)$, and the same partial- R^2 formula applies with the full control block in place of the lead block alone. The $\hat{\xi}^{Q, \nu} \approx 0.18$ reported in the main text refers to the lead block; when the correction is applied to (9), the full-block version should be computed and reported alongside.

Combined bias and residual-augmented correction

Combining Steps 1 $^\nu$ –3 $^\nu$,

$$\delta_h^{*, \nu} = \delta_h^{*, \nu, 1st} + \delta_h^{*, \nu, LP} + \delta_h^{*, \nu, Q}, \quad * \in \{W, \varepsilon\}.$$

The residual-augmented correction

$$\hat{\beta}_h^{*, \nu, cor} = \hat{\beta}_h^{*, \nu} - \frac{1}{T} \hat{\delta}_h^{*, \nu}$$

inherits the properties of Proposition 3: the corrected estimator is unbiased to order $O(T^{-1})$, its asymptotic distribution is unchanged, and the correction is stable in the presence of informationally weak controls. In our empirical sample $\hat{\xi}^{Q, \nu} \approx 0.18$ for both BUS5 and BUS12, implying a many-controls correction factor of ≈ 1.02 that leaves the noise IRFs visually unchanged. □

Proof of Theorem 1

Throughout, X_t is the observable vector (the bivariate $(\Delta z_t, s_t)'$ in statements (ii)–(iii); any finite-dimensional stationary extension in statement (i)), $\mathcal{I}_t^{(k)} = \overline{\text{sp}}(X_s : s \leq t + k)$, and the signal is the benchmark $s_t = \varepsilon_t + \nu_t$, so that $\varepsilon_t \perp \mathcal{I}_{t-1}$. Let $X_t = \sum_{i \geq 0} \Psi_i \iota_{t-i}$ be the Wold representation, with $\Psi_0 = I$, ι_t the innovation, and $F = \text{Var}(\iota_t) \succ 0$ ($\sigma_u^2, \sigma_\nu^2 > 0$ guarantee a nonsingular spectral density and hence a nondegenerate innovation). Because ε_t is orthogonal to $\mathcal{I}_{t-1}$ and (under **A.1**) lies in the closed span of the process, its projection on the observables decomposes on the innovations basis as

$$\varepsilon_t = \sum_{m \geq 0} c_m F^{-1} \iota_{t+m} + e_t^\infty, \quad c_m = \mathbb{E}[\varepsilon_t \iota_{t+m}'], \quad (\text{A.9})$$

with e_t^∞ orthogonal to the entire observable process (and $e_t^\infty = 0$ exactly when ε_t is two-sided recoverable). Structural news IRFs of the system satisfy $\text{Cov}(X_{t+h}, \varepsilon_t) = \sum_{m=0}^h \Psi_{h-m} c_m'$ for $h \geq 0$.

Step 1: reduction to the innovations basis. Every specification in the class projects $y_{j,t+h}$ on a focus regressor residualised against controls, all measurable with respect to $\mathcal{I}_t^{(k)}$, with the lag block among the controls. As $p \rightarrow \infty$ the lag block spans $\mathcal{I}_{t-1}$, and the orthogonal decomposition $\mathcal{I}_t^{(k)} = \mathcal{I}_{t-1} \oplus \overline{\text{sp}}(\iota_t, \dots, \iota_{t+k})$ implies that the residualised focus regressor is an element

$$r_t = \sum_{m=0}^k w_m' \iota_{t+m}, \quad w_m \in \mathbb{R}^{\dim X},$$

and by Frisch–Waugh–Lovell the specification's slope at horizon h for variable j is $\text{Cov}(y_{j,t+h}, r_t) / \text{Var}(r_t)$.

Step 2: the triangular characterisation. Attainment requires, for some $\gamma \neq 0$ and all $h \geq 0$ (writing the system rows jointly),

$$\sum_{m=0}^{\min(h,k)} \Psi_{h-m} F w_m = \gamma \sum_{m=0}^h \Psi_{h-m} c_m', \quad (\text{A.10})$$

where the left side is $\text{Cov}(X_{t+h}, r_t)$ and the right side is $\gamma \text{Cov}(X_{t+h}, \varepsilon_t)$ (a harmless renormalisation of γ by $\text{Var}(r_t)$). Solve (A.10) forward. At $h = 0$: $F w_0 = \gamma c_0'$, so $w_0 = \gamma F^{-1} c_0'$. Assuming $w_i = \gamma F^{-1} c_i'$ for $i < h \leq k$, the terms $m < h$ cancel on both sides and $F w_h = \gamma c_h'$. Hence $w_m = \gamma F^{-1} c_m'$ for all $m \leq k$: the attaining regressor is proportional to the finite-lead smoother $P[\varepsilon_t | \mathcal{I}_t^{(k)}] = \sum_{m \leq k} c_m F^{-1} \iota_{t+m}$, and is unique up to scale. For $h > k$ the left side has no terms beyond $m = k$, and cancelling the common terms leaves $0 = \gamma F F^{-1} c_h'$ at $h = k + 1$, then recursively $c_h = 0$ for every $h > k$. Conversely, if $c_m = 0$ for all $m > k$, the smoother weights satisfy (A.10) at every horizon. This proves statement (i), and identifies the failure when $k < k^*$: the smoother specification matches $\Theta_{j,2}(h)$ for $h \leq k$ and produces spurious responses $-\gamma \sum_{k < m \leq h} \Psi_{h-m} c_m'$ at $h > k$ — the truncated tail.

Step 3: the lead support under (a), (b), and $\sigma_\nu = 0$. Under **A.3'**, $\varepsilon_t \in \overline{\text{sp}}(X_s : s \leq t)$, and since $\iota_{t+m} \perp \mathcal{I}_{t+m-1} \supseteq \mathcal{I}_t$ for $m \geq 1$, $c_m = 0$ for all $m \geq 1$; $c_0 \neq 0$ because $\varepsilon_t \perp \mathcal{I}_{t-1}$ places ε_t in $\overline{\text{sp}}(\iota_t)$ nondegenerately. Hence $k^* = 0$. The case $\sigma_\nu = 0$ is the special case $\varepsilon_t = s_t - P[s_t | \mathcal{I}_{t-1}] \in \overline{\text{sp}}(\iota_t)$. Under the single-horizon loading (a) in the bivariate system, ε_t enters the observables only through s_t (coefficient 1) and Δz_{t+q} (coefficient b_q). For $m \notin \{0, q\}$, both X_{t+m} and its best predictor given $\mathcal{I}_{t+m-1}$ involve ε_t only through predictors of *other* news shocks, each of which is estimated from its own signal and its own TFP arrival alone; with one-shot news no observation mixes two news shocks, estimation errors are uncorrelated across shocks, and $c_m = 0$. At $m = q$,

$$c_q^{(\Delta z)} = \mathbb{E}[\varepsilon_t (b_q \varepsilon_t + u_{t+q} - b_q P[\varepsilon_t | \mathcal{I}_{t+q-1}])] = b_q \text{Var}(\varepsilon_t - P[\varepsilon_t | \mathcal{I}_{t+q-1}]) > 0,$$

since the only information about ε_t in $\mathcal{I}_{t+q-1}$ is the noisy signal s_t , leaving the signal-extraction error $\sigma_\varepsilon^2 \sigma_\nu^2 / (\sigma_\varepsilon^2 + \sigma_\nu^2) > 0$. Hence $k^* = q$, and the smoothing error remains bounded away from zero at every window (the path is never recovered), which proves statement (ii).

Step 4: generic infinite support under a diffusion tail. Write the observables in state-space form with state α_t containing the active news shocks (and, for a rational diffusion polynomial, its autoregressive state), transition T , loading Z , and steady-state one-step prediction covariance P solving the Riccati equation; let K be the steady-state gain and $L = T - KZ$ the filter transition. A standard recursion gives

$$c_m = e_1' P (L')^m Z', \quad m \geq 0,$$

where e_1 selects ε_t in the state. The generating function $c(\zeta) = \sum_m c_m \zeta^m = e_1' P(I - \zeta L')^{-1} Z'$ is rational; the sequence $\{c_m\}$ terminates at a finite k^* if and only if the minimal realisation of $c(\zeta)$ is nilpotent, i.e. every non-nilpotent eigenmode of L' is either unreachable from Pe_1 or unobservable through Z' — a finite collection of polynomial restrictions on the model parameters $(b_q, \dots, \sigma_u, \sigma_\varepsilon, \sigma_\nu)$, since P is real-analytic in them. With one-shot news the restrictions hold identically (the learning dynamics are finite-memory: Step 3). With a diffusion tail they fail at the benchmark calibration of Remark 1, where $\|c_m\|$ decays geometrically without vanishing — each TFP observation $\Delta z_\tau = u_\tau + \sum_j b_j \varepsilon_{\tau-j}$ mixes adjacent news shocks, the filter's errors are correlated across shocks, and the non-nilpotent learning mode loads on ε_t . Because the restrictions are analytic and violated at one parameter point, the set where they hold is closed with empty interior, proving statement (iii). The geometric decay rate of the spurious tail is the spectral radius of the relevant block of L . $\square$

Remark (population verification). The characterisation was verified in population on the benchmark calibrations: with one-shot news the computed c_m vanish exactly outside $\{0, q\}$ and the window- q smoother specification reproduces the structural news IRFs to machine precision at every horizon; with the diffusion tail $(1, 0.6, 0.3)$ the computed c_m are nonzero at every lead with geometric decay, the window- q specification distorts the diffusion coefficients by -32% and -35% , and the window- $(q+2)$ specification is exact on the diffusion support while generating the spurious tail responses reported in Remark 1.

Proof of Corollary 2. Let g be any square-integrable function of $(X_t, \dots, X_{t+k})$ and decompose $g = g_{\text{lin}} + \delta$, where g_{lin} is the L^2 -projection of g onto $\mathcal{I}_t^{(k)}$ and $\mathbb{E}[\delta z] = 0$ for every $z \in \mathcal{I}_t^{(k)}$. Any system variable decomposes as $y_{j,t+h} = P[y_{j,t+h} | \mathcal{I}_t^{(k)}] + e_{j,h}$, where under joint Gaussianity the projection error $e_{j,h}$ is *independent* of $(X_t, \dots, X_{t+k})$, hence of g . Therefore $\text{Cov}(y_{j,t+h}, g) = \text{Cov}(P[y_{j,t+h} | \mathcal{I}_t^{(k)}], g) = \text{Cov}(P[y_{j,t+h} | \mathcal{I}_t^{(k)}], g_{\text{lin}}) = \text{Cov}(y_{j,t+h}, g_{\text{lin}})$, the middle equality because $P[y_{j,t+h} | \mathcal{I}_t^{(k)}]$ is linear and δ is orthogonal to all linear functionals of the window. The slope conditions for attainment with regressor g thus coincide, up to the scale factor $\text{Var}(g) \geq \text{Var}(g_{\text{lin}})$, with those for its linear part g_{lin} , after partialling the lag block an element of $\overline{\text{sp}}(\iota_t, \dots, \iota_{t+k})$; Theorem 1 then applies verbatim. $\square$

B Benchmark statistics, examples, and implementation details

This appendix collects the conventional news-to-signal statistics used as benchmarks in the main text, the comparison across all seven statistics, two illustrative examples, and implementation details for the choice of the lead window.

B.1 The stock statistic: exactness

The long-run stock statistic θ_h^{LR} of (20) is exact under general diffusion and loadings:

Proposition 5 (Exactness of the stock statistic under TFP diffusion). *Assume the TFP process (2)–(4) with $a(1) = 0$ and $b(1) \neq 0$, the general signal structure (16) with ν_t white noise, and mutual orthogonality of u , ε , and ν at all leads and lags. Then, for $h \geq q + H + n$,*

$$\theta_h^{LR} = \theta$$

exactly, for any diffusion polynomial $b(L)$ and any signal loadings $\{\ell_i\}_{i=0}^H$.

Proof. See Appendix A. $\square$

Two equivalent readings are derived in Appendix B: θ_h^{LR} equals the squared structural NSR of (B.1) plus a pure signal-side correction that vanishes in the simple-news case — so that Proposition 6 is the special case $H = 0$ — and, equivalently, one minus the share of the signal's long-run variance that is incoherent with TFP growth at frequency zero.

B.2 The simple news-to-signal ratio

The ratio of interest (see Forni et al. (2017a)) is defined as:

$$\alpha = \frac{\sigma_\varepsilon}{\sigma_s},$$

which measures the contribution of noise. A value of α close to zero implies that expected changes in TFP are almost entirely driven by noise. Conversely, a value close to one suggests that future technology changes (up to some lags) are nearly perfectly predictable. Furthermore, the relationship between the noise and news shock ratios is given by:

$$\frac{\sigma_\varepsilon}{\sigma_\nu} = \frac{r}{\sqrt{1-r^2}}, \quad \text{where } r = \frac{\sigma_\varepsilon}{\sigma_s}.$$

We propose a simple statistic to estimate this ratio by leveraging moment conditions on TFP and the signal. This statistic, referred to as the *news-to-signal ratio* for expected TFP at horizon h , is defined as:

$$\mathbf{nsr}_h^{\text{news}} = \sum_{j=0}^h \frac{E(s_t \Delta z_{t+j})}{\sqrt{V_{LR}(\Delta z_t)} \sqrt{V(s_t)}}, \quad (\text{B.1})$$

where $V_{LR}(\Delta z_t)$ is the long-run variance of the growth rate of TFP (defined as the sum of all autocovariances, or equivalently, the spectral density at frequency zero), and $V(s_t)$ is the variance of the signal.

This statistic requires estimating moments between the current signal s_t and realized TFP variations, both today and in future periods. These moment conditions exploit the fact that TFP includes an expected component that immediately affects the economy, as agents anticipate future improvements in technology. The partial sum $\sum_{j=0}^h E(s_t \Delta z_{t+j})$ measures the cumulative effect of news shocks on TFP (see the discussion below).

The following property holds:

Proposition 6. *If $h > q + n$ and $\frac{a(1)^2 \sigma_\eta^2}{b(1)^2 \sigma_\varepsilon^2} = 0$, then $\mathbf{nsr}_h^{\text{news}} = \alpha$.*

Proof. See Appendix A. □

Proposition 6 demonstrates that our statistic, $\mathbf{nsr}_h^{\text{news}}$, accurately uncovers the ratio of interest α , thereby providing a direct measure of the size of noise in aggregate TFP. The statistic leverages properties of various moments. Specifically, the partial sum $\sum_{j=0}^h E(s_t \Delta z_{t+j})$ equals $\sigma_\varepsilon^2 b_h(1) \equiv \sigma_\varepsilon^2 \sum_{j=0}^{\min(h-q, n)} b_j$, while the long-run variance $V_{LR}(\Delta z_t)$ is given by $b(1)^2 \sigma_\varepsilon^2$, and the variance of the signal $V(s_t)$ is $\sigma_\varepsilon^2 + \sigma_\nu^2$.

For $h \geq q + n$, we obtain $\sum_{j=0}^h E(s_t \Delta z_{t+j}) = b(1) \sigma_\varepsilon^2$, leading directly to $\mathbf{nsr}_h^{\text{news}} = \alpha$. Notably, this statistic relies solely on moments of the signal and TFP, without imposing parametric assumptions on the exact processes governing TFP and noise. This approach is expected to yield a more robust identification of noise and news in aggregate TFP.

B.3 The conditional R^2 and alternative formulations

The conditional R^2 and the role of system invertibility. A natural regression-based estimator of the flow target is the LR-corrected conditional R^2 ,

$$\tilde{R}_h^{2,LR} \equiv \frac{[\text{Cov}(\eta_t^s, z_{t+h} - z_{t-1})]^2}{\text{Var}(\eta_t^s) \cdot V_{LR}(\Delta z_t)}, \quad (\text{B.2})$$

where η_t^s is the signal innovation with respect to the lag block $X_{t-1}^{(p)}$. In the simple-news case $H = 0$, the signal is white and unpredictable, $\eta_t^s = s_t = \ell_0 \varepsilon_t + \nu_t$, and

$$\tilde{R}_h^{2,LR} \xrightarrow{h \geq q+n} \frac{\ell_0^2 \sigma_\varepsilon^2}{\ell_0^2 \sigma_\varepsilon^2 + \sigma_\nu^2} = \tilde{\theta}, \quad (\text{B.3})$$

for any diffusion polynomial: the long-difference covariance is a square-of-sum object, which factorises at frequency zero exactly as in (19). Under multi-horizon aggregation ($H \geq 1$), however, the signal carries anticipated shocks that past observables reveal only mixed with past noise; the projection on the lag block is a partial signal extraction, η_t^s retains anticipated content beyond the news-component innovation, and the square-of-sum overcounts it: in a population calculation with the loadings of our Monte Carlo designs, $\tilde{R}_h^{2,LR}$ converges to 0.87 against a target $\tilde{\theta} = 0.61$, irrespective of the diffusion polynomial. In short, $\tilde{R}_h^{2,LR}$ is exact only in the simple-news case, while the Kolmogorov statistic (22) is exact for the flow target under general aggregation; reporting both is a costless robustness check, their gap measuring the multi-horizon content of the signal.

Comparison with alternative R^2 formulations. Three earlier R^2 -style statistics also estimate θ , but each has a known bias. The *squared structural NSR*,

$$\mathbf{nsr}_h^{LR} \equiv \frac{[\text{Cov}(s_t, z_{t+h} - z_{t-1})]^2}{\text{Var}(s_t) \cdot V_{LR}(\Delta z_t)}, \quad (\text{B.4})$$

uses the square-of-sum form in the numerator: for same-signed loadings $(\sum_i \ell_i)^2 \geq \sum_i \ell_i^2$ (the pairwise cross-products are non-negative), so it is inflated whenever the signal aggregates news across several arrival dates — with uniform loadings the inflation factor is $H + 1$, and empirical values of $\mathbf{nsr}_h^{LR}$ can exceed one. The *general auxiliary R^2* from the OLS regression of s_t on $\{\Delta z_{t+0:h}\}$ and the lag block,

$$s_t = \sum_{k=0}^h \delta_k \Delta z_{t+k} + \gamma' X_{t-1}^{(p)} + e_t, \quad \mathbf{nsr}_h^{\text{adj}} \equiv 1 - \frac{T-1}{T-K_h-1} (1 - R^2), \quad (\text{B.5})$$

is contaminated by the transitory surprise through the OLS slope shrinkage. Augmenting (B.5) with the surprise-residual block restores invariance,

$$s_t = \sum_{k=0}^h \delta_k \Delta z_{t+k} + \sum_{k=0}^h \eta_k \hat{u}_{t+k} + \gamma' X_{t-1}^{(p)} + e_t, \quad \mathbf{nsr}_h^\varepsilon \equiv R^2, \quad (\text{B.6})$$

and gives an exact estimator of θ in population — but requires the explicit surprise-residual block as an additional control.

Remark 3 (Comparison of R^2 -style statistics). Under the general signal structure (16), the seven statistics have the following population limits as estimators of θ or $\tilde{\theta}$ at $h \geq q + H + n$ (limits stated under the single-horizon loading $b(L) = b_q$ except where indicated):

Statistic	Target	Population limit	Invariant to σ_u^2	No partialling
θ_h^{LR} (20)	θ	θ (any $b(L)$)	✓	✓
$\tilde{\theta}^{LR}$ (22)	$\tilde{\theta}$	$\tilde{\theta}$ (any $b(L)$; min-phase $\ell(L)$)	✓	✓
θ_h^{stat} (23)	θ	θ ; $\lambda \theta$ under diffusion (25)	✓	✓
$\tilde{R}_h^{2,LR}$ (B.2)	$\tilde{\theta}$	$\tilde{\theta}$ ($H = 0$); overstates $\tilde{\theta}$ if $H \geq 1$	✓	✓
$\mathbf{nsr}_h^\varepsilon$ (B.6)	θ	θ	✓	×
$\mathbf{nsr}_h^{LR}$ (B.4)	θ	$\frac{(\sum_i \ell_i)^2}{\sum_i \ell_i^2} \theta$ (any $b(L)$)	(✓)	✓
$\mathbf{nsr}_h^{\text{adj}}$ (B.5)	θ	$\kappa(\sigma_u^2) \theta \leq \theta$	×	✓

The top two rows are the recommended primary statistics, one per structural target: exact under TFP diffusion and multi-horizon aggregation, invariant to σ_u^2 , free of partialling, and requiring no knowledge of $b(L)$ (the flow statistic requiring minimum phase only for the structural-loading reading of its target). θ_h^{stat} is exact under the single-horizon loading and doubles as the plateau-based lead-window selector; for same-signed loadings it bounds θ from below while $\mathbf{nsr}_h^{LR}$ bounds it from above, so the pair brackets the stock target and their gap is a free diagnostic for the joint importance of diffusion and aggregation. $\tilde{R}_h^{2,LR}$ is exact in the simple-news case and overstates the flow share under multi-horizon aggregation; its gap with $\tilde{\theta}^{LR}$ measures the multi-horizon content of the signal. $\mathbf{nsr}_h^\varepsilon$ is exact but needs the surprise-residual partialling. The last two rows are dominated: $\mathbf{nsr}_h^{LR}$ overstates the stock under multi-horizon aggregation, $\mathbf{nsr}_h^{\text{adj}}$ understates it through the transitory-shock contamination of OLS.

In the empirical exercise of Section 6 we report θ_h^{stat} (with its bias-corrected variant and the selector h^*) and $\tilde{R}_h^{2,LR}$, together with $\mathbf{nsr}_h^{\varepsilon, \text{adj}}$ and $\mathbf{nsr}_h^{LR}$ for comparison. The two long-run statistics require only one additional HAC quantity, $\hat{V}_{LR}(s_t)$, plus the signal autocovariances; we recommend reporting θ_h^{LR} alongside θ_h^{stat} — their gap estimating the diffusion discount $1 - \lambda$ — and $\tilde{\theta}^{LR}$ alongside $\tilde{R}_h^{2,LR}$ — their gap measuring the news content that the lag-block projection fails to remove.

The empirical implementation of these statistics is detailed in Section 6.1 (Table 2).

B.4 Two Relevant Examples

Here, we consider two examples commonly used in the literature on news shocks. Both examples are consistent with the observed random walk behavior of TFP.

Example 1: A Singular News Shock

We assume that TFP is solely driven by news shocks that materialize q periods ahead:

$$\Delta z_t = \varepsilon_{t-q}. \quad (\text{B.7})$$

A similar specification has been employed in DSGE modeling (see Schmitt-Grohé and Uribe (2012), Fujiwara et al. (2011), and Khan and Tsoukalas (2012)). The signal is defined as before. The moment conditions between s_t and Δz_t are given by:

$$\begin{aligned} E(s_t \Delta z_t) &= 0 & \text{for } h < q, \\ &\vdots \\ E(s_t \Delta z_{t+q}) &= \sigma_\varepsilon^2 & \text{for } h = q, \\ E(s_t \Delta z_{t+q+n}) &= 0 & \text{for } h > q. \end{aligned}$$

Replacing these moment conditions into the partial sum, we obtain for $h \geq q$:

$$\sum_{j=0}^h E(s_t \Delta z_{t+j}) = \sigma_\varepsilon^2.$$

The long-run variance of Δz_t is given by $V_{LR}(\Delta z_t) = \sigma_\varepsilon^2$, and the variance of the signal is $V(s_t) = \sigma_\varepsilon^2 + \sigma_\nu^2$. Plugging these three statistics into (B.1), we obtain:

$$\mathbf{nsr}_h^{\text{news}} = \alpha.$$

Example 2: Persistent Transitory Unexpected and News Shocks in TFP

We adopt the specification proposed by Blanchard et al. (2013). Using our decomposition (2), the two components $z_t^{(1)}$ and $z_t^{(2)}$ of TFP changes are given by:

$$z_t^{(1)} = \rho_1 z_{t-1}^{(1)} + u_t,$$

$$\Delta z_t^{(2)} = \rho_1 \Delta z_{t-1}^{(2)} + \varepsilon_{t-q}.$$

As in BLL, we impose $\rho = \rho_1 = \rho_2$ and the restriction $\sigma_u^2 = \rho \sigma_\varepsilon^2 / (1 - \rho)^2$ to ensure that TFP follows a random walk. Under this assumption, the lag polynomials in (2) are given by $a(L) = b(L) = \sum_{i=0}^{\infty} \rho^i$.

We can deduce the moment conditions between s_t and Δz_t :

$$E(s_t \Delta z_{t+h}) = \sigma_\varepsilon^2 \rho^h,$$

and the partial sum:

$$\sum_{j=0}^h E(s_t \Delta z_{t+j}) = \sigma_\varepsilon^2 \sum_{j=0}^h \rho^j.$$

As $h \rightarrow \infty$, we obtain:

$$\sum_{j=0}^h E(s_t \Delta z_{t+j}) = \frac{\sigma_\varepsilon^2}{1 - \rho}.$$

The long-run variance of Δz_t is given by $V_{LR}(\Delta z_t) = \frac{\sigma_\varepsilon^2}{(1 - \rho)^2}$, and the variance of the signal is $V(s_t) = \sigma_\varepsilon^2 + \sigma_\nu^2$. Substituting these three statistics into (B.1), we obtain:

$$\mathbf{nsr}_h^{\text{news}} = \alpha.$$

B.5 Implementation

- The long-run variance can be estimated either parametrically, by fitting an $\text{AR}(p)$ process, or non-parametrically, using the Newey-West approach. Alternatively, one can use the long-run variance implied by the VAR representation itself: letting $\hat{A}_1, \dots, \hat{A}_p$ denote the estimated VAR coefficient matrices and $\hat{\Sigma}$ the residual covariance matrix, the VAR-based long-run variance is

$$\hat{V}_{LR}^{\text{VAR}} = \hat{C}(1) \hat{\Sigma} \hat{C}(1)', \quad \hat{C}(1) = \left(I - \sum_{i=1}^p \hat{A}_i \right)^{-1}.$$

The $(1, 1)$ element of $\hat{V}_{LR}^{\text{VAR}}$ provides the long-run variance of TFP growth. This estimator has the advantage of being internally consistent with the VAR specification used to compute the statistic, though it may be more sensitive to the choice of lag length and the number of variables included in the VAR.

- The signal can be obtained from confidence indexes, the Survey of Professional Forecasters, or the conditional expectation of TFP, as in Chahrour and Jurado (2022).
- The lag length h : The selection of h in the statistic can be based on an information criterion. The proposed empirical statistic is defined as:

$$\mathbf{nsr}_{h,T}^{\text{news}} = \sum_{j=0}^h \left(\frac{1}{T-j} \sum_{t=1}^{T-j} \frac{s_t \Delta z_{t+j}}{\sqrt{V_{LR}(\Delta z_t)} \sqrt{V(s_t)}} \right),$$

Under the null hypothesis of no news content ($\alpha = 0$), each normalized sample cross-covariance $\hat{\gamma}_{\text{norm}}(j)$ is $O_p(1/\sqrt{T})$ with variance approximately $1/T$. Since the terms are approximately uncorrelated across j under the null, the variance of the cumulated statistic is

$$\text{Var}(\mathbf{nsr}_{h,T}^{\text{news}}) \approx \frac{h+1}{T},$$

so that $\mathbf{nsr}_{h,T}^{\text{news}} = O_p(\sqrt{h}/\sqrt{T})$ in probability.

We consider several strategies for selecting the truncation horizon h^* .

(i) Penalized criterion. A direct approach selects h^* as the maximizer of a penalized criterion. Given the stochastic order derived above, the appropriate penalty rate is $\sqrt{h}/\sqrt{T}$:

$$SC_{\text{pen}}(h) = \mathbf{nsr}_{h,T}^{\text{news}} - c \times \frac{\sqrt{h+1}}{\sqrt{T}},$$

where $c > 0$ is a tuning constant.

(ii) Information criteria on the squared statistic. A more principled approach exploits the bias-variance decomposition of $[\mathbf{nsr}_{h,T}^{\text{news}}]^2$. Under the null, $E[\mathbf{nsr}_{h,T}^2] = \alpha_h^2 + (h+1)/T$, where $\alpha_h = E[\mathbf{nsr}_{h,T}]$. The squared statistic is therefore biased upward by $(h+1)/T$, and information criteria correct for this bias with different penalty constants. The general form is

$$SC_{\text{IC}}(h) = [\mathbf{nsr}_{h,T}^{\text{news}}]^2 - c_{\text{IC}} \times \frac{h+1}{T},$$

with $h^* = \arg \max_{0 \leq h \leq H} SC_{IC}(h)$, and the penalty constant given by:

- Akaike (AIC): $c_{AIC} = 2$;
- Bayesian (BIC): $c_{BIC} = \ln(T)$;
- Hannan–Quinn (HQ): $c_{HQ} = 2 \ln(\ln(T))$.

For a typical macroeconomic sample of $T = 200$ quarters, the penalty constants are $c_{AIC} = 2$, $c_{HQ} \approx 3.35$, and $c_{BIC} \approx 5.30$. The AIC is the most liberal (favoring larger h), the BIC the most conservative (favoring smaller h), and the HQ lies in between. Both BIC and HQ are consistent: they select the true anticipation horizon as $T \rightarrow \infty$.

(iii) Sequential marginal significance test. Under the news model, the marginal cross-covariance

$$\gamma(j) = \text{Cov}(\varepsilon_{s,t}, \Delta z_{t+j})$$

is strictly positive for j up to the true anticipation horizon and zero beyond. This motivates a sequential testing procedure: for each horizon h , compute the t -statistic

$$t_h = \frac{|\hat{\gamma}(h)|}{\widehat{se}(\hat{\gamma}(h))},$$

where $\widehat{se}(\hat{\gamma}(h))$ is a HAC standard error based on the Bartlett kernel with bandwidth $\lfloor h/2 \rfloor + 1$. The selected horizon is

$$h^* = \max\{h : t_h \geq c_\alpha \text{ and fewer than } K \text{ consecutive } t_j < c_\alpha \text{ for } j > h\},$$

with default values $c_\alpha = 1.645$ (10% level) and $K = 3$. This criterion directly tests when the signal ceases to predict future TFP growth, without imposing a parametric penalty structure.

(iv) Plateau detection. From Proposition 6, $\mathbf{nsr}_h^{\text{news}}$ converges monotonically to the true α as $h \rightarrow \infty$. Analogous to truncation lag selection in spectral density estimation, the plateau criterion selects the smallest horizon at which the statistic has reached a fraction τ of its peak value:

$$h^* = \min\left\{h : |\mathbf{nsr}_{h,T}^{\text{news}}| \geq \tau \times \max_{0 \leq k \leq H} |\mathbf{nsr}_{k,T}^{\text{news}}|\right\},$$

with a default threshold $\tau = 0.90$. This approach identifies the earliest horizon at which the cumulated covariance has effectively converged, avoiding the noise amplification that occurs at large h .

Online Appendix

(Not for print publication)

C Simulations with a Canonical RBC Model

The model used for simulation purposes is the *canonical RBC model*. The *representative agent* maximizes the following intertemporal utility function:

$$U_0 = E_0 \sum_{t=0}^{\infty} \beta^t [\ln C_t + \psi_t \ln(1 - N_t)]$$

where β is the discount factor, ψ_t is an exogenous process measuring the agent's relative preference for leisure over consumption, C_t is aggregate consumption, and N_t is normalized hours worked, with total available time set to 1, so that leisure equals $1 - N_t$. The model is specified at a *quarterly frequency*.

The *resource constraint* is given by:

$$C_t + I_t \leq Y_t$$

where I_t is aggregate investment and Y_t is total output. Output is generated using a *Cobb-Douglas production function*:

$$Y_t = K_t^{\alpha_K} (A_t N_t)^{1-\alpha_K}$$

with K_t as the capital stock and A_t representing total factor productivity (TFP). Capital accumulation follows the process:

$$K_{t+1} = (1 - \delta)K_t + I_t$$

where δ is the depreciation rate.

TFP evolves according to the process:

$$\ln A_t = \mu + \ln A_{t-1} + \varepsilon_t^{\text{surprise}} + \varepsilon_{t-q}^{\text{news}}$$

This specification includes two types of shocks: a *surprise shock* $\varepsilon_t^{\text{surprise}}$, which is unanticipated and has an immediate effect, and a *news shock* $\varepsilon_{t-q}^{\text{news}}$, which is anticipated and affects productivity q periods later. Thus, in the absence of informational frictions, the agent observes the news shock $\varepsilon_{t-q}^{\text{news}}$ at time $t - q$, and its effect materializes in period t .

The shocks $\varepsilon_t^{\text{surprise}}$ and $\varepsilon_t^{\text{news}}$ are distributed as $\mathcal{N}(0, \sigma_{\text{surprise}}^2)$ and $\mathcal{N}(0, \sigma_{\text{news}}^2)$, respectively.

The *preference parameter* ψ_t follows an AR(1) process:

$$\psi_t = (1 - \rho_{\psi})\bar{\psi} + \rho_{\psi}\psi_{t-1} + \varepsilon_t^{\psi}$$

with $\varepsilon_t^{\psi} \sim \mathcal{N}(0, \sigma_{\psi}^2)$.

A *consumer confidence series* is also added, influenced by the model's shocks:

$$\text{Conf}_t = \rho_{\text{conf}}\text{Conf}_{t-1} + \varepsilon_t^{\text{surprise}} + \varepsilon_t^{\text{news}} + \varepsilon_t^{\text{conf}}$$

where $\varepsilon_t^{\text{conf}} \sim \mathcal{N}(0, \sigma_{\text{conf}}^2)$ is a *sentiment shock* specific to confidence. This series is *idiosyncratic*, meaning it has no impact on other variables in the model. It is included to allow for VAR representations that incorporate such a variable.

A *level VAR* is estimated on the simulated series $X_t = (a_t, c_t, n_t, \text{conf}_t)'$, where lowercase letters denote the natural logarithm of the corresponding uppercase variables.

The objective is to *identify the TFP shocks*, $\varepsilon_t^{\text{surprise}}$ and $\varepsilon_t^{\text{news}}$, using the *Barsky and Sims (2011)* identification strategy based on two assumptions:

1. The *surprise shock* is the only one to contemporaneously affect TFP;
2. The *news shock* is the one that explains the largest share of the variance of TFP over a ten-year horizon, aside from the surprise shock.

The structural parameter values used are presented in *Table 1*.

Monte Carlo Results

We summarize the Monte Carlo evidence from simulations of the RBC model introduced above. For each scenario, we simulate N samples of length T , estimate a VAR on $X_t = (\text{TFP}_t, \text{conf}_t, c_t, n_t)'$ (possibly augmented with npc principal components), and apply the FGLS and BFG identification strategies to recover impulse responses to news and noise shocks. We stress-test the estimators along five dimensions: (i) the anticipation horizon $q \in \{1, 4\}$, (ii) the sample length $T \in \{250, 10,000\}$, (iii) the presence of a surprise TFP shock, (iv) the number of principal components $\text{npc} \in \{0, 2\}$ included as additional controls, and (v) the persistence of the surprise shock ρ_v .

Baseline: no surprise shock, $q = 1$. Figures 5–8 report the baseline experiment in which only the news shock drives TFP ($q = 1$). In the large-sample environment ($T = 10,000$), both FGLS and BFG track the true news- and noise-shock IRFs almost perfectly: the Monte Carlo means lie on top of the true IRFs, and the ± 1 -std. dev. bands are tight. In the short-sample case ($T = 250$), the point estimates remain essentially unbiased, but the dispersion widens noticeably, confirming that the identification is valid asymptotically and that most of the finite-sample deterioration is variance-driven rather than bias-driven.

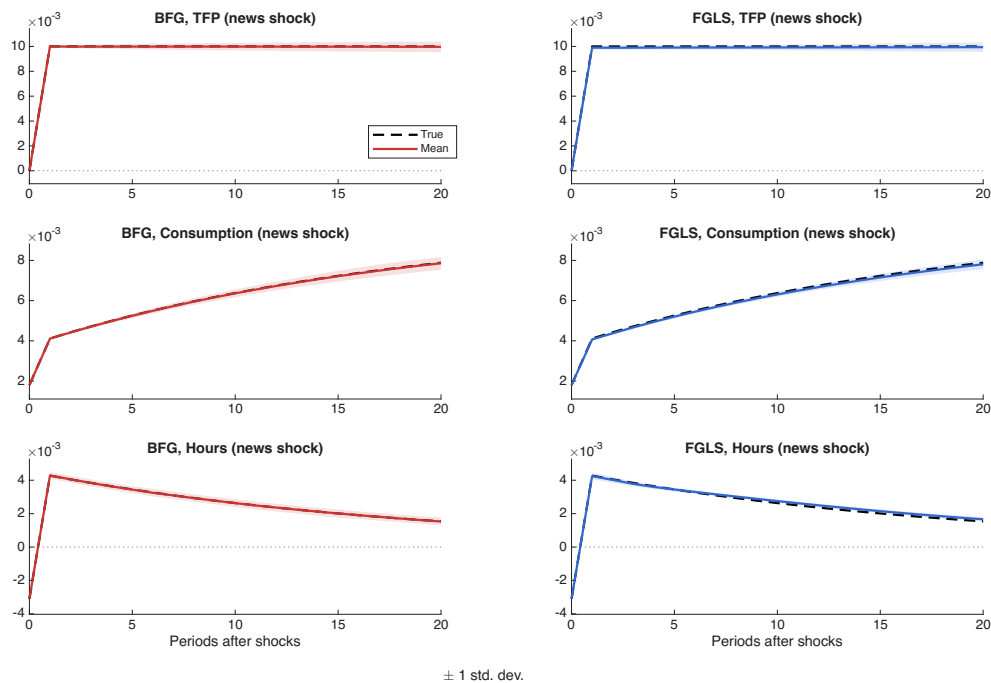


Figure 5: Estimated impulse responses to a unit news shock in the canonical RBC simulation, with the surprise channel inactive ($\sigma_u = 0$). Sample length $T = 10,000$; news horizon $q = 1$. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

Longer anticipation horizon, $q = 4$. Figures 9–12 increase the anticipation horizon to $q = 4$ quarters. With $T = 10,000$ both methods still recover the true dynamic responses, although the variance of the BFG local-projection estimator is somewhat larger at long horizons. For $T = 250$ the anticipation horizon is long relative to the effective sample size, and the confidence bands for both news and noise widen further, especially on impact. Nevertheless, the *shape* of the IRFs is well captured and both estimators deliver the right qualitative picture: a permanent news effect on TFP and a purely transitory noise effect.

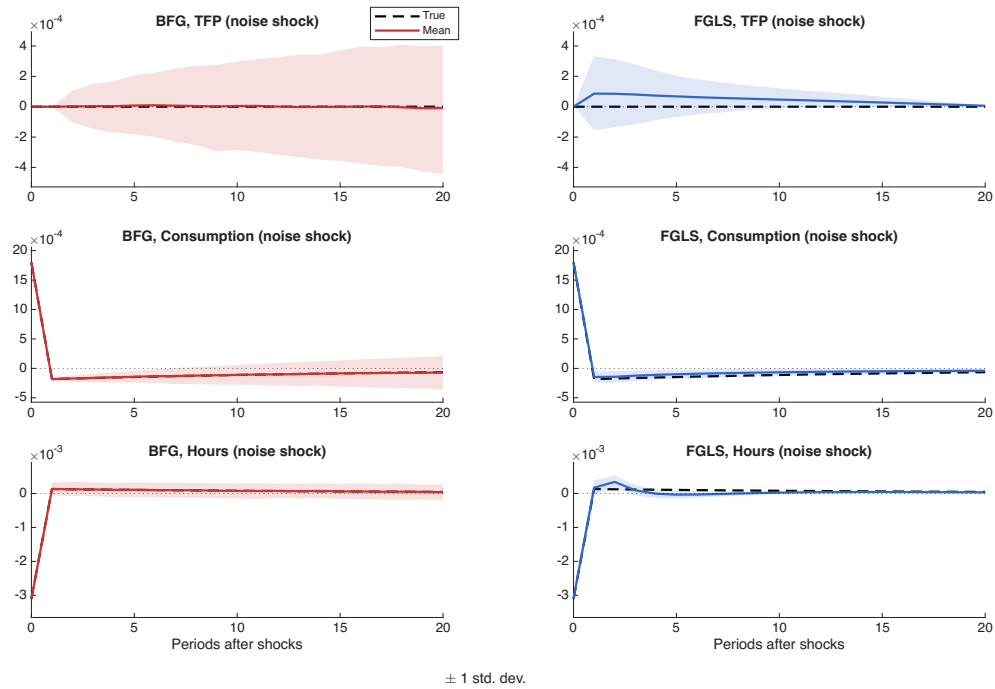


Figure 6: Estimated impulse responses to a unit noise shock in the canonical RBC simulation, with the surprise channel inactive ($\sigma_u = 0$). Sample length $T = 10,000$; news horizon $q = 1$. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

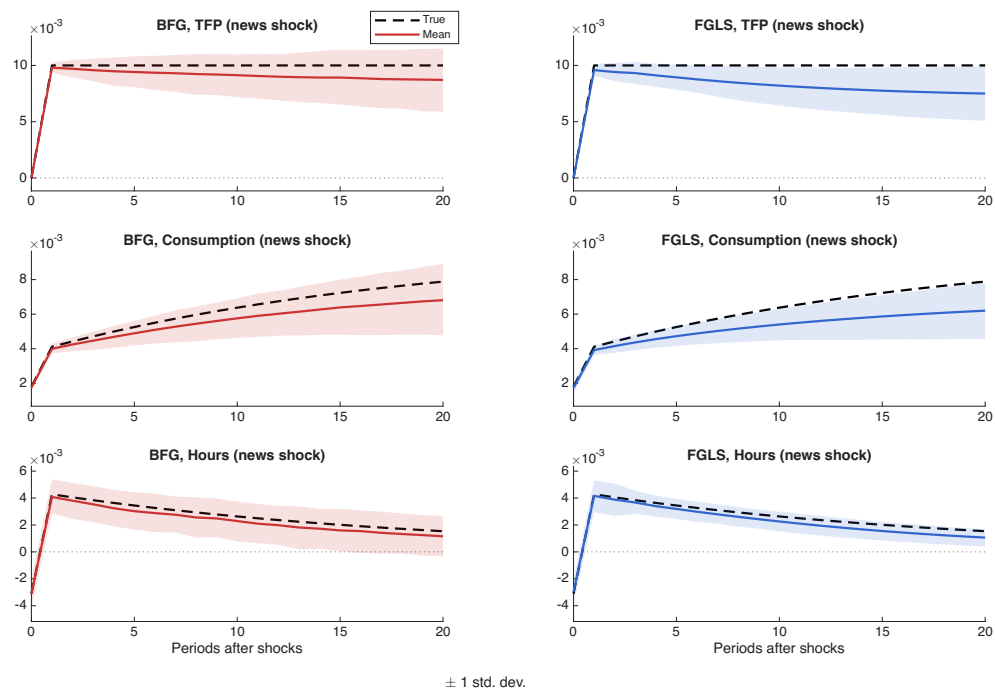


Figure 7: Estimated impulse responses to a unit news shock in the canonical RBC simulation, with the surprise channel inactive ($\sigma_u = 0$). Sample length $T = 250$; news horizon $q = 1$. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

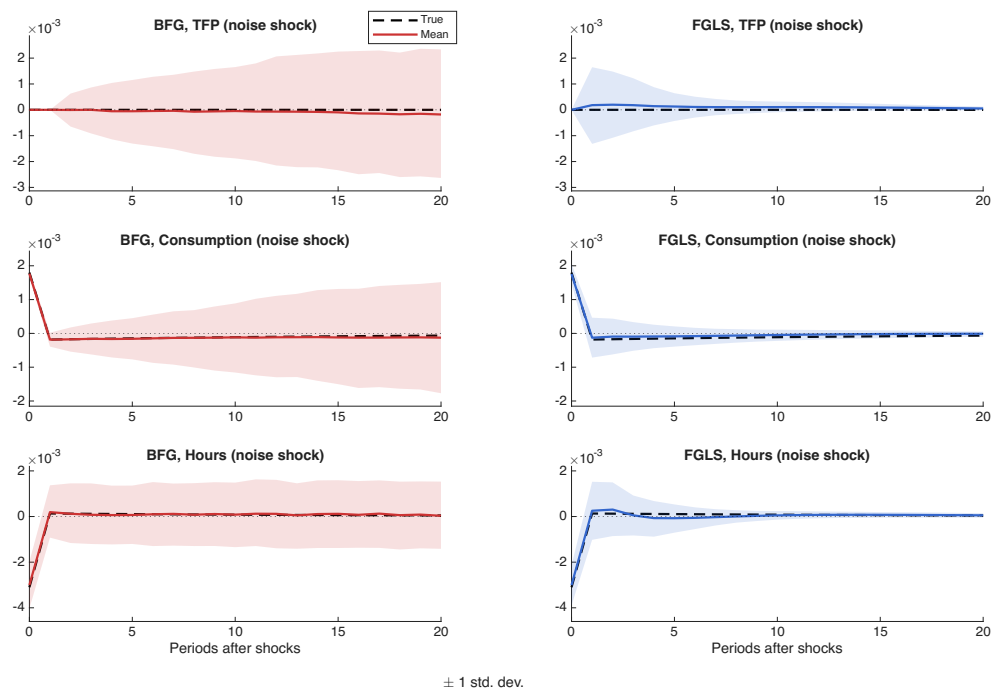


Figure 8: Estimated impulse responses to a unit noise shock in the canonical RBC simulation, with the surprise channel inactive ($\sigma_u = 0$). Sample length $T = 250$; news horizon $q = 1$. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

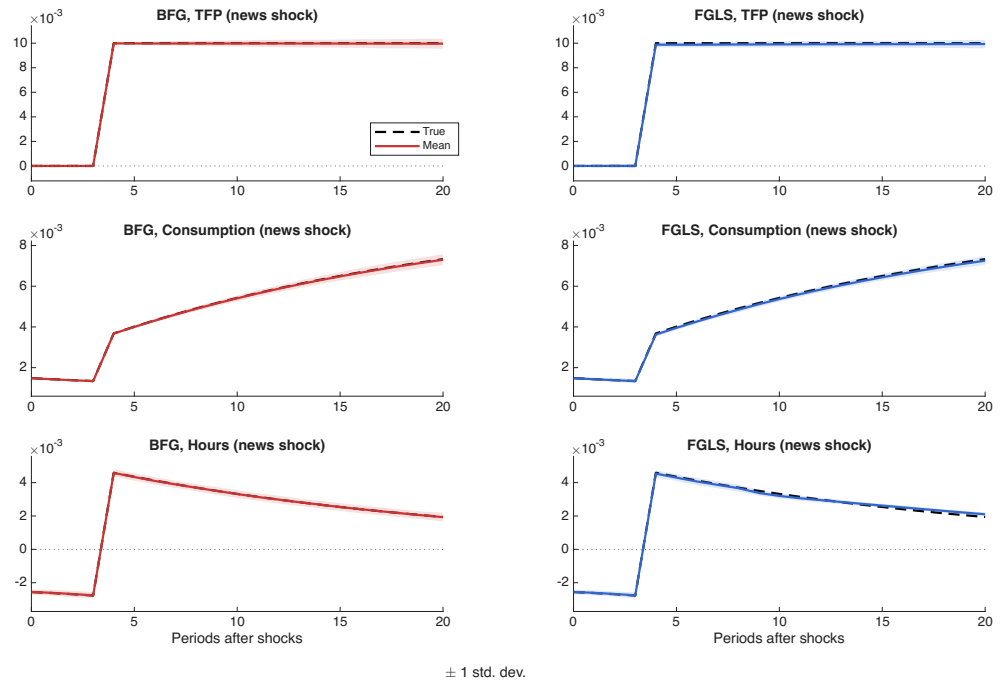


Figure 9: Estimated impulse responses to a unit news shock in the canonical RBC simulation, with the surprise channel inactive ($\sigma_u = 0$). Sample length $T = 10,000$; news horizon $q = 4$. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

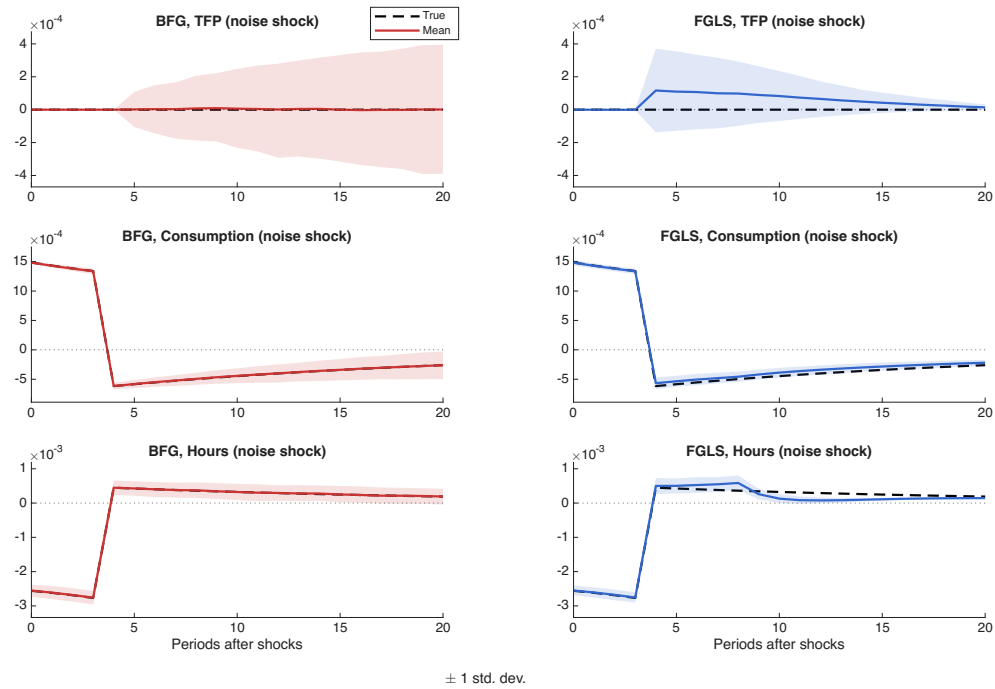


Figure 10: Estimated impulse responses to a unit noise shock in the canonical RBC simulation, with the surprise channel inactive ($\sigma_u = 0$). Sample length $T = 10,000$; news horizon $q = 4$. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

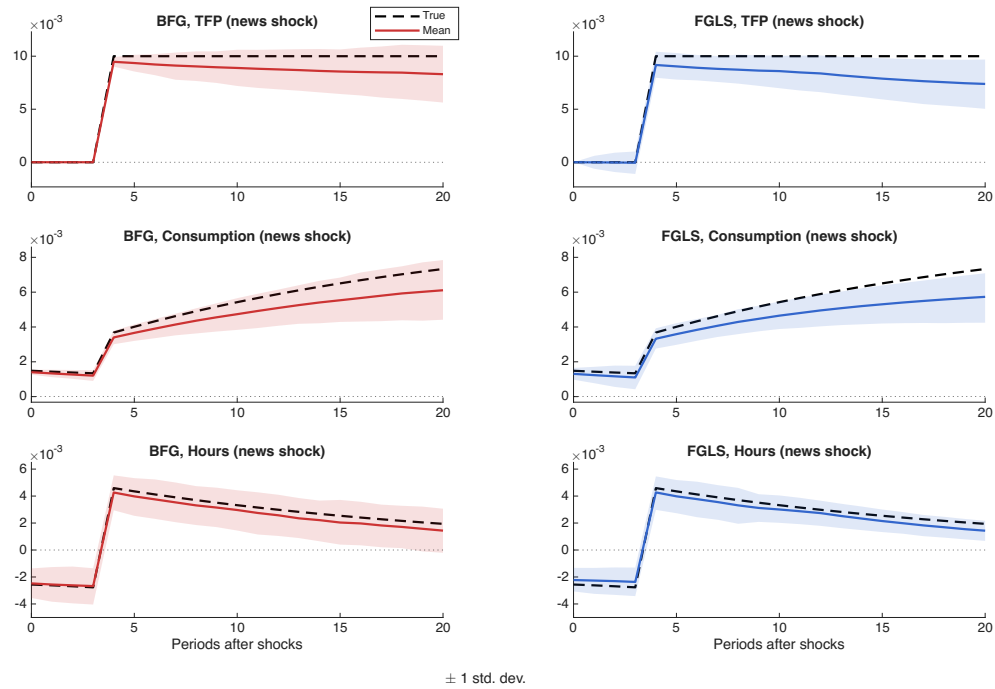


Figure 11: Estimated impulse responses to a unit news shock in the canonical RBC simulation, with the surprise channel inactive ($\sigma_u = 0$). Sample length $T = 250$; news horizon $q = 4$. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

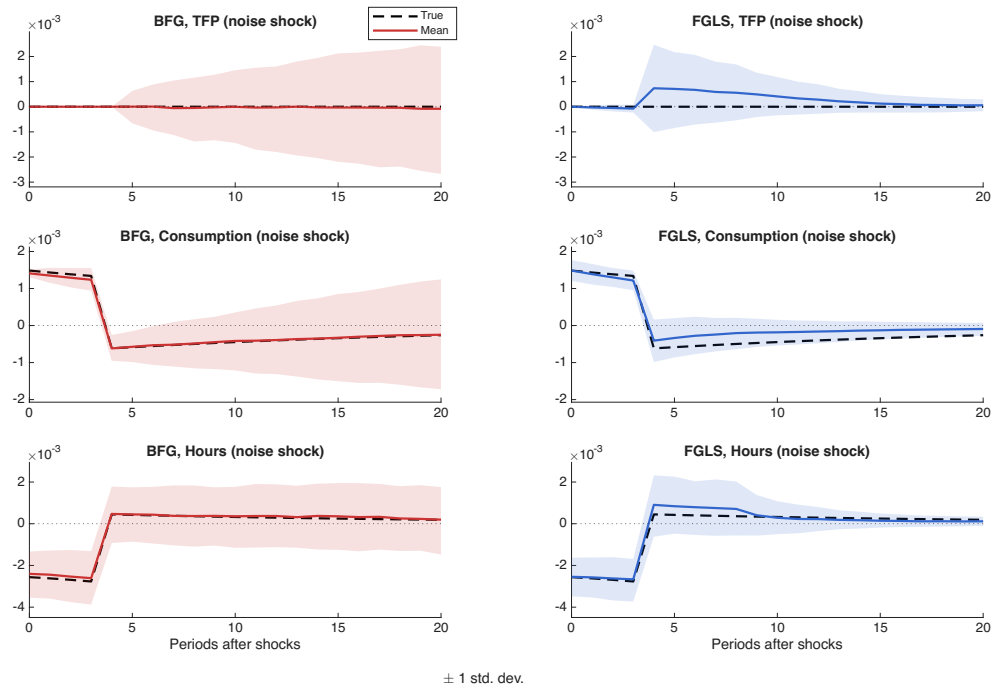


Figure 12: Estimated impulse responses to a unit noise shock in the canonical RBC simulation, with the surprise channel inactive ($\sigma_u = 0$). Sample length $T = 250$; news horizon $q = 4$. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

Adding a surprise TFP shock. Figures 13–14 add a contemporaneous surprise shock to TFP while omitting principal-component augmentation ($\text{npc} = 0$). In that non-augmented VAR, the innovation variance is partly absorbed by the surprise shock, generating a mild attenuation of the estimated news response and a slightly noisier noise response. Figures 15–18 show that augmenting the VAR with two principal components ($\text{npc} = 2$) restores the accuracy of both FGLS and BFG: in the large-sample case the estimated responses are again indistinguishable from the true ones, and in the short-sample case the finite-sample variance is materially reduced relative to the $\text{npc} = 0$ specification. This corroborates the empirical practice of enlarging the information set with FRED-MD factors when a surprise component is present.

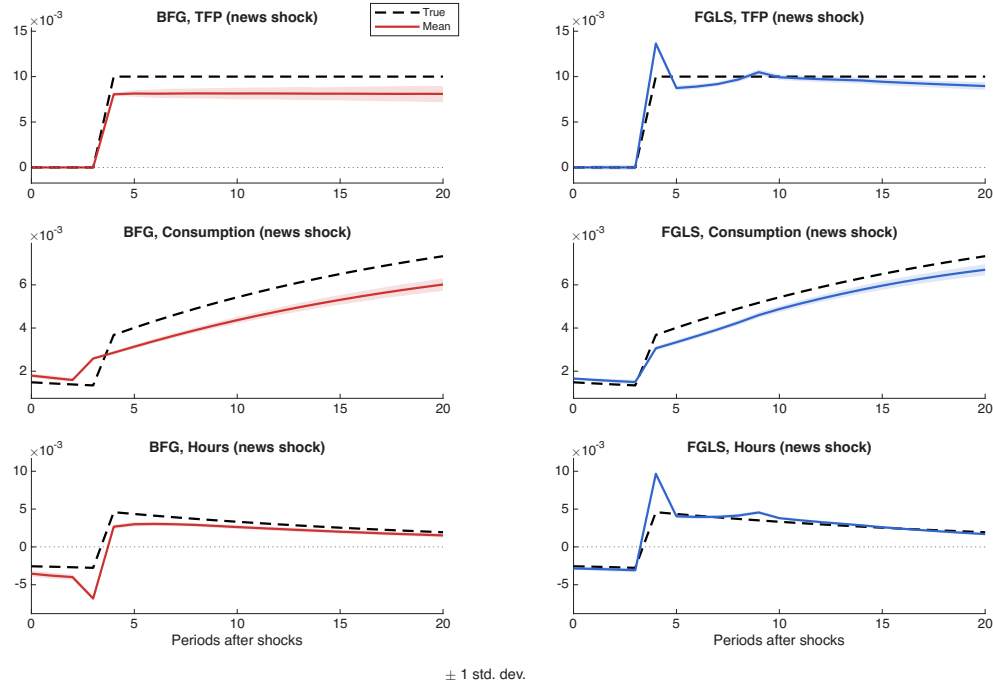


Figure 13: Estimated impulse responses to a unit news shock in the canonical RBC simulation with the surprise channel active. Sample length $T = 10,000$; news horizon $q = 4$; $\text{npc} = 0$ principal components added to the VAR. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

Persistent surprise shock, $\rho_v = 0.75$. Figures 19–22 make the surprise shock persistent ($\rho_v = 0.75$). With $T = 10,000$ and $\text{npc} = 2$, both estimators continue to recover the true news and noise IRFs, showing that persistence in the non-anticipated component does not confound the identification once the information set is sufficiently rich. In the $T = 250$ case, the short-sample variance is larger, but the Monte Carlo mean remains close to the true IRF for both shocks, which is reassuring given the size of typical empirical samples.

Summary of Monte Carlo findings. Across all experiments, FGLS and BFG deliver consistent identification of news and noise shocks as long as the information set is rich enough (principal-component augmentation when a surprise shock is present). Finite-sample performance is mostly governed by the sample size and the anticipation horizon: bias is modest even at $T = 250$, while the variance of the estimated impulse responses scales sharply with q and the persistence of the surprise shock. These Monte Carlo experiments provide a supportive small-sample benchmark for the empirical results of Section 6 and motivate the factor-augmented VAR specifications used there.

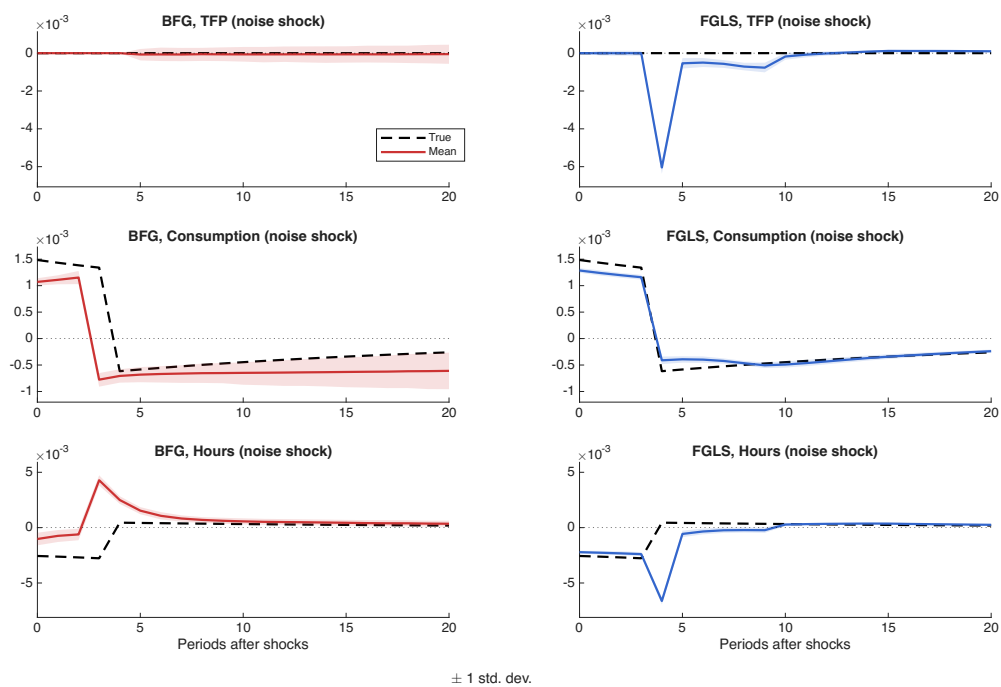


Figure 14: Estimated impulse responses to a unit noise shock in the canonical RBC simulation with the surprise channel active. Sample length $T = 10,000$; news horizon $q = 4$; npc = 0 principal components added to the VAR. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

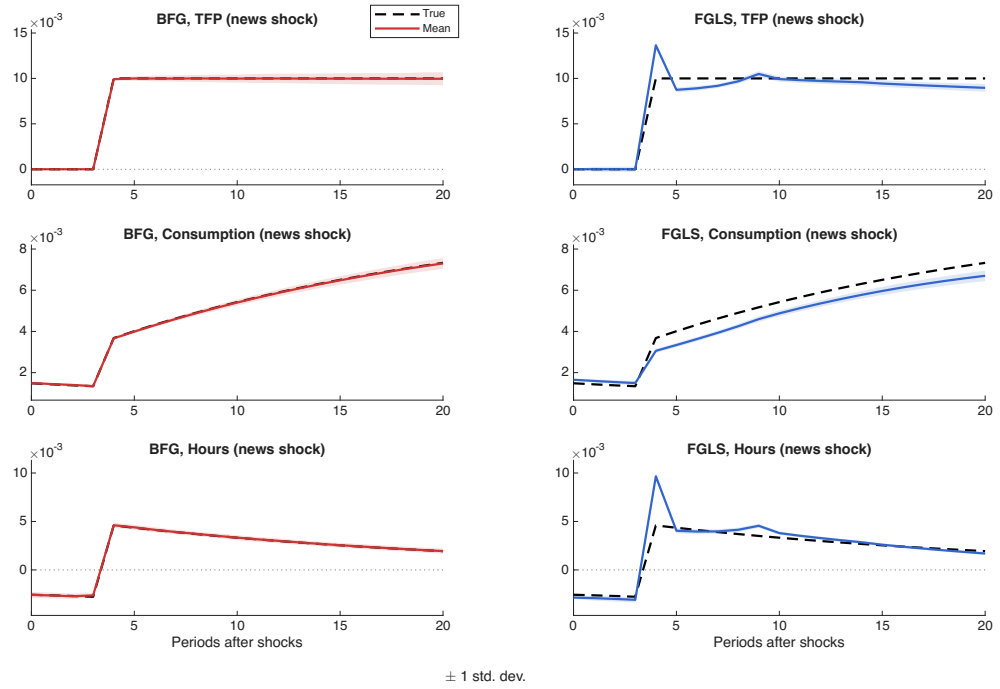


Figure 15: Estimated impulse responses to a unit news shock in the canonical RBC simulation with the surprise channel active. Sample length $T = 10,000$; news horizon $q = 4$; npc = 2 principal components added to the VAR. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

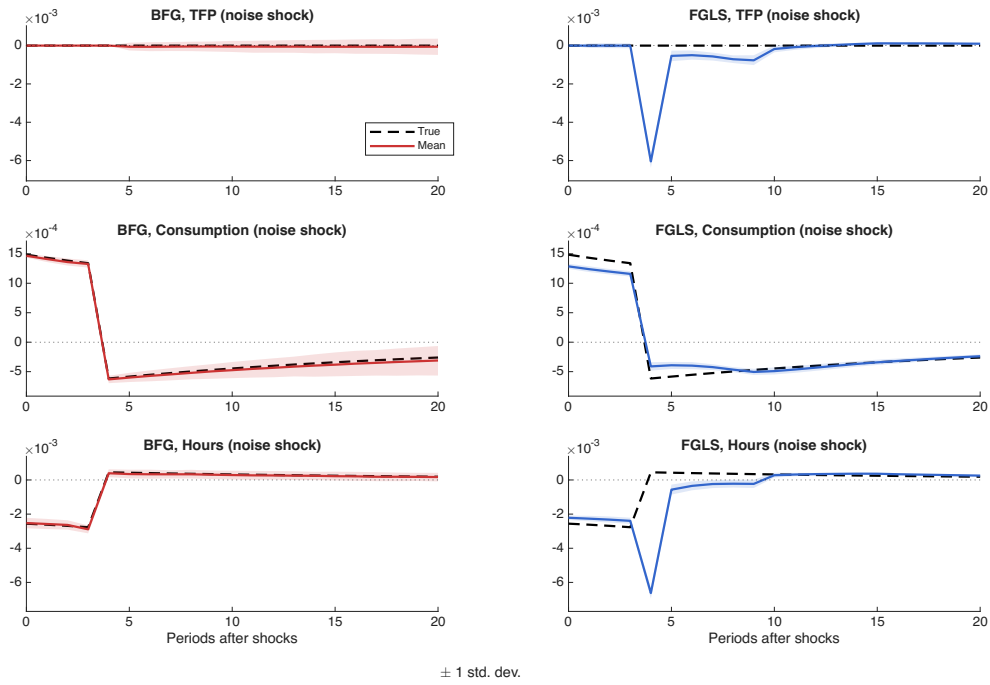


Figure 16: Estimated impulse responses to a unit noise shock in the canonical RBC simulation with the surprise channel active. Sample length $T = 10,000$; news horizon $q = 4$; npc = 2 principal components added to the VAR. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

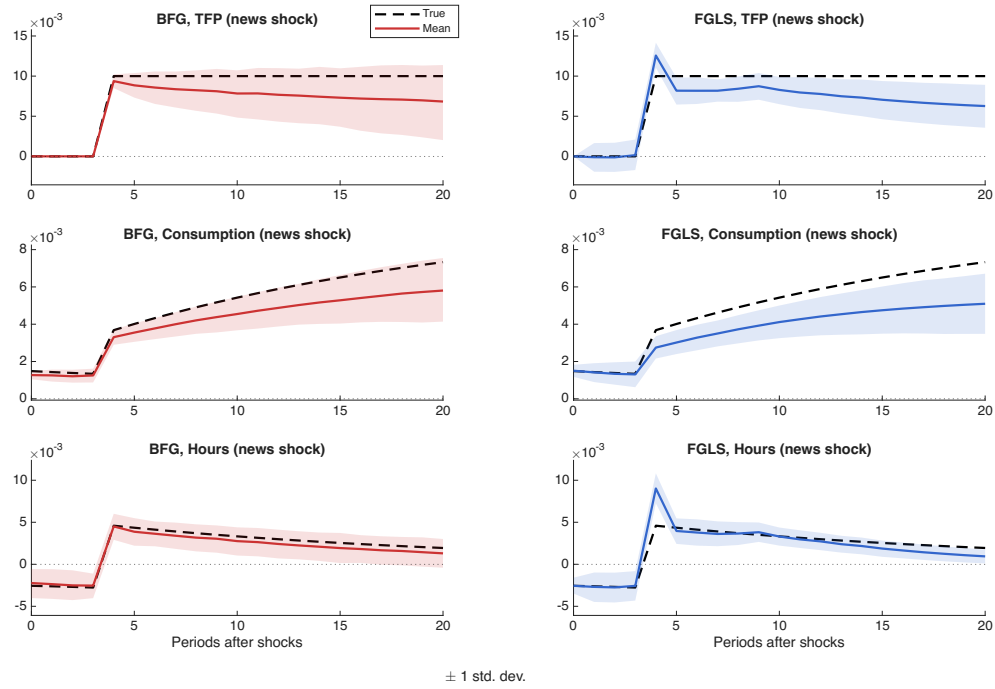


Figure 17: Estimated impulse responses to a unit news shock in the canonical RBC simulation with the surprise channel active. Sample length $T = 250$; news horizon $q = 4$; npc = 2 principal components added to the VAR. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

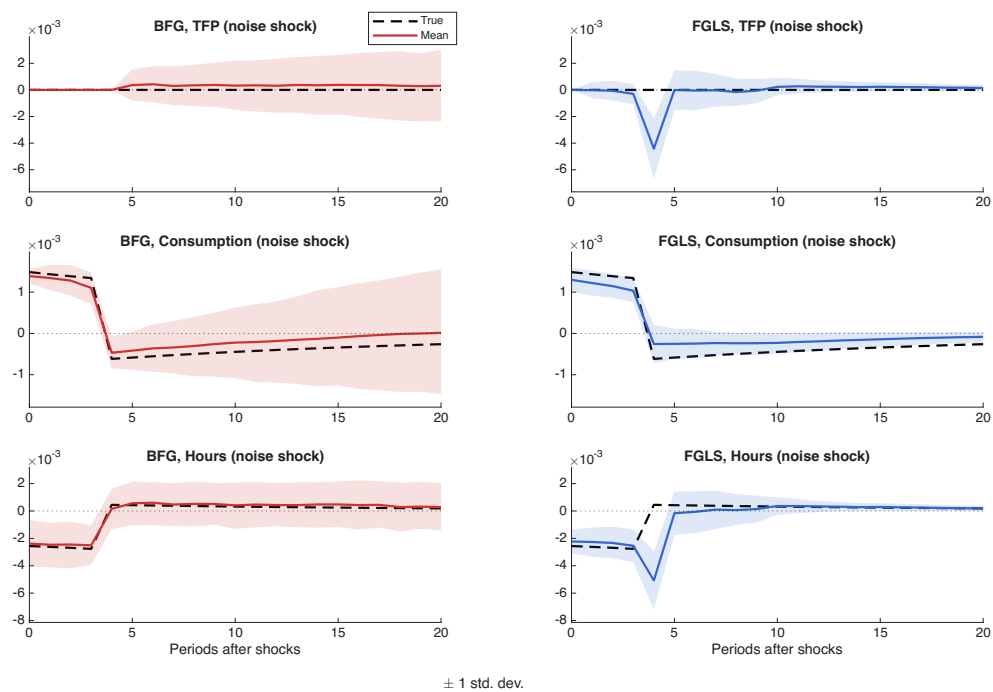


Figure 18: Estimated impulse responses to a unit noise shock in the canonical RBC simulation with the surprise channel active. Sample length $T = 250$; news horizon $q = 4$; npc = 2 principal components added to the VAR. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

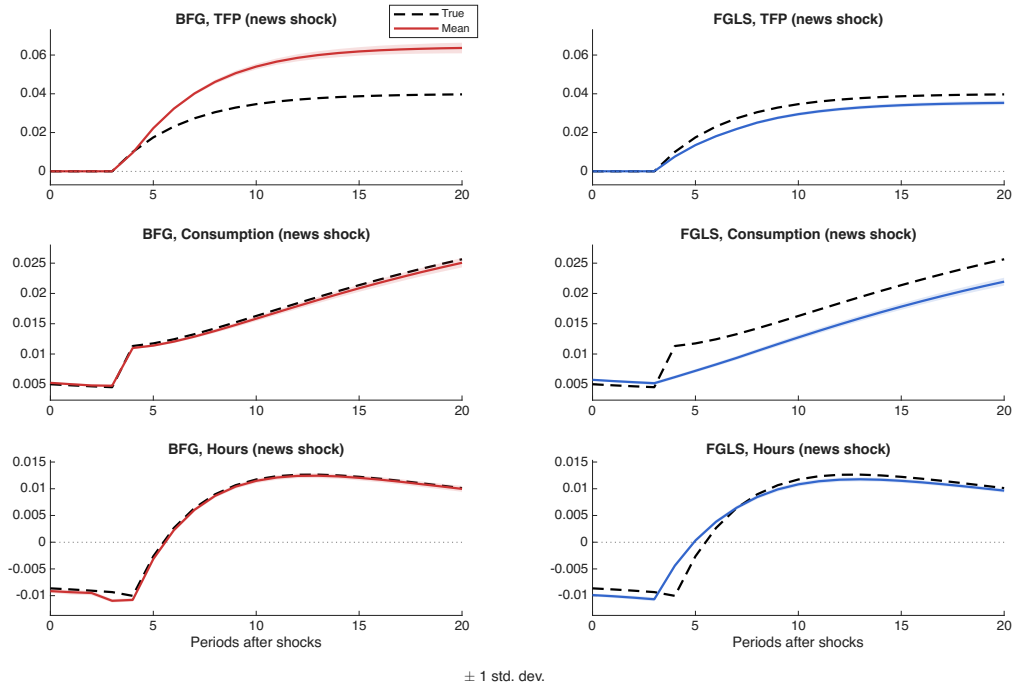


Figure 19: Estimated impulse responses to a unit news shock in the canonical RBC simulation under persistent noise ($\rho_v = 0.75$). Sample length $T = 10,000$; news horizon $q = 4$; npc = 2 principal components. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

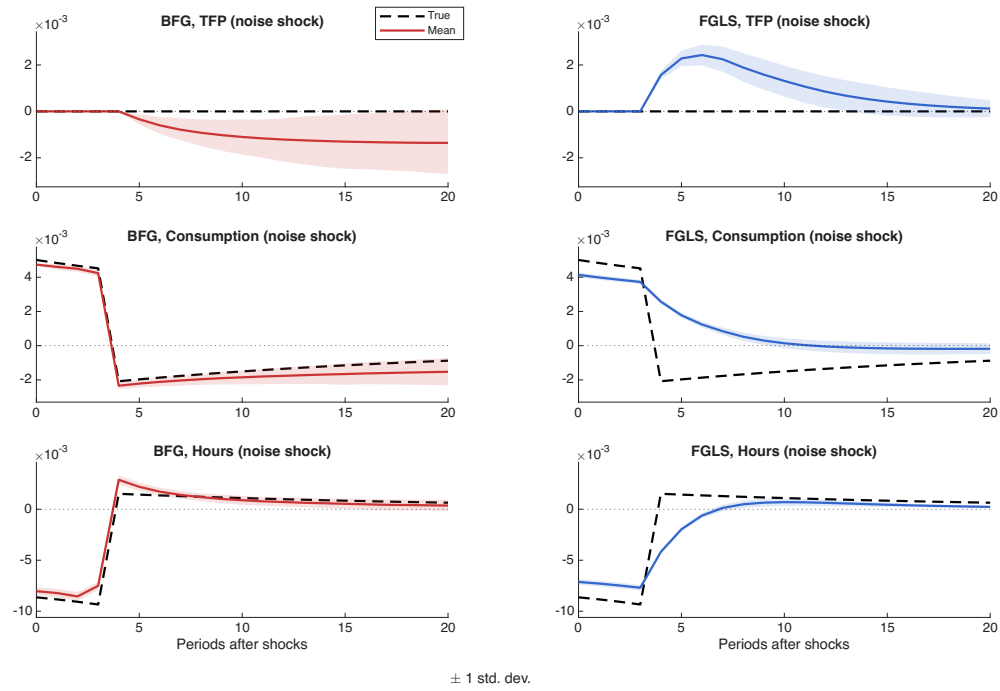


Figure 20: Estimated impulse responses to a unit noise shock in the canonical RBC simulation under persistent noise ($\rho_v = 0.75$). Sample length $T = 10,000$; news horizon $q = 4$; npc = 2 principal components. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

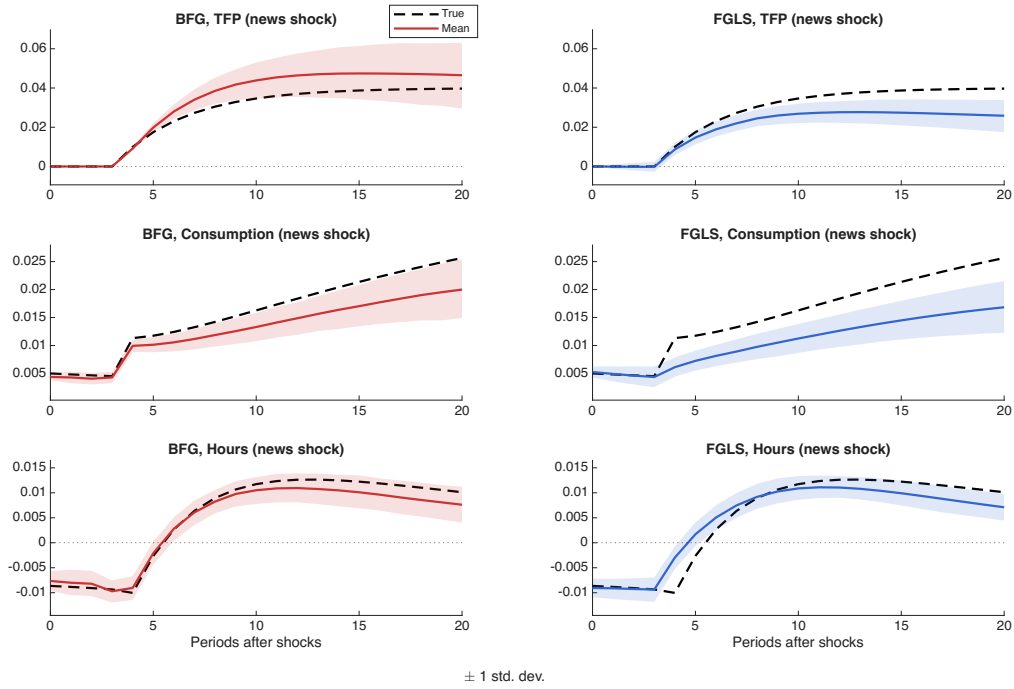


Figure 21: Estimated impulse responses to a unit news shock in the canonical RBC simulation under persistent noise ($\rho_v = 0.75$). Sample length $T = 250$; news horizon $q = 4$; npc = 2 principal components. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

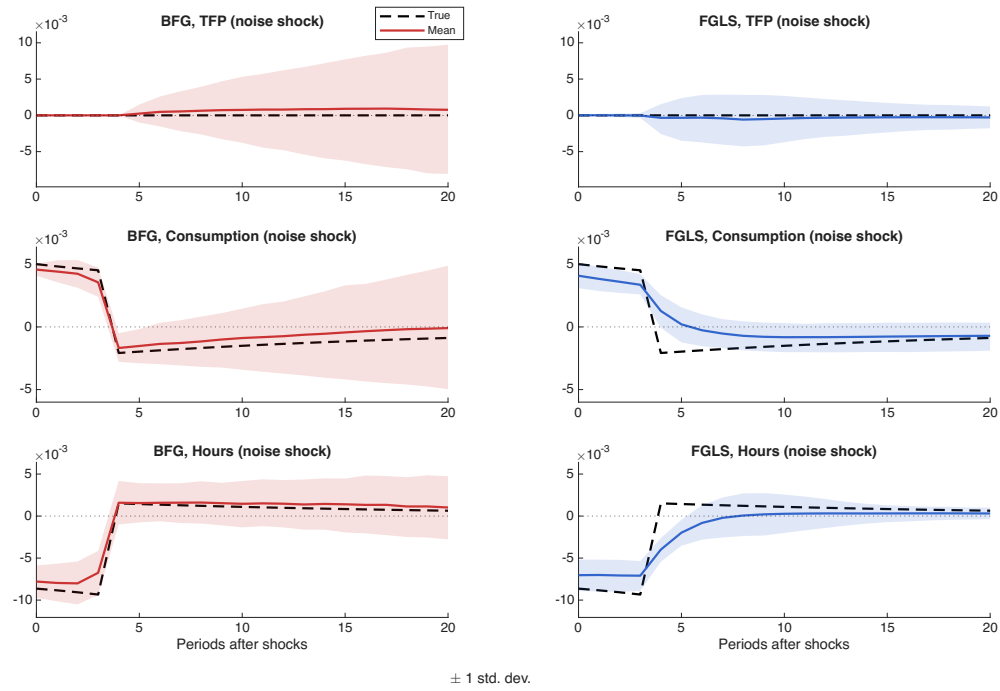


Figure 22: Estimated impulse responses to a unit noise shock in the canonical RBC simulation under persistent noise ($\rho_v = 0.75$). Sample length $T = 250$; news horizon $q = 4$; npc = 2 principal components. Solid lines: median across Monte Carlo replications; shaded regions: ± 1 standard deviation; dashed line: structural truth.

First- and second-set experiments: FG and FGLS summary tables

The four tables below report the FG and FGLS point estimates and RMSEs obtained in the simpler statistical DGPs of the first and second sets of experiments (Sections 5.1–5.2). All experiments use $T = 250$ observations and 10,000 Monte Carlo draws. Table 4 reports the $\delta = 0$ specification of the first set at $q = 1$ and $q = 4$; Table 5 extends the same DGP to $q = 8$; Table 6 considers the $\delta = 1$ specification at $q = 1$; and Table 7 reports the BLL specification of the second set with $\rho = 0.5$ and $q = 1$.

Table 4: Simulation Results: $\delta = 0$, $q \in \{1, 4\}$

$\sigma_\nu/\sigma_\varepsilon$						
	10	5	1	.2	.1	.01
$\sigma_\varepsilon/\sigma_s$						
	.0995	.1961	.7071	.9806	.9950	1.000
q=1						
FGLS						
Estimates	.1040	.1929	.7030	.9540	.7390	.0929
RMSE	.0888	.0921	.0638	.2086	.7080	1.3516
FG						
Estimates	.1047	.1931	.7036	.9816	.9929	1.0009
RMSE	.0911	.0938	.0736	.0865	.0892	.0917
q=4						
FGLS						
Estimates	.1039	.1853	.6869	.9192	.7295	.1090
RMSE	.1400	.1336	.0941	.3378	.7251	1.3320
FG						
Estimates	.1055	.1873	.7071	.9957	1.0275	1.0387
RMSE	.1449	.1359	.1250	.1751	.1907	.1858

D BLL model: CJ (2022) calibration and Monte Carlo

This appendix reports Monte Carlo evidence on the finite-sample performance of the BFG and FGLS estimators applied to data generated from the estimated medium-scale DSGE model of Chahrour and Jurado (2022), which uses the BLL information structure of Chahrour and Jurado (2018). The conceptual framing — two equivalent representations of the same agent beliefs, observational equivalence on the agent side, and the role of realised TFP as the “missing link” that breaks the equivalence — is set out in Section 2.5 and not repeated here. The simulation has three purposes: (i) to verify that BFG $q=40$ implements the technology-side Chahrour and Jurado (2018) target both in population and in finite samples; (ii) to illustrate the simplicity-and-flexibility property of Section 2.5 numerically (a single local-projection framework, with the choice of controls as the only tuning knob, identifies the relevant structural object); and (iii) to make

Table 5: Simulation Results: $\delta = 0, q = 8$

$\sigma_\nu/\sigma_\varepsilon$						
	10	5	1	.2	.1	.01
$\sigma_\varepsilon/\sigma_s$						
	.0995	.1961	.7071	.9806	.9950	1.000
q=8						
FGLS						
Estimates	.1043	.1859	.6799	.9273	.7164	.1131
RMSE	.1888	.1812	.1189	.3012	.7381	1.3288
FG						
Estimates	.1103	.1944	.7222	1.0459	1.0641	1.0796
RMSE	.2046	.1359	.1801	.2721	.2731	.2738

Table 6: Simulation Results: $\delta = 1, q = 1$

$\sigma_\nu/\sigma_\varepsilon$						
	10	5	1	.2	.1	.01
$\sigma_\varepsilon/\sigma_s$						
	.0995	.1961	.7071	.9806	.9950	1.000
q=1						
FGLS						
Estimates	.0991	.1966	.6574	.8571	.8673	.8672
RMSE	.0874	.0883	.0788	.1301	.1340	.1386
FG						
Estimates	.1006	.1980	.7066	.9809	1.0018	1.0077
RMSE	.0887	.0886	.0756	.0956	.0959	.0990

Table 7: Simulation Results: BLL specification, $\rho = 0.5$, $q = 1$

	$\sigma_\nu/\sigma_\varepsilon$					
	10	5	1	.2	.1	.01
	$\sigma_\varepsilon/\sigma_s$					
	.0995	.1961	.7071	.9806	.9950	1.000
	q=1					
FGLS						
Estimates	.0923	.1812	.6672	.8938	.9012	.9051
RMSE	.1761	.1781	.1286	.1226	.1523	.1507
FG						
Estimates	.0906	.1787	.6742	.9296	.9517	.9537
RMSE	.1953	.2074	.1587	.1418	.1380	.1316

visible the gap between the technology-side Chahrouh and Jurado (2018) target and the structural permanent shock ε^μ of the underlying DSGE — the recoverability ceiling of Chahrouh and Jurado (2022).

D.1 The variance decomposition that motivates the noise-representation view

The single technical fact about Chahrouh and Jurado (2018) that we use in what follows — and that is not repeated from Section 2.5 — is that, in the noise representation, the variance of any endogenous variable y_t admits a unique decomposition into four mutually orthogonal components:

$$\text{Var}(y_t) = \underbrace{\sum_{j \geq 1} \alpha_j^2}_{\text{past fundamentals}} + \underbrace{\alpha_0^2}_{\text{present fundamentals}} + \underbrace{\sum_{j < 0} \alpha_j^2}_{\text{future fundamentals}} + \underbrace{\sum_{j \geq 0} \beta_j^2}_{\text{noise}},$$

where α_j and β_j are the impulse-response coefficients to fundamental and noise shocks. The decomposition is identifiable across all observationally equivalent noise representations (Corollary 1 of Chahrouh and Jurado, 2018), which is what makes the noise component a well-defined measurable quantity. Even when the structural noise variance in a model is small ($\sigma_\varepsilon \rightarrow 0$), the noise component of this decomposition can be substantial: the BLL DSGE features a transitory technology component that creates a non-trivial signal-extraction problem and so generates a non-zero β_j block in the canonical representation, regardless of the size of the explicit measurement error.

D.2 Data-generating process

The DGP is the equilibrium model estimated in Chahrouh and Jurado (2022), which builds on the Blanchard et al. (2013) (BLL) framework augmented with the information structure of Chahrouh and Jurado (2018). The model features nominal wage and price rigidities (Calvo), external consumption habits, investment adjustment costs, variable capital utilization, and a Taylor rule for monetary policy. Four structural shocks are active: permanent and transitory technology, government spending, and monetary policy. Preference, investment-specific, and markup shocks are set to zero, consistent with a four-variable VAR.

D.2.1 Technology and information structure

The level of labor-augmenting technology Z_t satisfies $z_t = \log Z_t$ with

$$\Delta z_t = \Delta \mu_t + \eta_t - \eta_{t-1}, \quad (D.1)$$

where μ_t is the permanent component and η_t is the transitory component. Their dynamics are

$$\Delta \mu_t = \rho \Delta \mu_{t-1} + \varepsilon_t^\mu, \quad \varepsilon_t^\mu \sim \mathcal{N}(0, \sigma_\mu^2), \quad (D.2)$$

$$\eta_t = \rho \eta_{t-1} + \varepsilon_t^\eta, \quad \varepsilon_t^\eta \sim \mathcal{N}(0, \sigma_\eta^2), \quad (D.3)$$

where both processes share a common persistence parameter ρ . The permanent component μ_t is a random walk with first-order autoregressive growth; the transitory component η_t is stationary.

Agents do not separately observe μ_t and η_t . Instead, they observe TFP growth Δz_t and a noisy signal about the permanent component,

$$s_t = \mu_t + \xi_t, \quad \xi_t \sim \mathcal{N}(0, \sigma_\xi^2). \quad (D.4)$$

Agents solve a Kalman filtering problem to extract estimates of the unobserved state from the history of observables $\{\Delta z_\tau, s_\tau\}_{\tau \leq t}$. Even though $\sigma_\xi \approx 0$ (the signal is almost perfectly informative about μ_t), agents face a non-trivial signal extraction problem: observing $\Delta z_t = \Delta \mu_t + \eta_t - \eta_{t-1}$ does not allow them to separately identify the permanent innovation ε_t^μ from the transitory innovation ε_t^η . This inability to distinguish permanent from transitory movements in TFP is the fundamental source of noise in the model.

The steady-state Kalman gain defines a *noise representation* in the sense of Chahrour and Jurado (2018). In this representation, the three original structural innovations $(\varepsilon_t^\mu, \varepsilon_t^\eta, \xi_t)$ are rotated into two canonical shocks:

1. A *technology innovation* $\varepsilon_t^{\text{tech}}$: the Wold innovation of the TFP process. This shock captures all fundamental movements in productivity—both the permanent (ε_t^μ) and transitory (ε_t^η) components—as seen through the lens of the univariate spectral density of Δz_t .
2. A *noise innovation* $\varepsilon_t^{\text{noise}}$: the component orthogonal to TFP fundamentals at all leads and lags. This shock affects agents' beliefs (and hence their consumption, investment, and labor decisions) without changing actual TFP at any horizon. It arises because agents, upon observing a particular realization of Δz_t and s_t , cannot perfectly determine whether the underlying cause is permanent or transitory. Their forecast errors about future TFP generate real economic fluctuations through the model's forward-looking equilibrium conditions.

The noise representation is the state-space form that we solve using the method of Chahrour and Jurado (2022). In the estimated BLL model, the noise innovation has a standard deviation of approximately 0.23—an economically meaningful magnitude—even though $\sigma_\xi \approx 0$. This illustrates the key point of Chahrour and Jurado (2018): the importance of belief-driven fluctuations depends on the economic environment (forward-looking behavior, rigidities) and not merely on the size of the explicit noise in the signal.

D.2.2 Exogenous disturbances

In addition to the two technology innovations $(\varepsilon_t^\mu, \varepsilon_t^\eta)$ and the noise shock (ξ_t) , the model features:

- A government spending shock: $g_t = \rho_g g_{t-1} + \varepsilon_t^g$, $\sigma_g = 0.25$, $\rho_g = 0.95$.
- A monetary policy shock: $r_t = \rho_r r_{t-1} + (1 - \rho_r)[\phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t] + \varepsilon_t^r$, $\sigma_r = 0.34$.

D.2.3 Parameter values

All parameters are set to the maximum likelihood estimates from column 1 (“BLL info”) of Table 6 in the online appendix of Chahrour and Jurado (2018), as used in the Monte Carlo of Chahrour and Jurado (2022). The key parameters are:

Parameter	Description	Value
ρ	Common persistence of $\Delta\mu_t$ and η_t	0.8911
σ_η	Std. dev. of transitory tech shock	1.2295
σ_ξ	Std. dev. of signal noise	0.0001
α_K	Capital share	0.36
h	Habit persistence	0.7066
χ	Investment adjustment cost	15.000
ξ	Utilization cost curvature	0.0079
θ_p	Calvo price stickiness	0.8645
θ_w	Calvo wage stickiness	0.8708
ϕ_π	Taylor rule: inflation	3.8640
ρ_r	Interest rate smoothing	0.4540
σ_r	Std. dev. of monetary shock	0.2792
ρ_g	Government spending persistence	0.95
σ_g	Std. dev. of government shock	0.25

The near-zero value of σ_ξ implies that the signal is almost perfectly informative about μ_t , yet a non-trivial noise component arises in the noise representation because agents cannot distinguish permanent from transitory technology based on TFP growth alone. In the resulting noise representation, the noise innovation has a standard deviation of approximately 0.23, which generates economically meaningful expectation-driven fluctuations.

D.3 Observables and the signal variable

Following Chahrour and Jurado (2022), we simulate four macroeconomic observables—TFP, consumption, GDP, and hours—plus the agents’ signal s_t defined in (D.4). All series are expressed in annualized percentage changes (multiplied by 400). The level variables (TFP, consumption, GDP) are cumulated from growth rates; hours are in levels.

The signal s_t plays a central role in the identification:

- Both the BFG and FGLS estimators use the vector $[\Delta z_t, s_t, \Delta c_t, \Delta y_t]$ in a four-variable VAR.
- In the BFG local projection, current and lagged values of s_t enter as regressors alongside TFP leads, helping to separate the technology and noise components.
- Hours are recovered via a linear projection of n_t on $[z_t, s_t, c_t, y_t]$ (in levels), and the estimated coefficients are applied to the impulse responses of TFP, signal, consumption, and GDP.

D.4 Estimators

D.4.1 BFG estimator

The BFG estimator implements the local-projection identification of Section 2.5 in a single linear regression with controls. The VAR is estimated on first differences $X_t = [\Delta z_t, s_t, \Delta c_t, \Delta y_t]'$ with $p = 4$ lags, and the technology and noise IRFs are obtained from two separate local projections.

Technology IRF. For each variable j and horizon h , the technology response is the OLS slope β_h^{tech} in The technology IRF is identified by specification (10) of Proposition 1 (the LP_W form, with focus on Δz_t and lagged controls only). By Frisch–Waugh–Lovell, β_h^{tech} is the response of $y_{j,t+h}$ to the Wold innovation of Δz_t in the BFG VAR — the technology-side Chahrour and Jurado (2018) target discussed in Section 2.5.

This is the “without surprise controls” specification of the §2.5 flexibility argument: no estimated surprise innovation $\hat{u}_{t+j}$ enters (10). The simulation results below verify that β_h^{tech} recovers the Wold-innovation IRF essentially exactly in population.

Noise IRF. The noise response is the OLS slope β_h^{noise} in

$$y_{j,t+h} = \beta_h^{\text{noise}} s_t + \sum_{k=0}^q \delta_k \Delta z_{t+k} + \gamma'_h \text{lags} + u_{t+h}, \quad (\text{D.5})$$

with $q = 40$ TFP leads. Since the noise component of the signal is orthogonal to realised TFP at all leads by construction, partialling out $\Delta z_{t+0:q}$ leaves only the part of s_t uncorrelated with future TFP. The slope therefore identifies the response of $y_{j,t+h}$ to a unit noise innovation. Larger q improves the population approximation at the cost of degrees of freedom; we use $q = 40$ throughout.

Hours. Hours n_t are not included in the BFG VAR. Their IRFs are recovered by linear projection of n_t on $[z_t, s_t, c_t, y_t]$ in levels, applied to the IRFs of TFP, signal, consumption, and GDP.

D.4.2 FGLS estimator

The FGLS estimator follows Barsky and Sims (2011). It is applied to the levels VAR $[z_t, s_t, c_t, y_t]$ with $p = 4$ lags. The identification proceeds as follows:

1. A Cholesky decomposition with TFP ordered first separates the TFP innovation (shock 1) from the signal innovation orthogonal to TFP at impact (shock 2).
2. A Blaschke filter applied to the TFP response to shock 2 isolates the “news” component—the part of the signal that predicts future TFP—from the “noise” component.
3. By construction, the FGLS news shock has *zero contemporaneous impact* on TFP. This restriction is appropriate when the technology shock is purely anticipated (as in the standard news model), but may induce bias when technology has both an immediate and a long-run effect on TFP.

D.5 On bootstrap bias correction for BFG

Local projection estimators are known to suffer from finite-sample bias, particularly with persistent time series (Herbst and Johannsen, 2024; Kilian and Kim, 2011), and a natural remedy is a residual-bootstrap bias correction at the impulse-response level. We implemented and tested such a correction — estimate the VAR, resample the residuals to generate pseudo-data, re-estimate the BFG IRFs on each bootstrap replication, and apply the standard plug-in correction $\hat{\theta}^{\text{BC}}(h) = 2\hat{\theta}(h) - \frac{1}{B} \sum_b \hat{\theta}^{*b}(h)$.

In our simulations, this correction works well for some IRFs (e.g., the GDP response to a noise shock, where the bootstrap mean reproduces the original sampling bias) but *over-corrects* for variables whose response depends on long-horizon belief dynamics (notably the consumption response to a noise shock). The mechanism is that the residual bootstrap simulates from the estimated VAR, which by construction has the lag structure the VAR estimated; this eliminates the source of the original finite-sample bias in the orthogonalised-signal regression. The bootstrap mean then approaches the truth, the plug-in formula reads this as evidence that the original was downward biased, and subtraction pushes the corrected estimate below the truth, sometimes near zero.

For the figures and tables reported below we therefore drop the bootstrap-BC column and instead report the uncorrected BFG $q=40$ estimator alongside the residual-augmented correction of Proposition 3. As Table 10 and Figure 24 show, the residual-augmented correction reduces the noise-IRF median bias from -16% to within $\pm 3\%$ of the structural truth at the peak horizons, and the plug-in correction factor $1/[1 - (q+1)/T \cdot \hat{\xi}_Q]$ is essentially deterministic across replications (median 1.205 with sub-percent variation across 1,000 replications). This is exactly the stability property that distinguishes the residual-augmented correction from the bootstrap-BC: the plug-in $\hat{\xi}_Q$ depends only on the second-moment structure of X_t and does not require the bootstrap re-estimator to reproduce the original finite-sample bias — a property that fails in the BLL regime, as documented above. A full diagnostic of the bootstrap-BC behaviour for the noise-to-consumption response is provided in the simulation code (`simulate_CJ/diagnose_noise_consumption_mc.m`).

D.6 Monte Carlo design

We conduct two experiments:

1. **Population.** A single long sample of $T = 100,000$ observations to verify that the estimators recover the target impulse responses in population.
2. **Finite sample.** $N = 1,000$ independent samples of $T = 284$ observations (matching the sample size used in Chahrour and Jurado (2022)).

For the noise shock, the target “truth” against which BFG and FGLS are compared is the structural noise IRF computed from the state space of the solved DSGE model: it is well defined and identical across the news and noise representations of Chahrour and Jurado (2018) since noise has zero effect on realised TFP at all horizons. For the technology shock, the natural target is the Wold-innovation IRF of Δz_t in the BFG VAR — the technology-side Chahrour and Jurado (2018) target that the BFG technology regression (10) identifies. We also report the structural permanent-shock IRF ε^μ from the DSGE state space as a secondary reference, to make visible the gap between the two distinct objects.

D.7 Results

D.7.1 Population

Table 8 reports the population impulse responses of GDP to the noise shock. The BFG estimator with $q = 40$ leads matches the structural noise IRF almost perfectly at all horizons, confirming that the noise identification (D.5) is consistent and the truncation at $q = 40$ is adequate. By contrast, the FGLS estimator is severely biased: it yields near-zero (and eventually negative) responses, reflecting the misspecification induced by the zero-impact restriction in a model where TFP has both permanent and transitory components.

Table 8: Noise shock $\rightarrow$ GDP: Population estimates ($T = 100,000$)

Horizon	Truth	BFG ($q = 40$)	FGLS
0	1.702	1.701	0.241
1	2.692	2.697	0.203
2	3.190	3.195	0.052
3	3.358	3.358	−0.137
4	3.310	3.319	−0.347
5	3.126	3.126	−0.550
8	2.245	2.239	−0.998
10	1.630	1.636	−1.144

Table 9 reports the corresponding population responses of GDP to the technology shock, splitting the “truth” column into the Wold-innovation IRF (the technology-side Chahrour and Jurado (2018) target that BFG $q=40$ identifies in (10)) and the structural permanent-shock IRF ε^μ from the DSGE state space (a distinct object that the finite-lead projection cannot recover). BFG $q=40$ sits on top of the Wold IRF column at every horizon: the technology-side target is recovered exactly. The structural ε^μ IRF is markedly different and exhibits a hump-then-dip pattern that the finite-lead projection does not see — a numerical illustration of the recoverability point of Chahrour and Jurado (2022) discussed in Section 2.5.

D.7.2 Finite sample

Table 10 reports the median impulse response of GDP to the noise shock across the 1,000 simulations. The uncorrected BFG estimator at $q = 40$ captures the hump-shaped dynamics of the true response with a moderate downward bias of approximately 15–20% at the peak. The residual-augmented correction of Proposition 3 essentially eliminates this bias: at the peak ($h = 3$), the uncorrected median is −16.1% below truth, the corrected median is +1.0% above truth. The plug-in $\hat{\xi}_Q$ has median 0.990 across replications with an interquantile range smaller than 0.003, so the correction factor

Table 9: Technology shock $\rightarrow$ GDP: Population estimates ($T = 100,000$)

Horizon	Wold (CJ 2018)	ε^μ (struct.)	BFG ($q = 40$)	FGLS
0	2.248	0.000	2.248	1.832
1	3.020	1.707	3.020	3.213
2	3.453	1.443	3.453	4.205
3	3.633	0.774	3.633	4.876
5	3.577	-0.877	3.577	5.534
8	2.962	-2.531	2.962	5.797
10	2.484	-2.756	2.484	5.661
20	1.003	1.741	1.003	7.716

Table 10: Noise shock $\rightarrow$ GDP: Finite-sample median ($N = 1,000$, $T = 284$). The corrected column applies the residual-augmented Q-correction of Proposition 3.

Horizon	Truth	BFG ($q = 40$)	BFG ($q = 40$, corrected)	FGLS
0	1.702	1.483	1.787	0.494
1	2.692	2.322	2.797	0.831
2	3.190	2.714	3.270	0.988
3	3.358	2.816	3.392	1.040
4	3.310	2.716	3.272	1.066
5	3.126	2.519	3.033	1.081
8	2.245	1.669	2.011	0.968
10	1.630	1.138	1.371	0.915

$1/[1 - (q + 1)/T \cdot \widehat{\xi}_Q] \approx 1.205$ is essentially deterministic — a direct numerical illustration of the stability claim of Proposition 3(iii). FGLS, by contrast, fails to reproduce the dynamics of the structural noise IRF.

Two features stand out:

1. **BFG correctly identifies the hump shape.** BFG $q=40$ captures the characteristic hump-shaped response of GDP to the noise shock, peaking around horizon 3–4 and declining thereafter. This shape reflects the transitory nature of the noise shock: agents initially respond to what they perceive as a technology improvement, but as they learn the truth, the effect dissipates. The downward bias of about 15–20% at the peak is a finite-sample feature of the finite-lead projection; it diminishes asymptotically (in population, $T = 100,000$, BFG $q=40$ reproduces the structural noise IRF essentially exactly — see Figure 23).
2. **FGLS is fundamentally misspecified.** The FGLS estimator produces responses that are one-third to one-half of the truth and, more importantly, fails to capture the hump-shaped pattern. Instead, it yields a monotonically increasing then roughly flat response—qualitatively inconsistent with the transitory nature of the noise shock. This is a direct consequence of the zero-impact restriction: by forcing the identified technology shock to have no contemporaneous effect on TFP, FGLS misattributes part of the noise effect to technology and vice versa.

Figure 23 displays the population comparison for three variables (GDP, consumption, and hours), and Figures 24 and 25 show the finite-sample distributions with 68% probability bands.

E Conventional Benchmarks: News-to-Signal Ratio and FGLS

This online appendix reports the two conventional benchmarks of the empirical analysis (Section 6) on the same information set as the headline statistics.

E.1 Benchmark I: the conventional news-to-signal ratio

For comparison with the prior literature, we also report the conventional statistics; their higher implied news shares reflect the estimands and bias properties documented in Appendix B, and they are consistent with the transmission-side findings above. We first compute the semi-parametric benchmark statistic $\mathbf{nsr}_h^{\text{news}}$ of Appendix B, which provides a direct estimate of the news-to-signal ratio $\alpha = \sigma_\varepsilon / \sigma_s$ in the single-arrival benchmark. The signal innovation is extracted from a VAR(p) estimated on the vector $X_t = (\Delta z_t, s_t, F_{1t}, \dots, F_{7t})'$, where Δz_t is TFP growth, s_t is the consumer confidence index, and $F_{1t}, \dots, F_{7t}$ are the FRED-MD factors. The signal innovation corresponds to the VAR residual from the signal equation.

The long-run variance of TFP growth is estimated parametrically via an AR(4) model with a constant.

Table 11 reports the estimated $\mathbf{nsr}_h^{\text{news}}$ at selected horizons for both signals, using a VAR with 8 lags in TFP growth and 7 FRED-MD factors.

Table 11: News-to-signal ratio $\mathbf{nsr}_h^{\text{news}}$ at horizons $h = 4, 8, 12, 16, 20, 24$ quarters, computed from a VAR with $p = 8$ lags in $\Delta z_t, s_t$, and 7 FRED-MD principal components. A value of 1 corresponds to a perfectly informative signal; the statistic stabilises near 0.90 at $h = 20$ for both signals. The estimand is the correlation-type ratio α of the simple-news benchmark (B.1), exact when the signal carries a single fresh news shock; under multi-horizon news aggregation it overstates the news content (Remark 3), and the headline news-share estimates of this paper are the level and flow statistics of Table 2. Sample: 1960:Q1–2019:Q4.

	$h = 4$	$h = 8$	$h = 12$	$h = 16$	$h = 20$	$h = 24$
ICS 5-Year (BUS5)	0.18	0.34	0.57	0.74	0.90	0.90
ICS 12-Month (BUS12)	0.15	0.21	0.46	0.68	0.91	0.97

For both signals, the statistic increases monotonically with the horizon and converges to approximately $\alpha \approx 0.90$ around $h = 20$ quarters. This indicates that the bulk of the signal variance is attributable to the news component, with a modest noise component.

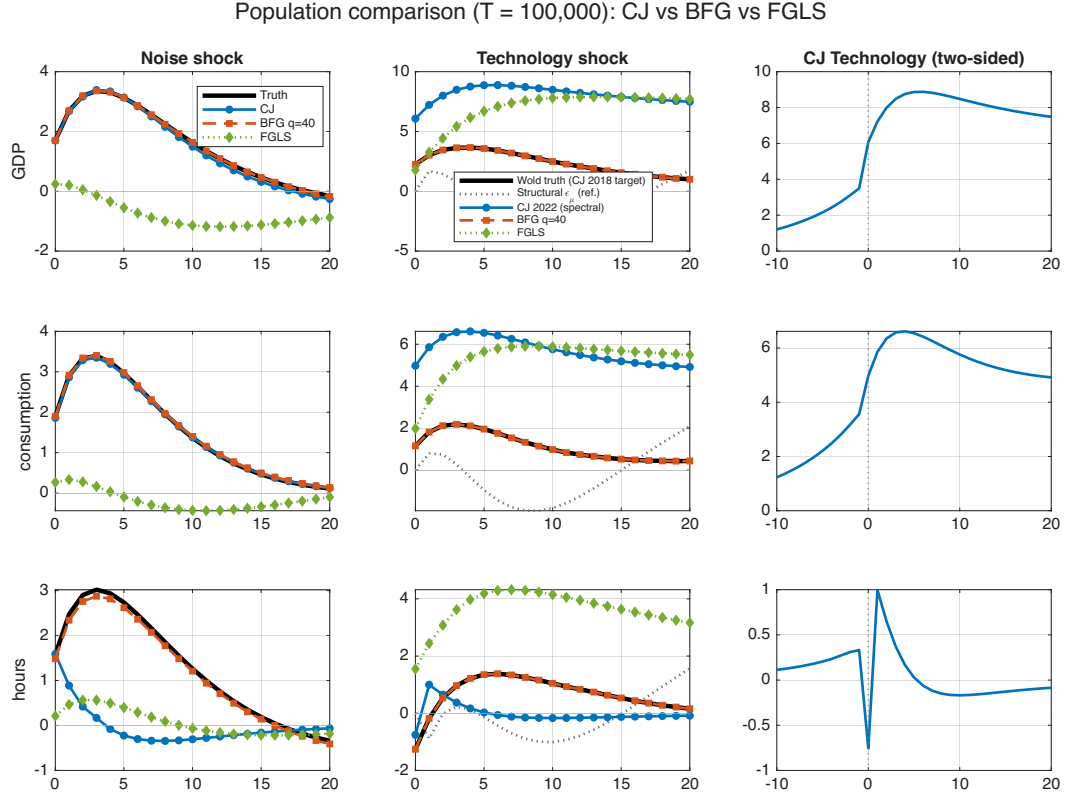


Figure 23: Population impulse responses ($T = 100,000$). Left column: noise shock for BFG with $q = 0$ (blue dotted), $q = 40$ (red solid), FGLS (green dash-dot), and structural truth (black dashed). Center: technology shock — the solid black line is the *Wold-innovation IRF* (the technology-side Chahrour and Jurado (2018) target); the gray dotted line is the structural permanent-shock IRF ε_μ for reference; BFG $q=40$ sits on the Wold IRF by construction in population. Right: CJ-spectral two-sided technology IRF.

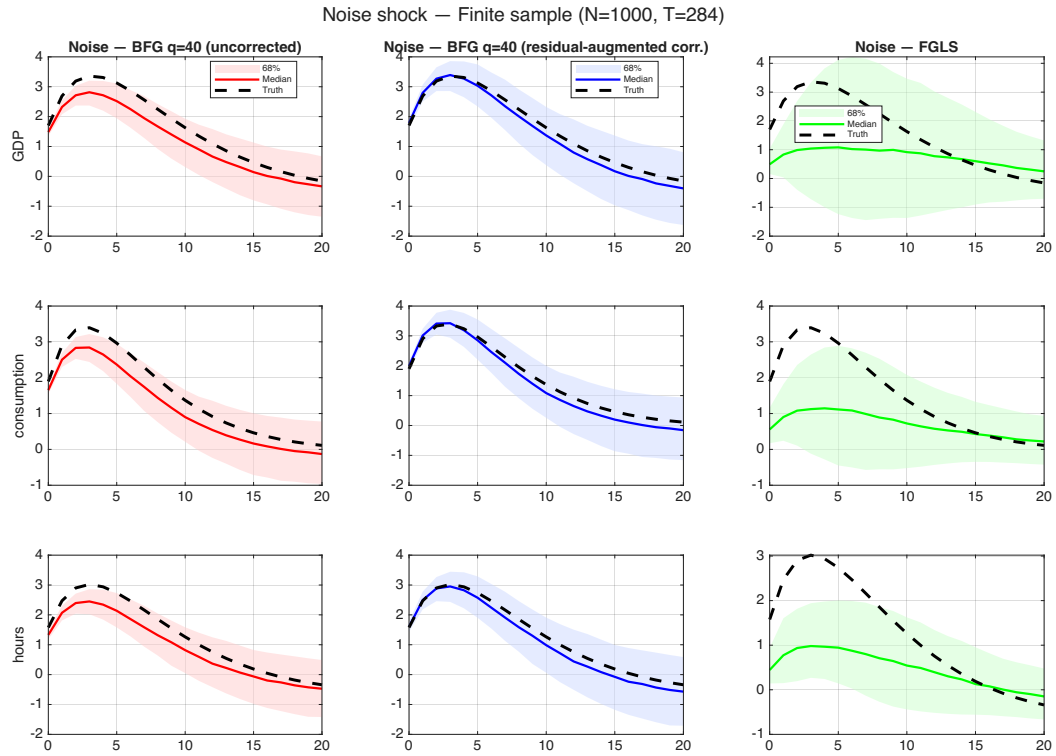


Figure 24: Noise shock: finite-sample distributions ($N = 1,000, T = 284$). Left: BFG $q = 40$ uncorrected. Middle: BFG $q = 40$ with the residual-augmented correction of Proposition 3. Right: FGLS. Shaded regions: 68% bands; solid line: median; dashed line: structural truth.

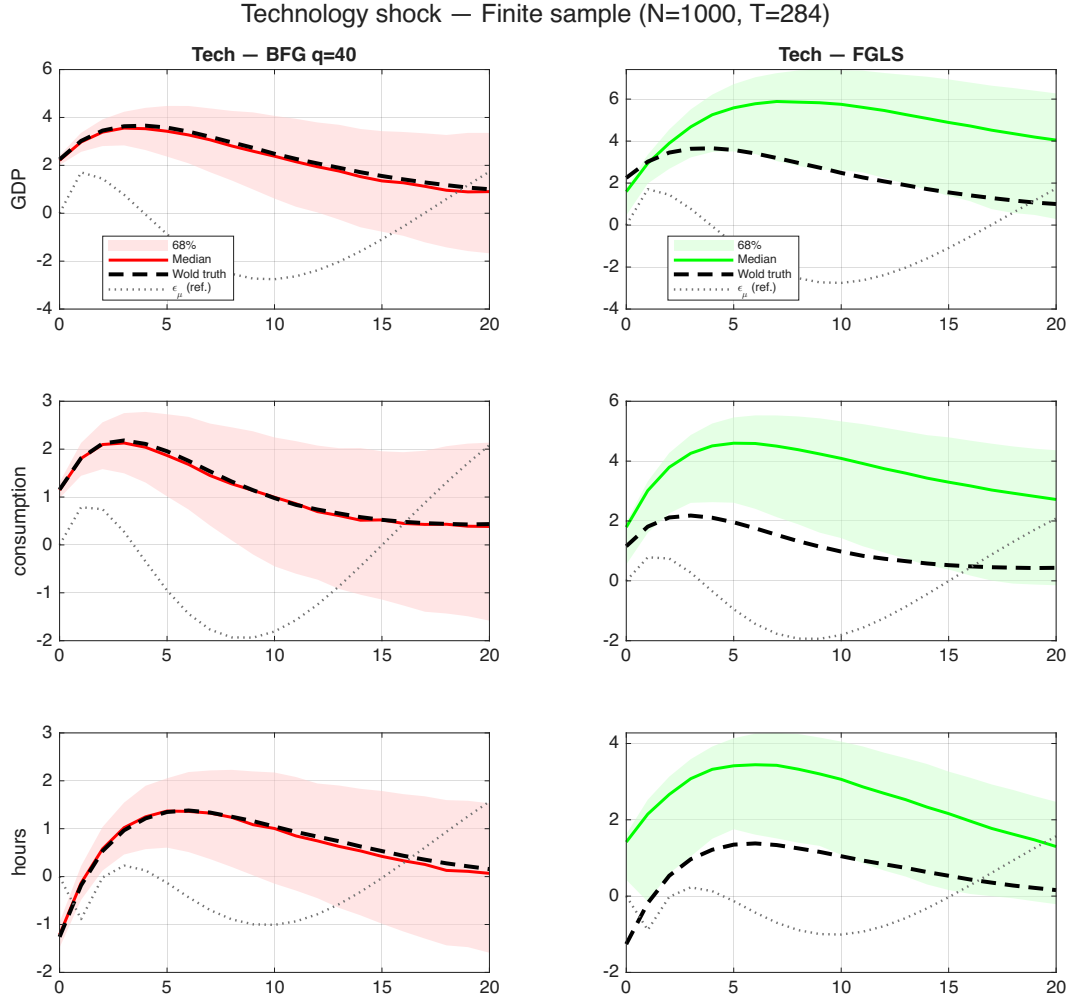


Figure 25: Technology shock: finite-sample distributions ($N = 1,000, T = 284$). Left: BFG $q = 40$. Right: FGLS. The dashed black line is the Wold-innovation IRF (the technology-side Chahrour and Jurado (2018) target that BFG identifies); the gray dotted line is the structural permanent-shock IRF ε_μ shown as a secondary reference.

The robustness of this finding is confirmed across different specifications. Table 12 reports the $\mathbf{nsr}_{20}^{\text{news}}$ for four configurations varying the number of factors (7 versus 8) and the number of VAR lags (4 versus 8).

Table 12: Robustness of the news-to-signal ratio at $h = 20$ across four VAR specifications, varying the number of FRED-MD principal components (7 vs. 8) and the lag length (4 vs. 8). All values cluster around 0.90, confirming that the empirical decomposition is not sensitive to the auxiliary information set used to span the agent’s signal. Sample: 1960:Q1–2019:Q4.

Configuration	ICS 5-Year	ICS 12-Month
7 factors, 8 lags	0.90	0.91
8 factors, 8 lags	0.88	0.92
7 factors, 4 lags	0.90	0.90
8 factors, 4 lags	0.89	0.91

The estimates are remarkably stable across configurations, consistently yielding $\alpha \approx 0.90$.

Figure 26 displays the $\mathbf{nsr}_h^{\text{news}}$ and the selection criterion $SC(h)$ for the ICS 5-Year (BUS5) signal across all four specifications.

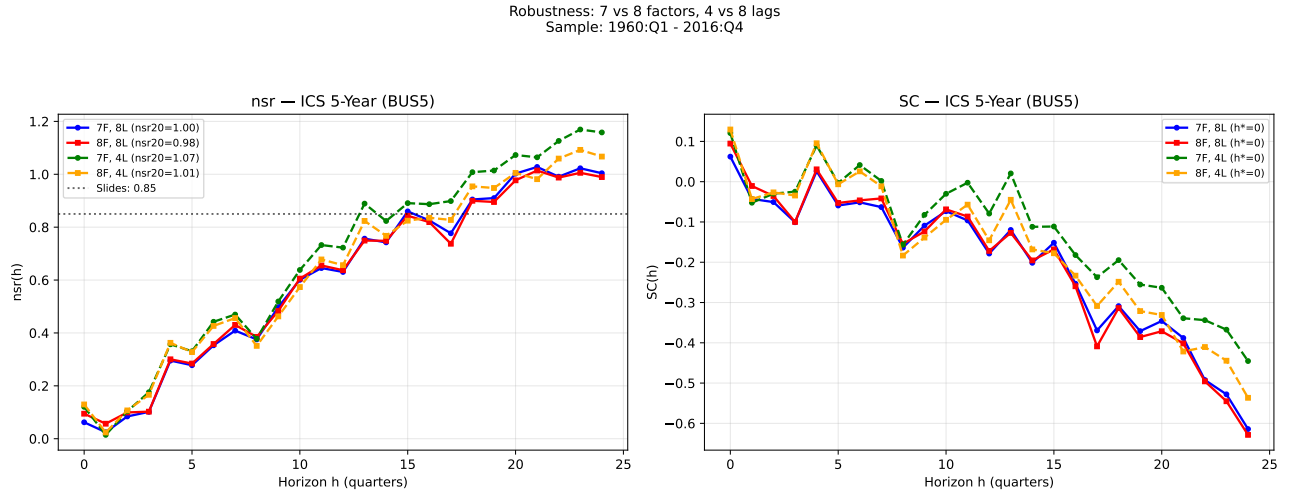


Figure 26: News-to-signal ratio and selection criterion for ICS 5-Year (BUS5). Four specifications are compared: 7 versus 8 factors, 4 versus 8 VAR lags. The dotted line marks the slides’ benchmark value of 0.85.

E.2 Benchmark II: FGLS estimates

We complement the semi-parametric analysis with the FGLS identification strategy based on dynamic rotations of the SVAR impulse responses. The procedure estimates a VAR with 8 lags, applies the Cholesky decomposition with TFP ordered first (identifying the surprise shock), and then rotates the second structural shock (associated with the signal) into news and noise components using the Blaschke factorization. We consider two VAR specifications: (i) TFP in *levels* (log TFP), corresponding to the approach used in Forni et al. (2017a), and (ii) TFP in *growth* (Δz_t).

Table 13 reports the FGLS estimates for both specifications with 7 factors.

The level specification yields $\alpha \approx 0.87$ – 0.88 for both signals, indicating a small but non-negligible noise component. These values are close to the benchmark of 0.85 reported in earlier work. The growth specification produces negative values of $\sigma_\varepsilon/\sigma_v$, indicating that the Cholesky-based FGLS rotation requires care when applied to a VAR in first differences: the sign of the ratio depends on the relative orientation of the surprise and signal impulse responses at long horizons.

Table 13: FGLS dynamic-rotation estimates of the news-to-signal ratio. The VAR is estimated with $p = 8$ lags and 7 FRED-MD principal components, with TFP ordered first (Cholesky); the second structural shock is rotated into a news / noise pair via Blaschke factorisation. Two TFP specifications are compared: log levels (*Level*, as in Forni et al., 2017a) and first differences (*Growth*). Both specifications, after sign normalisation, yield $|\alpha| \approx 0.90$, in agreement with the semi-parametric news-to-signal ratio of Table 11. Sample: 1960:Q1–2019:Q4.

Signal	TFP specification	$\sigma_\varepsilon/\sigma_v$	$\alpha = \sigma_\varepsilon/\sigma_s$
ICS 5-Year (BUS5)	Level	1.81	0.88
ICS 5-Year (BUS5)	Growth	−2.46	−0.93
ICS 12-Month (BUS12)	Level	1.78	0.87
ICS 12-Month (BUS12)	Growth	−11.31	−1.00

The FGLS-identified impulse responses from the level specification confirm that a news shock has a permanent, positive effect on TFP with a hump-shaped diffusion pattern, while the noise shock has no permanent effect on TFP—consistent with the theoretical restrictions.

F Additional Empirical Results

This online appendix collects the robustness and validation material for the empirical application of Section 6: the lead-window and single-lead robustness of the IRFs, the stability of all specifications in the information set, the news-to-signal level diagnostics for the two surveys, the VAR-implied-signal application, and the discussion of signal choice.

Robustness to the lead window q . The empirical results above use $q = 20$. This choice is supported *a priori* by the bias-corrected plateau-based selector (27): the adjusted statistic $\theta_h^{\text{stat,adj}}$ peaks at $h^* = 20$ for all three signals (BUS5, BUS12, and b_t), coinciding with the lead window used throughout the local-projection exercise; h^* estimates the materialisation horizon, and the discussion of the lead support k^* above is what supports treating that horizon as the window identification requires. We additionally re-run LP_W news and noise IRFs for $q \in \{1, 5, 10, 15, 20\}$ to verify empirically that the residual contamination disappears at the plateau-implied window. Figures 27 and 28 overlay the IRFs for the five q values for BUS5 and BUS12 respectively.

The two columns deliver very different robustness profiles. The *news* IRFs (left column) are highly stable for $q \geq 10$: TFP is essentially identical across all five q values, and the GDP, consumption, and hours profiles for $q \in \{10, 15, 20\}$ cluster tightly around the same hump-shaped trajectories. Only $q = 1$ and $q = 5$ depart visibly — with smaller and more erratic shapes — consistent with too narrow a lead block missing the bulk of the signal’s predictive content for future TFP. The *noise* IRFs (right column) are much more sensitive to q : the TFP panel is the smoking gun. By construction the noise shock has zero permanent effect on TFP, and only $q = 20$ delivers a numerically zero IRF; for $q = 1$ the IRF rises monotonically to about 1% at $h = 20$, indicating that a narrow lead block does not span the news content of s_t and the procedure mislabels news content as noise. With $q = 15$ the contamination is small but not zero; with $q = 20$ it disappears mechanically. The same pattern carries through to GDP, consumption, and hours: small q produces persistent (contaminated) noise IRFs, while $q \geq 15$ produces the expected hump-shaped transitory pattern.

The choice $q = 20$ is therefore doubly justified in our application. For news identification any $q \geq 10$ works; for noise identification q must extend up to the survey’s forecast horizon to fully orthogonalise s_t against future TFP. BUS5 and BUS12 are 5-year and 12-month consumer-confidence indices, but we use the same $q = 20$ for both — conservatively matching the longer survey horizon — which delivers a clean noise identification in both samples. The non-zero noise-on-TFP IRF at small q is itself a useful sample-level diagnostic: it measures how much news content the signal carries about TFP at horizons beyond q , and tells the practitioner whether the chosen lead window is large enough.

A complementary check uses a *single* TFP lead Δz_{t+q} as the only future-TFP control, instead of the full block $\{\Delta z_t, \dots, \Delta z_{t+q}\}$ (Figure 29 for BUS5; the BUS12 picture is similar). The contrast with the block specification confirms why the block is the structurally appropriate choice for noise identification. With a single lead at the survey horizon, the

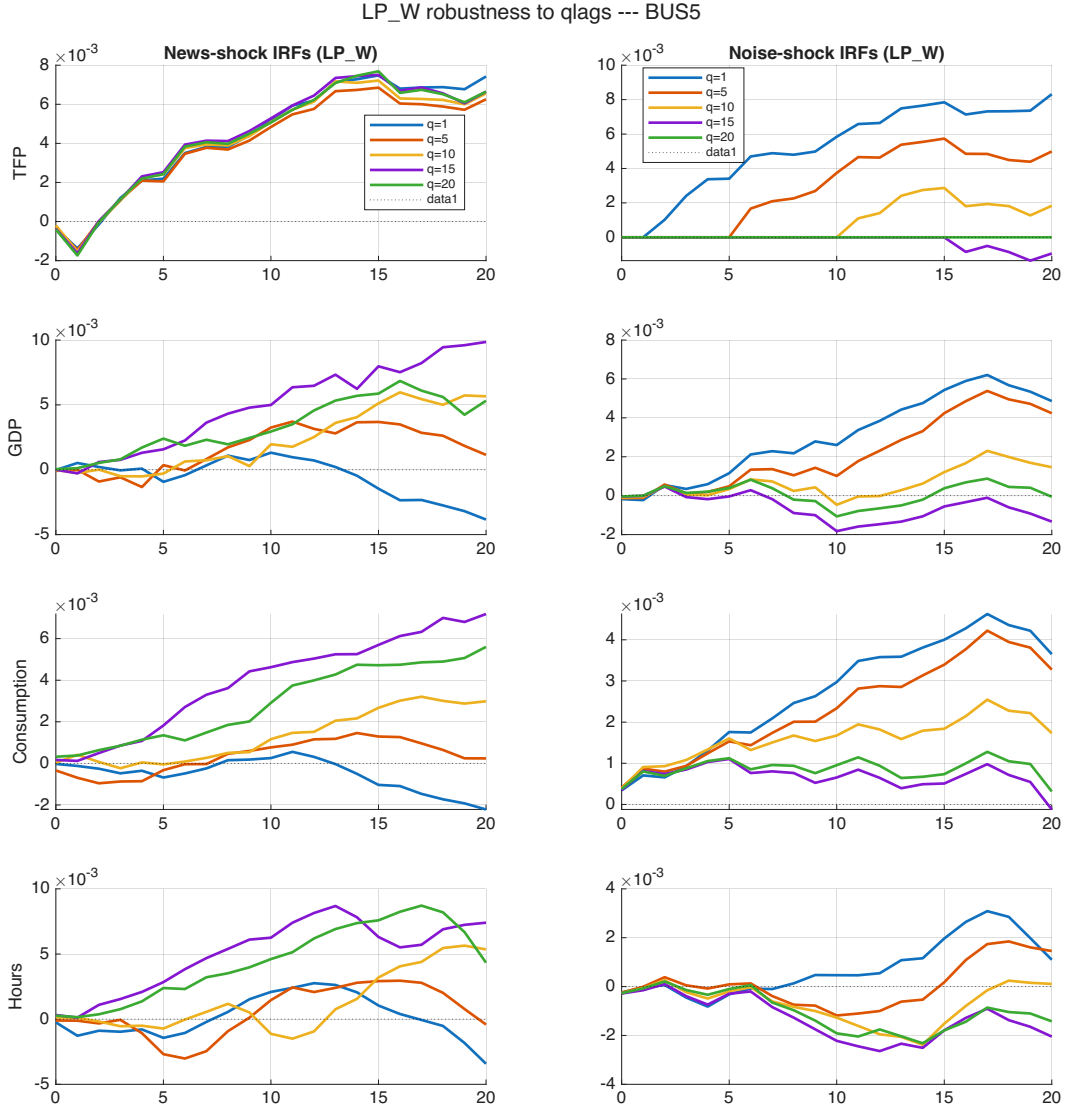


Figure 27: Robustness of LP_W news (left) and noise (right) IRFs to the lead window $q \in \{1, 5, 10, 15, 20\}$, signal = BUS5. Each panel overlays the five IRFs for the corresponding variable. News IRFs cluster tightly for $q \geq 10$; noise IRFs require $q \geq 15$ to avoid news contamination, and only $q = 20$ delivers a mechanically zero noise-on-TFP IRF.

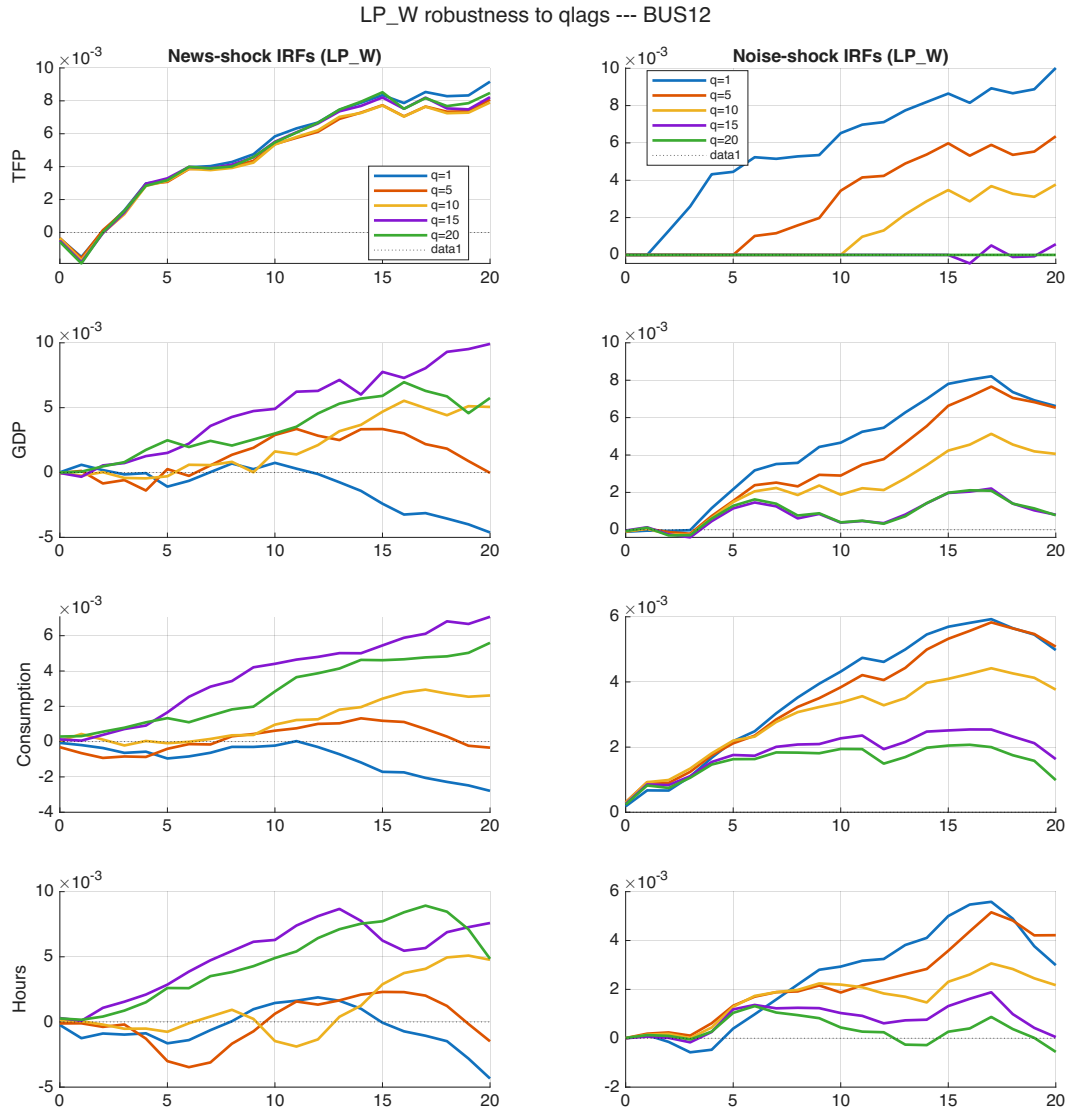


Figure 28: Robustness to q for the BUS12 signal. Conventions as in Figure 27.

noise IRF on TFP is contaminated at *every* value of q — including $q = 20$, where the IRF still rises monotonically to about 1% at $h = 20$ rather than the mechanically zero value the block specification delivers. Partialling out only Δz_{t+q} from s_t leaves the news content at horizons $0, 1, \dots, q - 1$ inside s_t , so the residualised regressor is not orthogonal to future TFP and the LP slope picks up that residual news channel. To make s_t orthogonal to the entire block of future TFP up to horizon q , the LP must include the entire block of leads, not just the lead at the endpoint. The same lesson does not apply to news identification: the news IRFs from the single-lead and block specifications agree closely for $q \geq 10$, since for the news regression both specs partial out the same lag block and the focus regressor is the same lead.

Stability of the structural news IRF as the information set grows. The identification theory ties the structural reading of the news IRFs to informational sufficiency, and makes its failure observable at the IRF level: on an insufficient information set the estimand is contaminated, so enriching the information set should keep moving the estimated IRFs, while on a sufficient one further enrichment should leave them unchanged. Figure 30 implements this check on the estimated $\mathcal{O}_2$ object itself, re-estimating the surprise-augmented news IRFs (LP_ε) with 0, 2, 4, and all 7 FRED-MD factors in the information set. The qualitative profile — a persistent, hump-shaped expansion — is invariant across information sets, but the poorer sets *overstate* the long-horizon responses, by up to two thirds of the peak on TFP in the factor-free system, and the estimated IRF sequence contracts monotonically toward the seven-factor baseline: the maximal deviation from that baseline falls to 9–18 percent of the peak response for GDP, consumption, and hours at four factors (22–26 percent for TFP), roughly halving with each enlargement. This convergence, together with the orthogonality test reported at the start of this section (which no longer rejects at seven factors) and the collapse of the FGLS noise shares in Section 6.4, constitutes the empirical case that the factor-augmented information set is rich enough for the structural reading — the IRF-level counterpart of the collapse of the lead support k^* under Assumption A.3'.

The ladder in the data: stability of the leads-only specifications. The same information-set experiment applied to the leads-only specifications completes the picture, and the contrast across the three exercises is the recoverability ladder made visible. The LP_W news IRFs (Figure 31) are essentially invariant to the factor block: on GDP and consumption the four information sets deliver responses within 10–20 percent of the seven-factor baseline even with no factors at all — consistent with Proposition 1(i) and the ladder of Section 3, which place the technology-side target at the bottom of the informational requirements, and with the near-unpredictability of TFP growth, which pins down its Wold innovation almost regardless of the conditioning set (the TFP row itself, estimated by the signal-focus variant, varies somewhat more). The leads-only *noise* IRFs (Figure 32) display the opposite pattern: the responses of GDP, consumption, and hours shrink monotonically — by a factor of three to six — as factors are added, converging to the small humps of the baseline *from above*. This is the observable signature of Corollary 1(ii): the leads-only estimand is a mixture whose mislabelled news content the richer information set progressively reclassifies, and its information-set fragility is precisely why the combined specification — whose residualised signal converges to the structural noise shock — is the paper's headline noise estimator. Read together, the three stability exercises reproduce the ordering of the ladder in the data: the Wold target is attainable low, the structural news target stabilises only as informational sufficiency is restored, and the inconsistent leads-only noise estimand never stops moving until it is.

Figure 33 reports the news-to-signal ratio nsr_h^{news} and the associated selection criterion $SC(h)$ for BUS5.

The statistic rises monotonically with the horizon and stabilizes at about $\alpha \approx 0.90$ around $h = 20$ quarters, in line with the NSR estimates reported in Online Appendix E (Table 11). This indicates that roughly 90% of the variance in BUS5 innovations comes from news about future TFP, while the residual 10% reflects genuine noise.

Figure 34 displays the news-to-signal ratio for BUS12.

As for BUS5, the ratio rises monotonically with h and stabilizes near 0.9 at long horizons, with the selection criterion pointing to roughly the same horizon. The overall conclusion is robust across the two signals: most of the innovation in the Michigan survey-based confidence index is attributable to genuine news about future productivity, and only a modest fraction reflects noise. The macro impulse responses differ in magnitude but agree in sign and persistence across BUS5 and BUS12, which lends additional credibility to the news–noise decomposition obtained with the FGLS, CJ, and BFG strategies.

Application 3: VAR-implied signal $b_t = \mathbb{E}_t[z_{t+h}] - z_t$

A natural alternative to a survey signal is to construct s_t as the h -period-ahead rational forecast of TFP itself, following Chahrour and Jurado (2022). We estimate a parsimonious 3-variable VAR with $p = 4$ lags on TFP growth (Δz_t), real

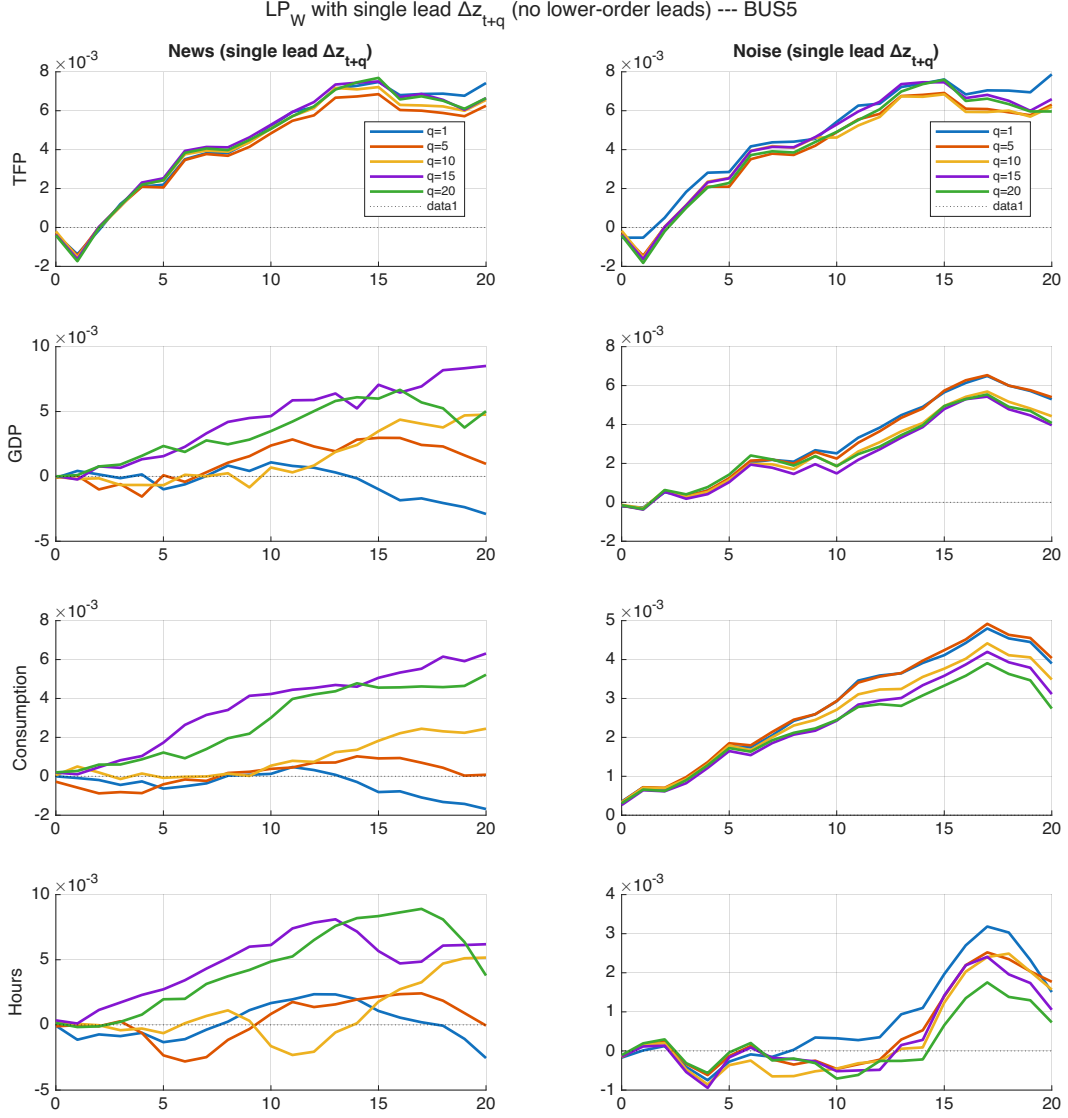


Figure 29: LP_W single-lead variant: only Δz_{t+q} enters as control instead of the full block $\{\Delta z_t, \dots, \Delta z_{t+q}\}$, for $q \in \{1, 5, 10, 15, 20\}$, BUS5. The noise IRF on TFP is contaminated at every q (including $q = 20$, top right): partialling out a single future-TFP lead does not span the news content of s_t at horizons $0, \dots, q - 1$, and the residualised s_t is not orthogonal to future TFP. Compare with Figure 27 (block specification), where the noise IRF on TFP is mechanically zero at $q = 20$.

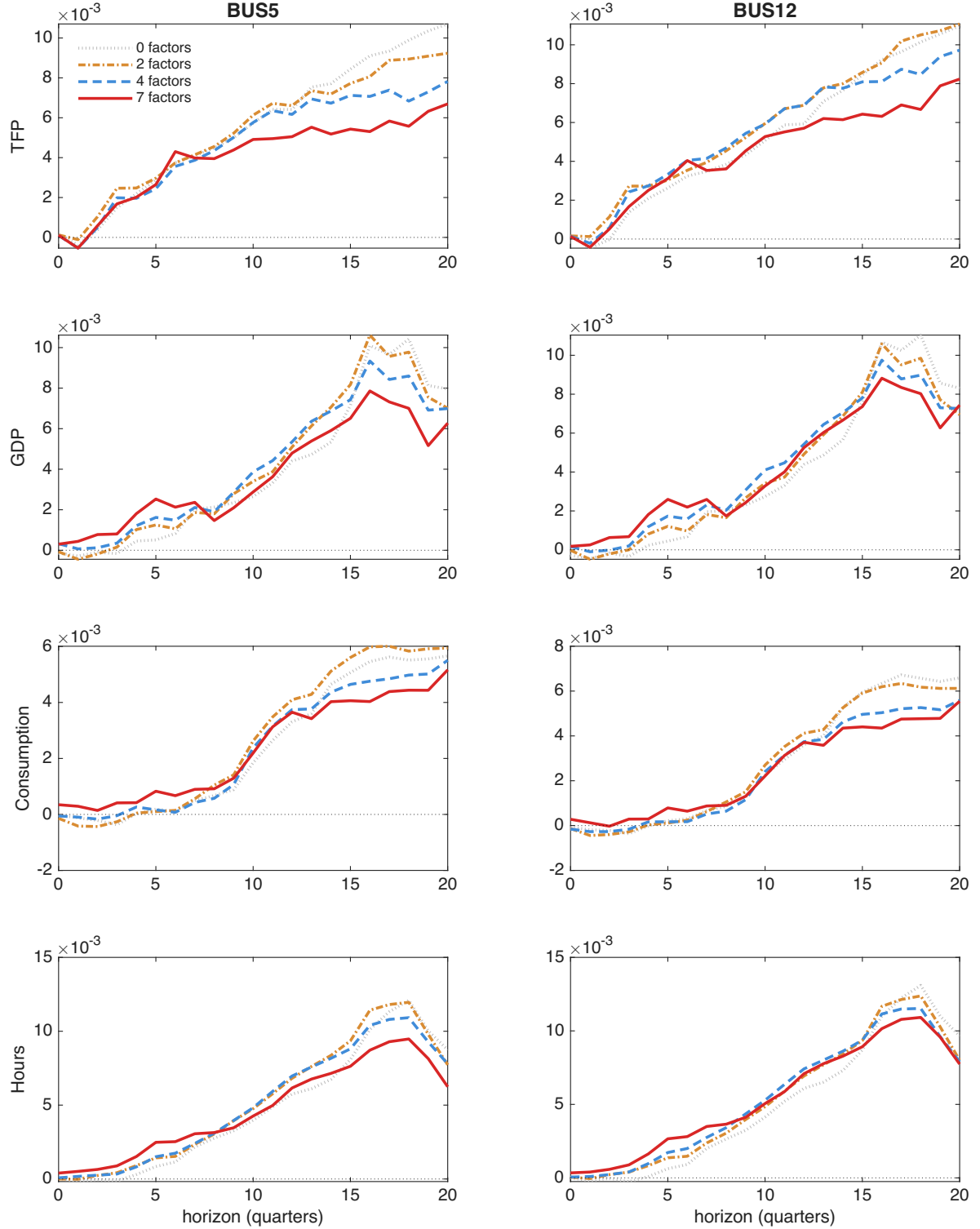


Figure 30: Stability of the structural news IRFs in the information set. Surprise-augmented news IRFs (LP_ε , point estimates) re-estimated with 0, 2, 4, and 7 FRED-MD factors in the lag block; $p = 4$ lags, $q = 20$, $H = 20$ quarters, sample 1960:Q1–2019:Q4. Rows: TFP, GDP, consumption, hours (level IRFs); columns: BUS5 and BUS12 signals. The estimated IRFs converge monotonically toward the seven-factor baseline as the factor block grows — the IRF-level counterpart of the informational-sufficiency requirement, whose orthogonality test stops rejecting at the full factor block.

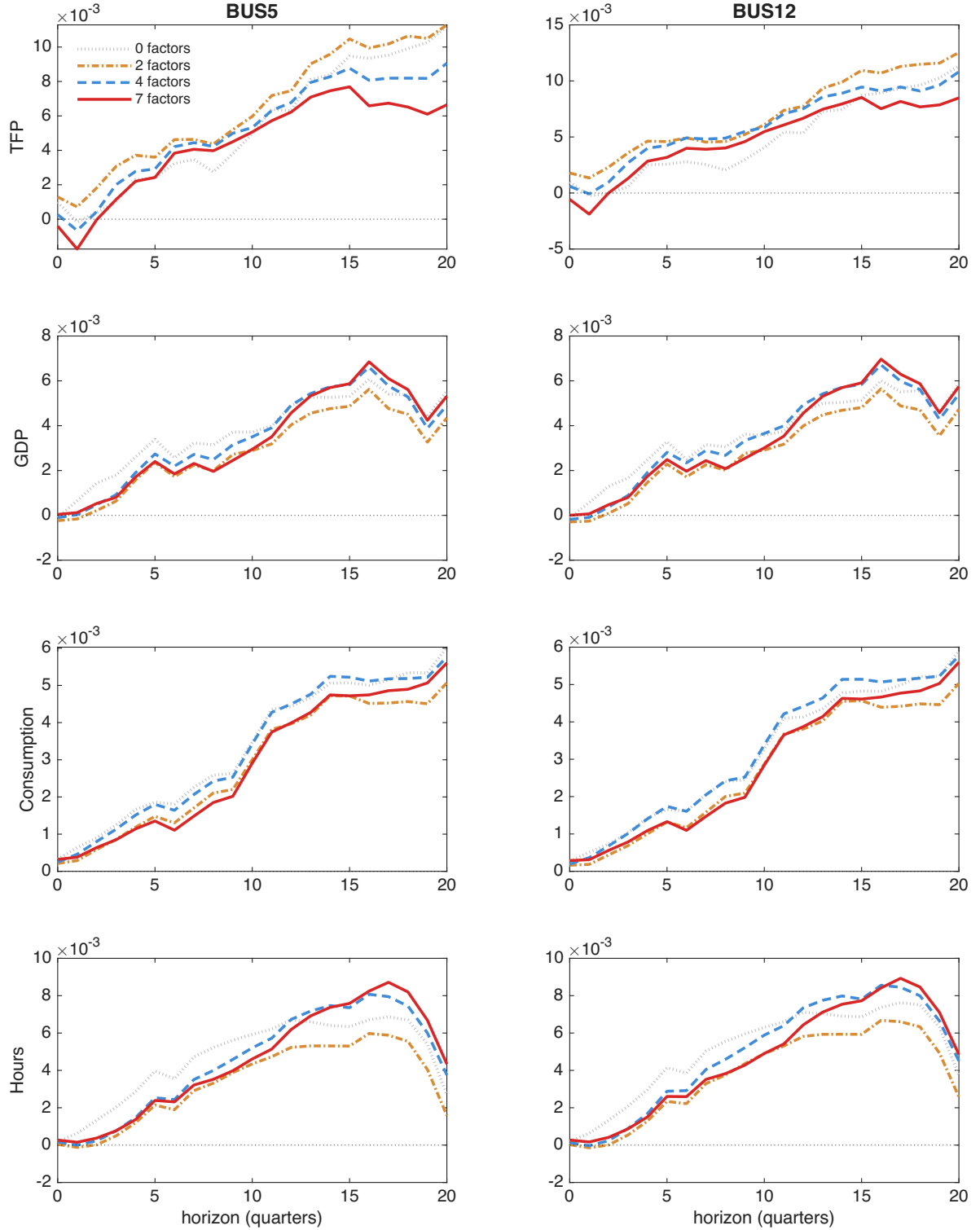


Figure 31: Stability of the LP_W news IRFs in the information set. Leads-only news IRFs (point estimates) re-estimated with 0, 2, 4, and 7 FRED-MD factors in the lag block; $p = 4$, $q = 20$, $H = 20$, sample 1960:Q1–2019:Q4. Rows: TFP, GDP, consumption, hours; columns: BUS5 and BUS12. In contrast with the structural news IRFs of Figure 30, the technology-side target is essentially invariant to the information set, as Proposition 1(i) predicts.

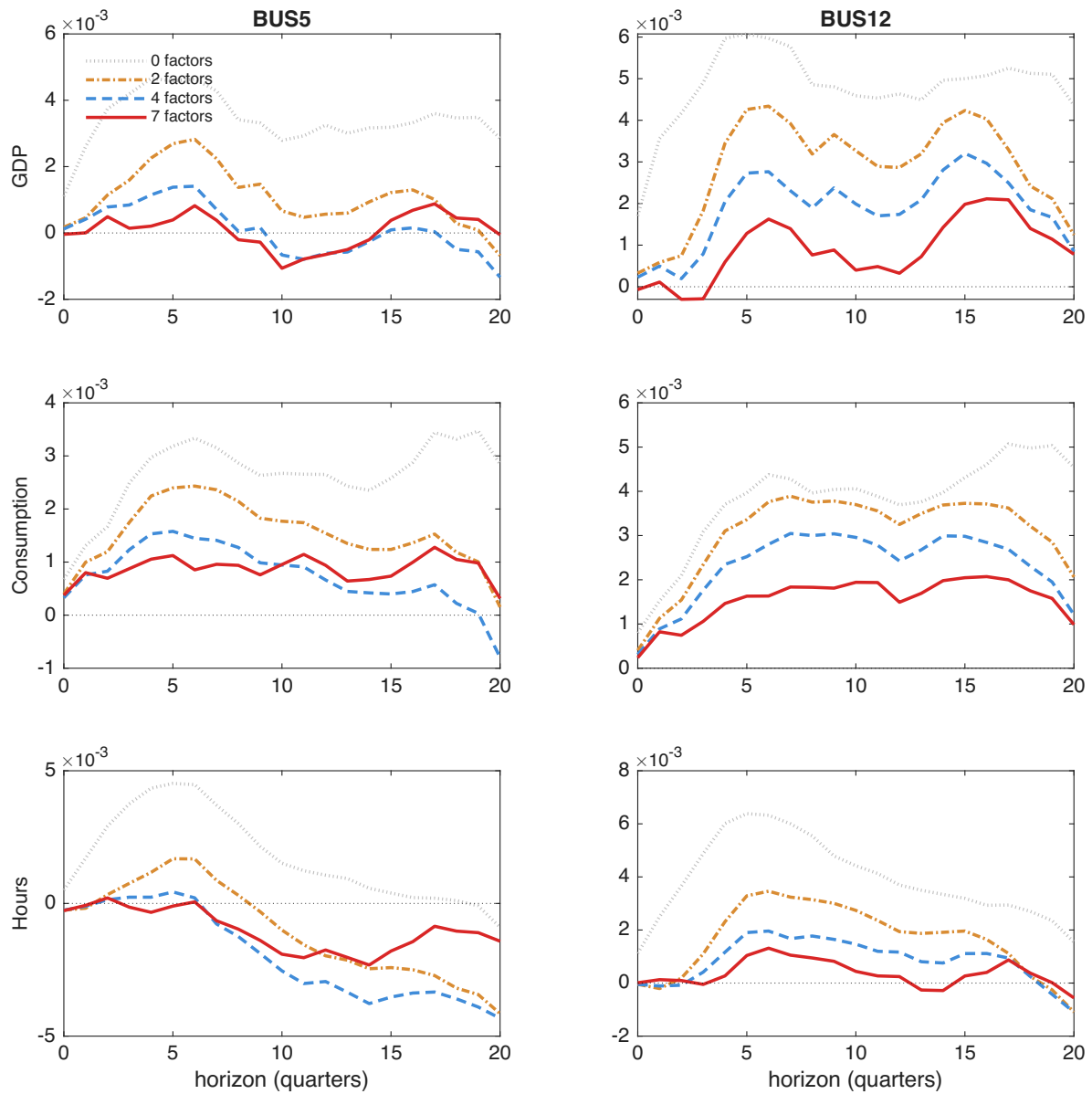


Figure 32: Instability of the leads-only noise IRFs in the information set. Leads-only noise IRFs (11) re-estimated with 0, 2, 4, and 7 FRED-MD factors; conventions as in Figure 31 (the TFP row, mechanically zero at $q = 20$, is omitted). The responses shrink toward the seven-factor baseline from above as factors reclassify the mislabelled news content of the residualised signal — the observable signature of the mixture result of Corollary 1(ii), and the empirical case for the combined specification used in the headline results.

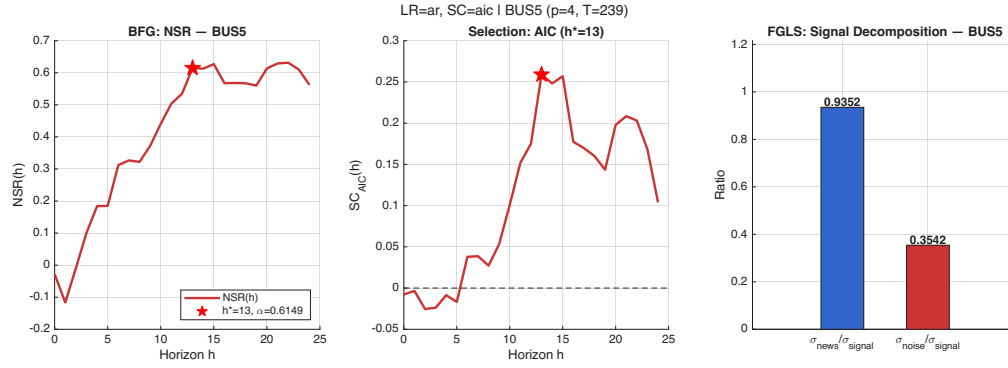


Figure 33: News-to-signal ratio and selection criterion for Michigan ICS 5-Year (BUS5). The statistic increases monotonically with h and stabilizes near $\alpha \approx 0.90$ at long horizons.

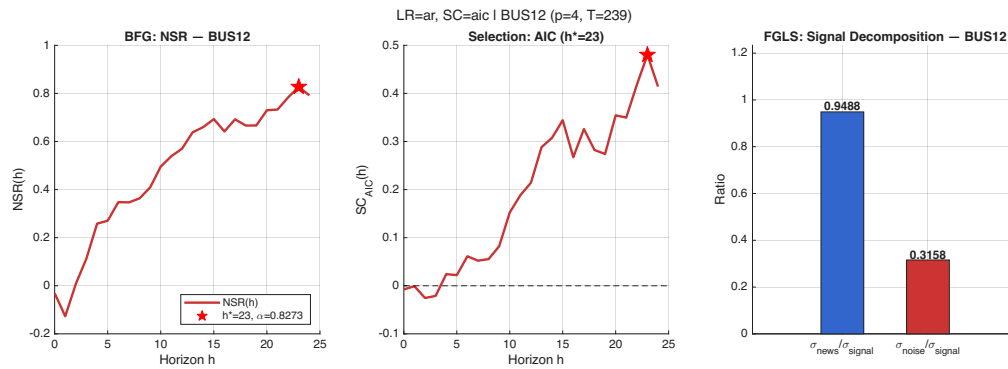


Figure 34: News-to-signal ratio and selection criterion for Michigan ICS 12-Month (BUS12). The statistic again stabilizes around $\alpha \approx 0.90$ at long horizons.

S&P 500 growth (Δsp_t , monthly real-stock-price series aggregated to quarterly by averaging), and annual GDP-deflator inflation (π_t), and iterate it forward to compute

$$b_t = \sum_{k=1}^h \mathbb{E}_t[\Delta z_{t+k}] = \mathbb{E}_t[z_{t+h}] - z_t,$$

with $h = 20$ to match the lead window $q = 20$ used throughout.

The forecast VAR includes only variables that are absent from the LP information set: Δsp_t and π_t are not directly in the LP, and the FRED-MD principal components in the LP do not separately span them.⁷ The LP and FGLS information sets are otherwise identical to Applications 1 and 2: Δz_t , b_t , Δy_t , Δc_t , N_t , R_t , augmented with the seven FRED-MD principal components. The effective sample is 1962:Q1–2019:Q4 ($T_b = 233$ quarters), reflecting the loss of four quarters to the YoY inflation construction and four more to the VAR’s initial conditions.

The news- and noise-shock IRFs for b_t are reported in Figures 35–36. The qualitative picture matches the survey-based applications: news shocks generate persistent positive responses in TFP, GDP, consumption, and hours; noise shocks have small transitory effects on real activity that decay back to zero. Quantitatively, the news IRFs on real activity reach approximately 0.5% at $h = 20$ — the same order of magnitude as the BUS5 and BUS12 cases — and the noise IRFs on real activity stay within ± 0.002 in absolute value at long horizons. This is direct empirical evidence that our LP machinery and the spectral factorisation of Chahrour and Jurado (2022) target the same news/noise object: the time-domain partialling carried out by LP_W (TFP leads as controls) and LP_ε (TFP leads + surprise residuals as controls) recovers the same decomposition that Chahrour and Jurado (2022) obtain by constructing b_t as the rational forecast and applying their lower-triangular spectral factorisation.

F.1 The choice of signal under the random-walk approximation of TFP

The contrast between the adjusted news content of survey signals ($\theta_{20}^{\text{stat,adj}} \in [0.42, 0.46]$) and of the VAR-implied rational forecast ($\theta_{20}^{\text{stat,adj}} \approx 0.03$) reported in point (4) above invites a short discussion of how the empirical properties of Fernald (2014) TFP shape the choice of signal in a noisy-news identification.

TFP as an approximate random walk. Over 1960:Q1–2019:Q4, Δz_t is statistically indistinguishable from a white noise: Ljung–Box statistics fail to reject the white-noise null at every standard horizon ($L = 4, 8, 12, 20$ with p -values of 0.45, 0.65, 0.73, 0.43), the Newey–West long-run-to-short-run variance ratio is 0.94, and the largest sample autocorrelation in absolute value over the first 20 lags is 0.12. This well-documented near-random-walk property of utilisation-adjusted TFP (see Section 1 above and the original Fernald (2014) discussion) bears directly on the construction of $b_t = \sum_{k=1}^q \mathbb{E}_t[\Delta z_{t+k} \mid \mathcal{I}_t^{\text{VAR}}]$: when Δz is unforecastable from past macro observables, the population variance of b_t is close to zero, so in finite samples b_t tends to be dominated by VAR estimation noise. Empirically $\text{Var}(b_t) \approx 3 \times 10^{-4}$, six orders of magnitude smaller than $\text{Var}(\text{BUS5}) \approx 3.5 \times 10^2$, and $\theta_{20}^{\text{stat,adj}}(b_t) \approx 0.03$ confirms that little of the residual sample variance reflects structural news content.

A complementary diagnostic for the recoverability framework. This empirical observation is fully consistent with the spirit of the recoverability literature. Chahrour and Jurado (2022) (henceforth CJ22) develop a general spectral framework for identifying news and noise from the joint dynamics of (z_t, s_t) , and explicitly emphasise that recoverability requires the signal s_t to span news content not already in the joint history. Our adjusted statistic $\theta_h^{\text{stat,adj}}$ provides a simple finite-sample diagnostic for this informational content: a signal with $\theta_h^{\text{stat,adj}}$ close to zero carries little information about realised future TFP movements, so the resulting news-noise decomposition is largely uninformative about agents’ true belief signals. Conversely, signals such as the Michigan consumer-confidence series, which by construction draw on information outside the econometrician’s VAR — private beliefs of households, micro-level signals, and unstructured information not summarised in macro time series — attain $\theta_{20}^{\text{stat,adj}} \in [0.42, 0.46]$ and yield identifications whose news-noise decompositions remain informative. In this respect our diagnostic complements rather than substitutes

⁷When the forecast VAR shares variables with the LP, b_t is mechanically explained by the LP’s lag block and the slope on b_t is undefined or numerically degenerate. The 3-variable VAR chosen here passes information from financial markets and the price level into the LP via the constructed signal, mirroring the spirit of Chahrour and Jurado (2022)’s spectral construction without inducing the collinearity that would arise from a kitchen-sink forecast VAR.

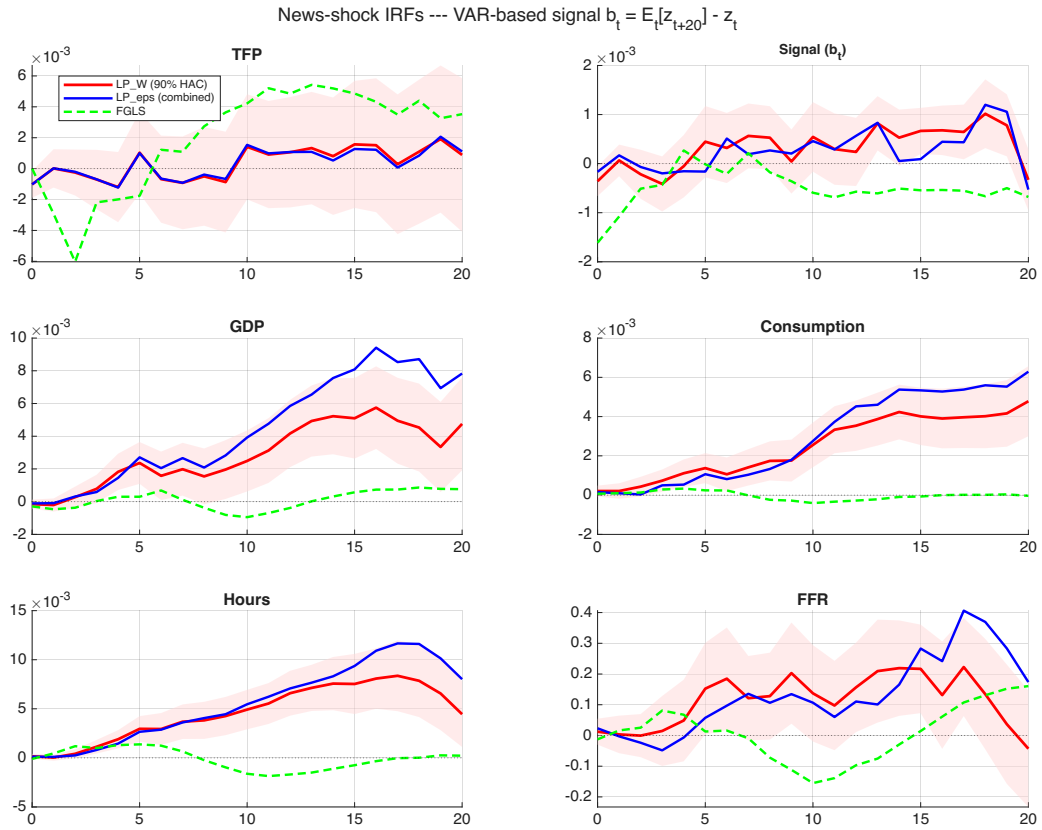


Figure 35: News-shock IRFs to the VAR-implied signal $b_t = \mathbb{E}_t[z_{t+20}] - z_t$, constructed from a 3-variable VAR $(\Delta z_t, \Delta sp_t, \pi_t)$ with $p = 4$ lags. Same LP / FGLS overlay conventions as Figure 1. Sample: 1962:Q1–2019:Q4 ($T_b = 233$).

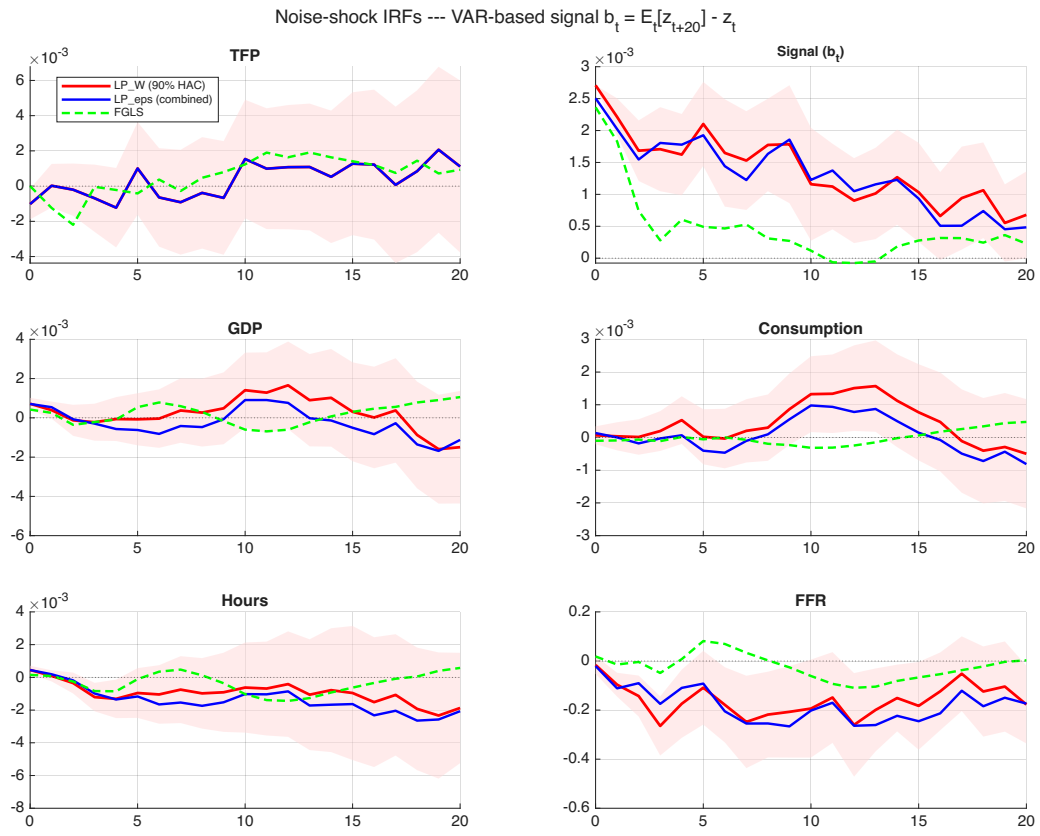


Figure 36: Noise-shock IRFs to the VAR-implied signal $b_t = E_t[z_{t+20}] - z_t$. Same conventions as Figure 35.

for the spectral identification of CJ22: it operationalises in a model-free way the signal-side requirement that their framework already articulates structurally — and when the informational conditions of Theorem 1 hold, the local projections of Section 2 deliver the same news/noise decomposition without the spectral machinery, as Application 3 verifies directly on the VAR-implied signal.

Relation to other signal choices. Several existing studies use signals constructed from $\mathcal{I}_t^{\text{VAR}}$, including the VAR-based rotations of Barsky and Sims (2011), Beaudry and Portier (2006), and Forni et al. (2017a). Each addresses the non-recoverability issue through different means — long-run restrictions, maximum-FEV criteria, factor augmentation — and each remains a legitimate approach when its identifying conditions are met. Our analysis does not invalidate any of these strategies; it instead suggests that the model-free $\theta_h^{\text{stat,adj}}$ can serve as a useful pre-test for any candidate signal, and that the relatively low adjusted news content of VAR-implied b_t over the present sample provides a complementary reason — alongside the standard measurement-error motivation — to favour direct expectation measures whenever they are available. Table 2 reports $\theta_h^{\text{stat,adj}}$ for all three signal choices used in our application; we report the figure rather than label any approach as preferred, leaving the comparison to the reader.

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